

NAND BOM Options

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335800169	1	NAND, H, 32GB, 16nm, TLC	U1701	CRITICAL	NAND_32G
335800182	1	NAND, H, 128GB, 16nm, TLC	U1701	CRITICAL	NAND_128G
335800156	1	NAND, H, 256GB, 3Dv3, TLC	U1701	CRITICAL	NAND_256G
13880867	6	QAP, X38, 110F, 29A, 6, 3V, 6.45NM, WRITL, 0402	C1748, C1713, C1716, C1718, C1701, C1733	CRITICAL	NAND_32G
138800003	6	QAP, X38, 110F, 29A, 6, 3V, 6.45NM, WRITL, 0402	C1748, C1713, C1716, C1718, C1701, C1733	CRITICAL	NAND_128G
138800003	6	QAP, X38, 110F, 29A, 6, 3V, 6.45NM, WRITL, 0402	C1748, C1713, C1716, C1718, C1701, C1733	CRITICAL	NAND_256G

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335800201	335800169	ALTERNATE	U1701	T, 15nm, MLC, 32GB
335800209	335800169	ALTERNATE	U1701	S, 16nm, MLC, 32GB
335800195	335800182	ALTERNATE	U1701	SS, 1Ynm, TLC, 128GB
335800180	335800182	ALTERNATE	U1701	T, 15nm, TLC, 128GB
335800179	335800182	ALTERNATE	U1701	SD, 15nm, TLC, 128GB
335800148	335800183	ALTERNATE	U1701	SD, 30v2, TLC, 256GB
335800190	335800183	ALTERNATE	U1701	SS, 30v3, TLC, 256GB

#22686038:See Radar

CAYMAN DDR Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339800254	339800253	ALTERNATE	ALL	DDR-H, 2G, B1
339800255	339800253	ALTERNATE	ALL	DDR-S, 2G, B1
339800257	339800253	ALTERNATE	ALL	DDR-H, 3G, B1
339800258	339800253	ALTERNATE	ALL	DDR-S, 3G, B1
339800256	339800253	ALTERNATE	ALL	DDR-M, 3G, B1

Magnesium Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
338S00173	338S00203	ALTERNATE	U2402	Larger Wafer (-25 Cлов) Magnesium

Carbon Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
338S00087	338S00226	ALTERNATE	U2401,U2404	Updated version of Carbon

Phosphorous Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
338S00230	338S00188	ALTERNATE	U2403	ST version of Phosphorus

UT LDO Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S00889	353S00015	ALTERNATE	U2501	ST, LDO REG, 3.325V, CSP 0.45x0.45

Mamba LDO Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353800932	353800576	ALTERNATE	U3801	ST, LDO REG, 2.75V

EEPROM Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00234	335S00233	ALTERNATE	U1101	10, 830PROM, 16000, 1, RV, 12C, M23P4, OR08001

Active Diode Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S00106	376S00047	ALTERNATE	Q2101	DIODES INC. ACT DIODE

DDR PLL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S00095	155S00068	ALTERNATE	FL1501	FEER 80, 100CMM, 154, 100MM, 20MM, 10105

Power Inductor Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152800118	152800075	ALTERNATE	ALL	301, PWR, 1002.1, 5 TP, 3.5A, 5.148 OHM, 2014
152800077	152800397	ALTERNATE	ALL	301, PWR, 1002.1, 5 TP, 2.35A, 5.151 OHM, 2014
152800121	152800081	ALTERNATE	ALL	301, PWR, 1002.1, 47 TP, 3.8A, 5.149 OHM, 2011
152800123	15281936	ALTERNATE	ALL	301, PWR, 1002.1, 19A, 3.70A, 5.151 OHM, 2023
152800402	152800366	ALTERNATE	ALL	301, M012, 1001.1, 1A, 4, 322 OHM, 0403
152800297	15281843	ALTERNATE	ALL	CUSTOM 2012 100
152800365	152800297	ALTERNATE	ALL	CUSTOM 2012 100
152800398	152800204	ALTERNATE	ALL	301, PWR, 5, 2020, 23A, 6, 7a, 250000, 2014
152800120	152800077	ALTERNATE	ALL	For Chassis Inductor only
152800117	152800074	ALTERNATE	ALL	301, PWR, 1002.1, 5 TP, 3.5A, 5.148 OHM, 2014

updated 11/12

updated 11/12

reverted 11/13

For Chestnut inductor; so it doesn't interfere with PMU inducotr Buck 7 alts

Except BUCK5 LX (BUCK5 LX is Taiyo only)

Global R/C Alternates

[illegible]

New in Carrier, per #25737217

Global Ferrite Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155800067	15580581	ALTERNATE	ALL	PERA, 2400MH, 0.360UM DCM, D21
15580581	155800067	ALTERNATE	ALL	PERA, 2400MH, 0.360UM DCM, D21
155800012	155800168	ALTERNATE	ALL	PLPR, 63 OHMS, 5050
155800194	15580610	ALTERNATE	ALL	PERA 80, 15500MH, T50
15580885	15580610	ALTERNATE	ALL	PERA 80, 15500MH, T50
152800489	152800456	ALTERNATE	ALL	PERA 80, 0.470H, TV

Global Varistor Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0168	377S0140	ALTERNATE	ALL	VARIATOR, 6.8V, 100PF, 01005

ACC BUCK Alternates

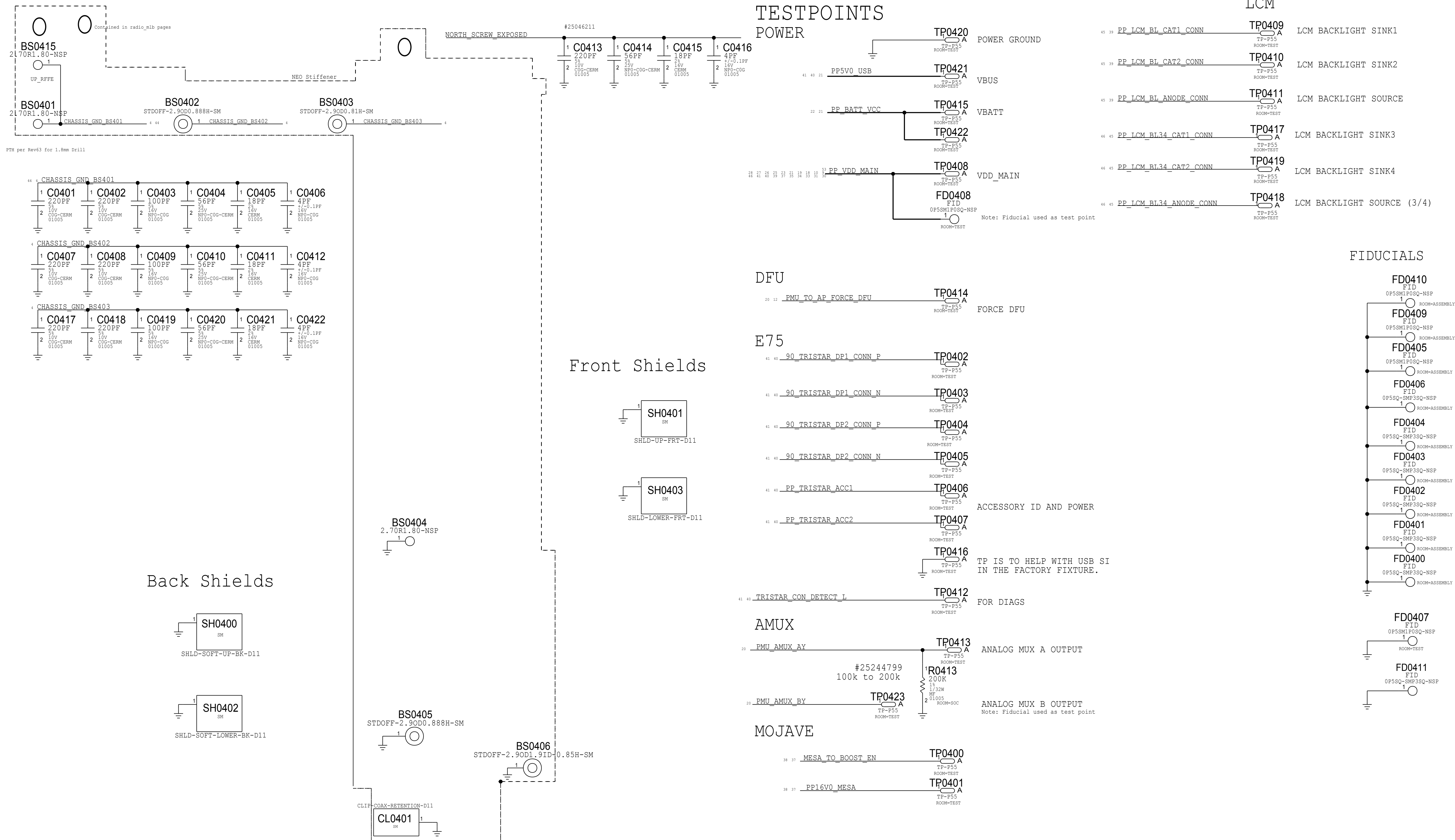
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S00087	371S00064	ALTERNATE	D2700	DIODE, SHOTTER, 30V, 200MA, 0201
152S00558	152S00557	ALTERNATE	L2700	IND, MLD, 0.470H, 2.5A, 80Mcho, 1608

8		7		6		5		4		3		2		1	
D111 EEEE CALLOUTS															
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION										
825-6838	1	EEEE CODE FOR 639-01824	EEEE_GY4R	CRITICAL	EEEE_D111_BEST										
825-6838	1	EEEE CODE FOR 639-01825	EEEE_GY4T	CRITICAL	EEEE_D111_SUPREME										
825-6838	1	EEEE CODE FOR 639-01826	EEEE_GY4V	CRITICAL	EEEE_D111_EXTREME										

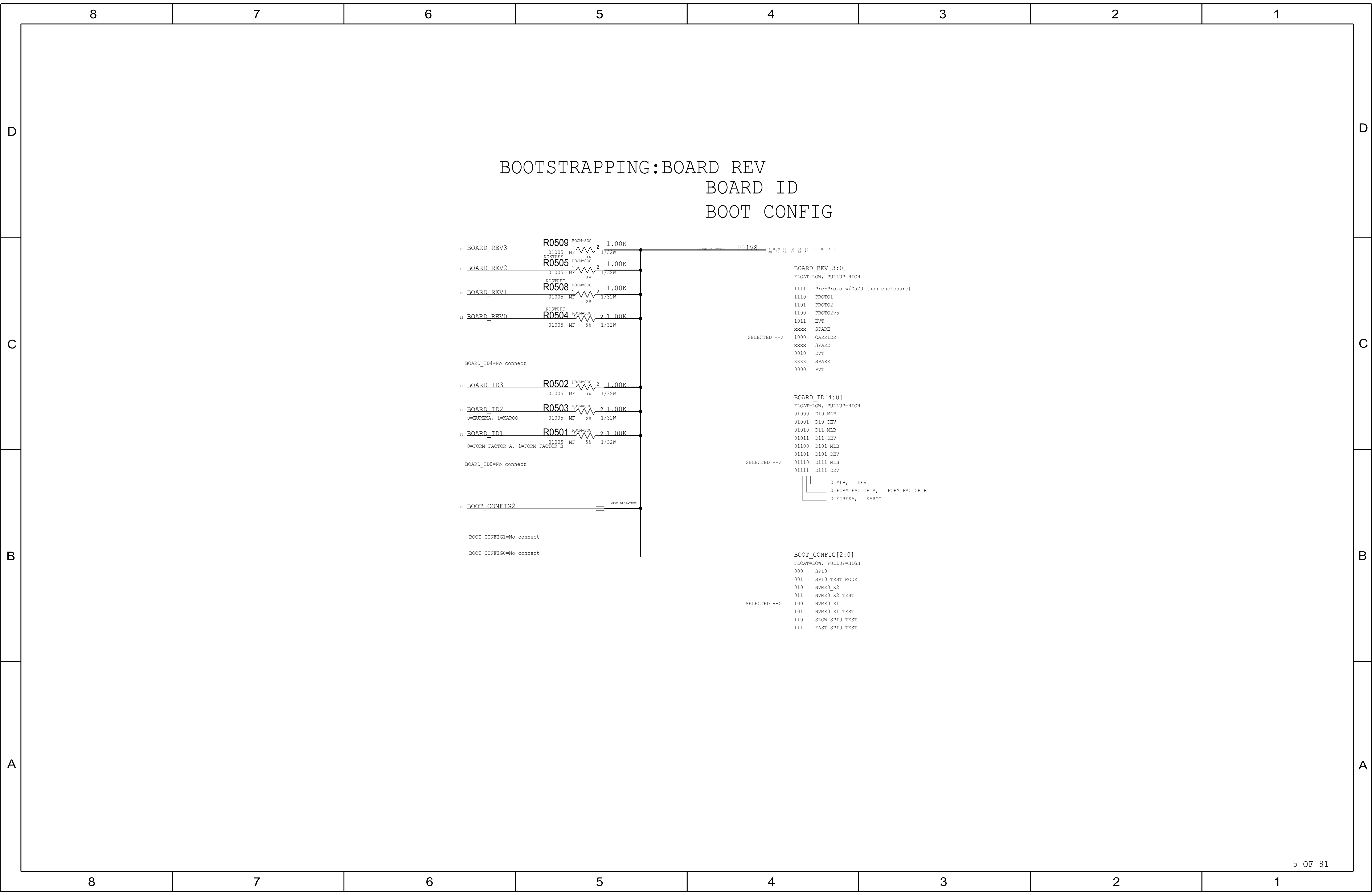
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138800049	138800032	ALTERNATE	ALL	000, 030, 050, 2, 200, 0100, 01, 010, 010000
13880831	138800032	ALTERNATE	ALL	000, 030, 050, 2, 200, 010, 0, 01, 010, 010000

3 OF 81

Current as of D11 MCO 056-01585 rev. 63



TOP SIDE



D

D

C

C

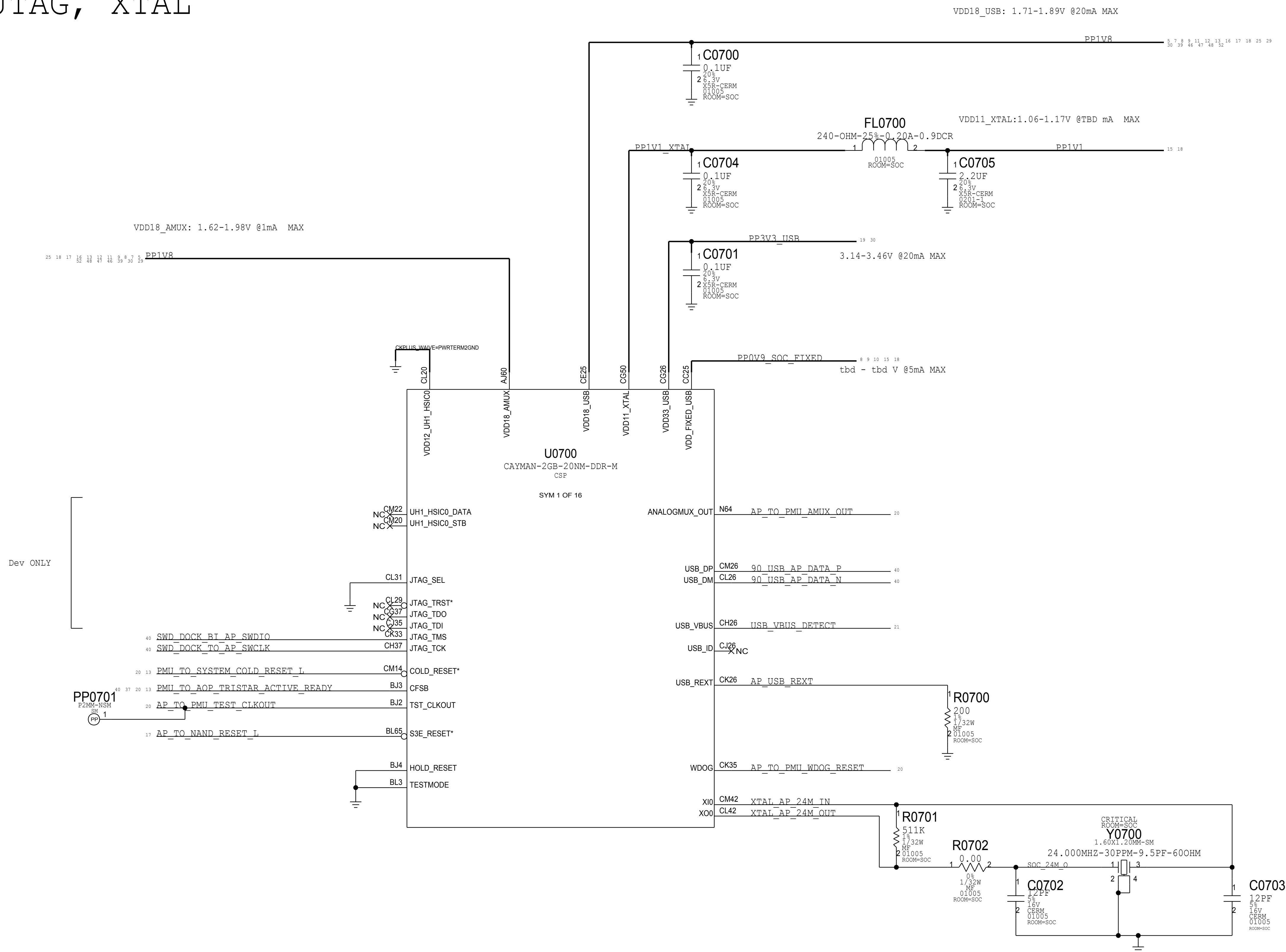
B

B

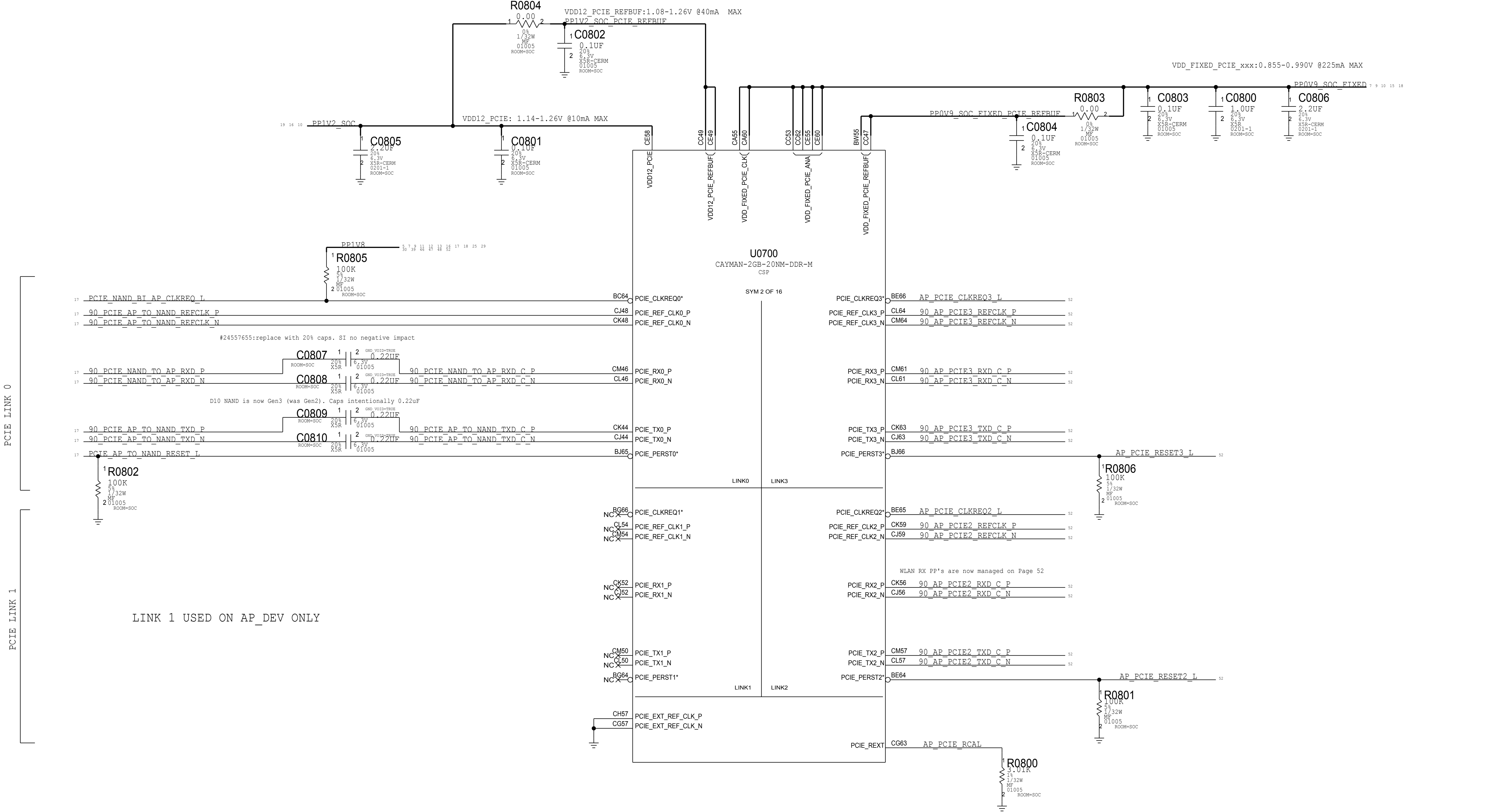
A

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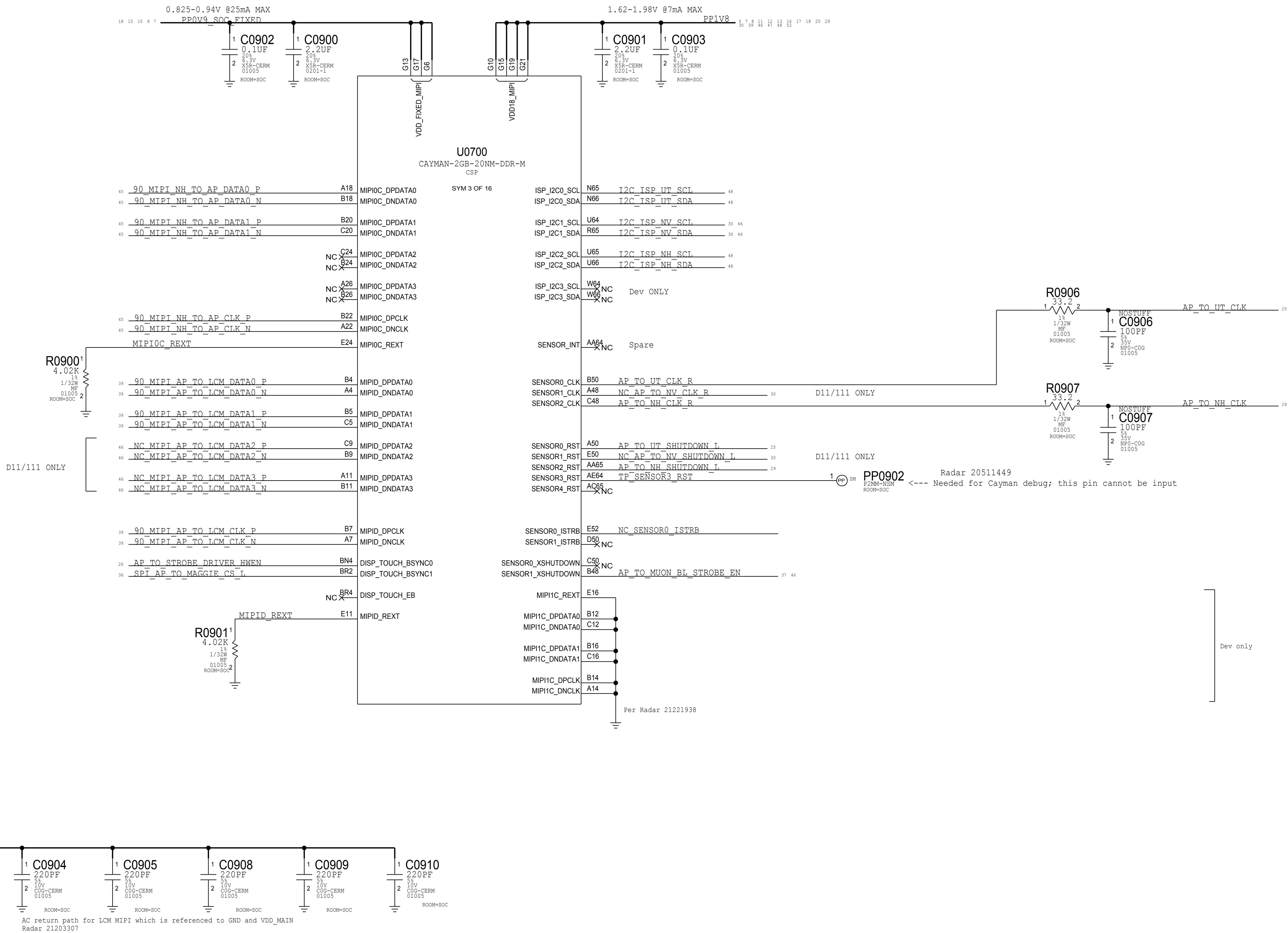
SOC - USB, JTAG, XTAL

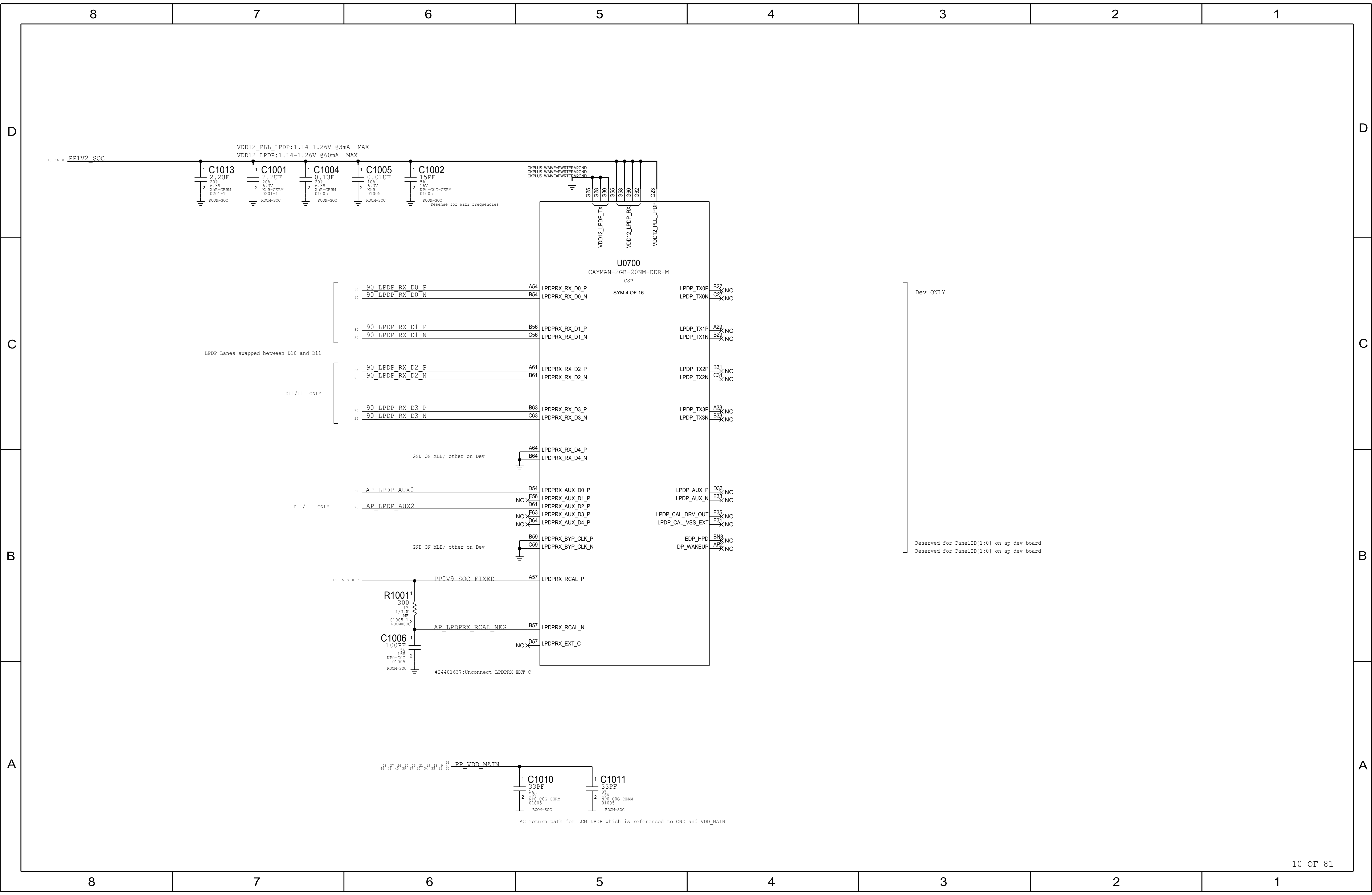


SOC - PCIE INTERFACES

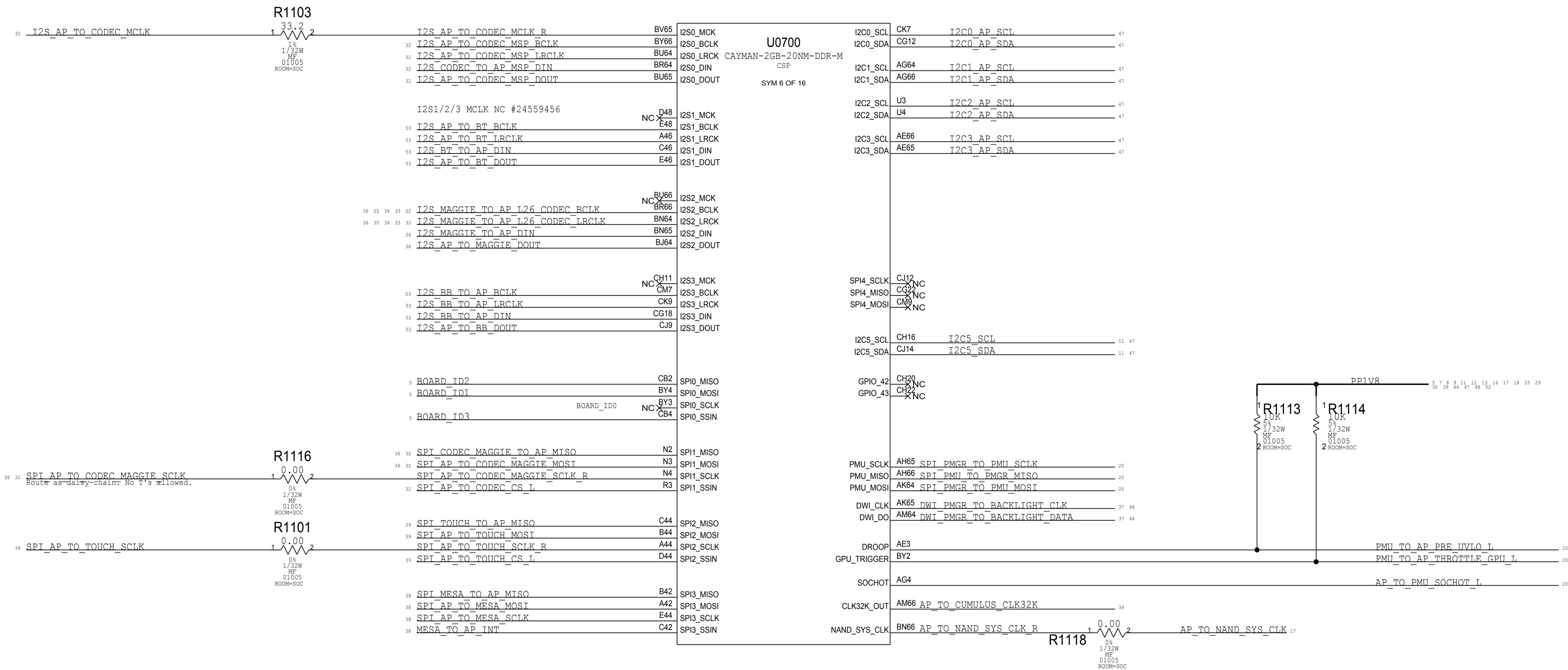


SOC - MIPI & ISP INTERFACES

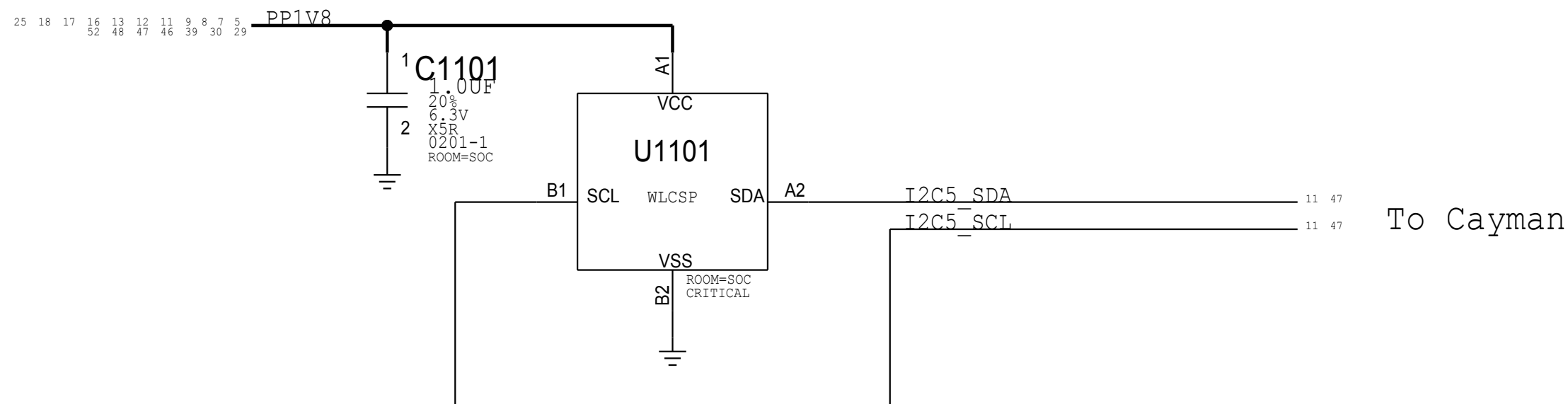




SOC - SERIAL INTERFACES



I2C5
See Radar#25316444 for Details



D

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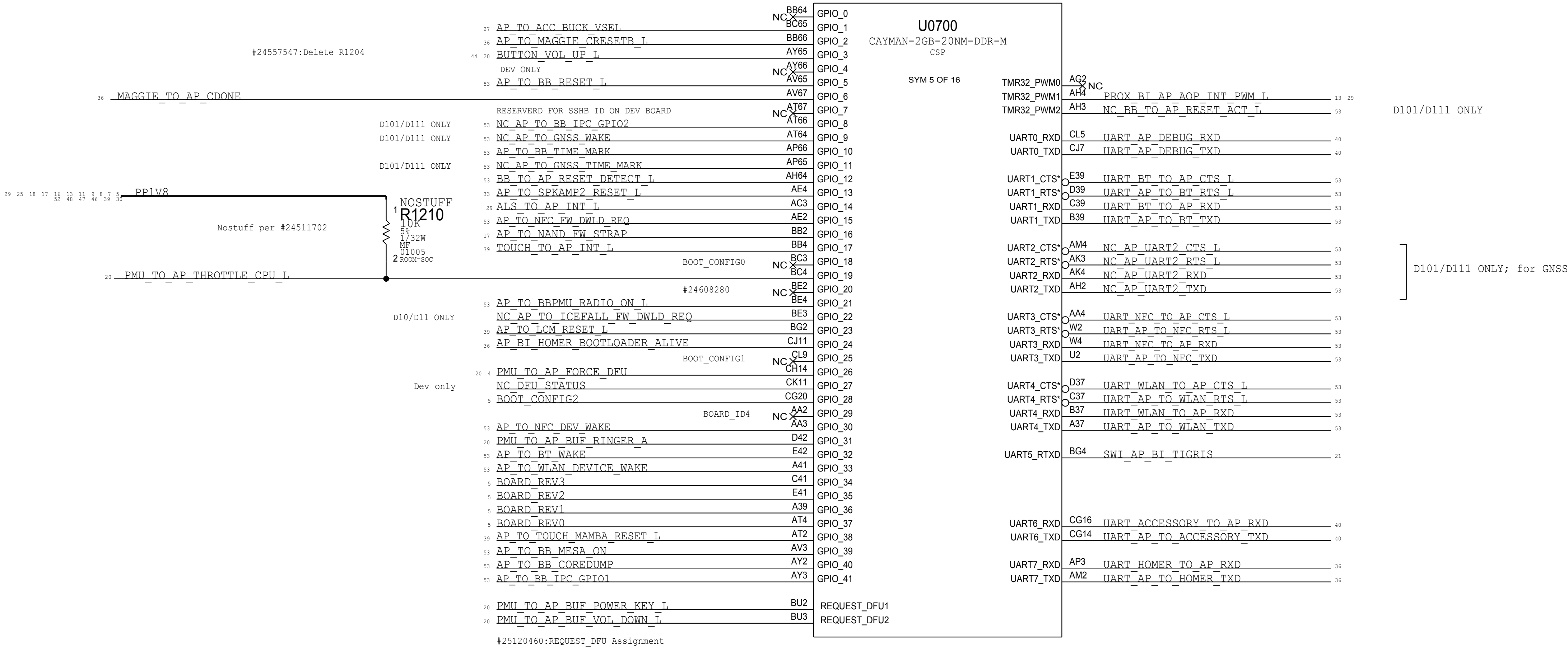
D

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SOC - GPIO INTERFACES



SOC - AOP

D

D

C

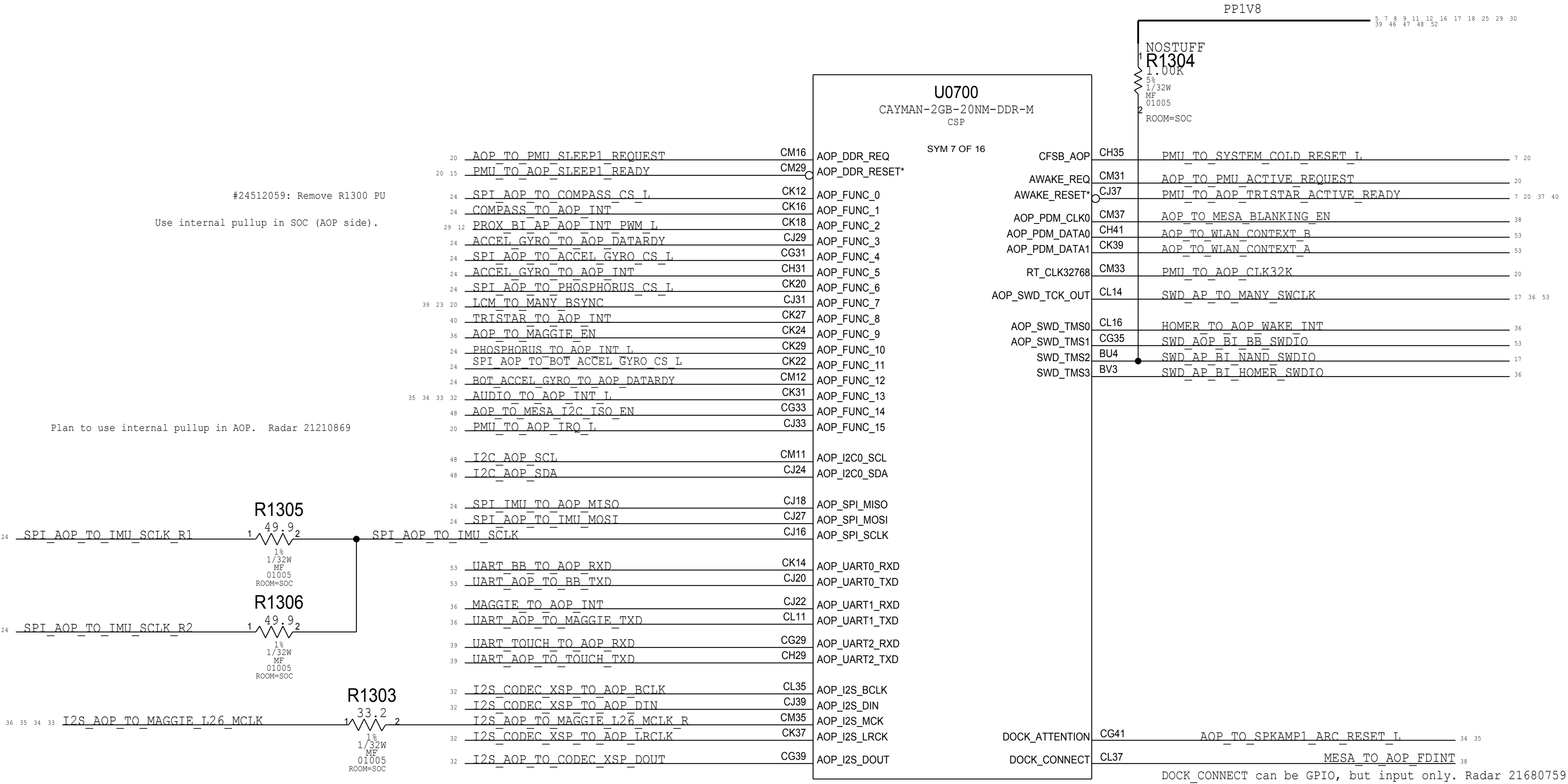
C

B

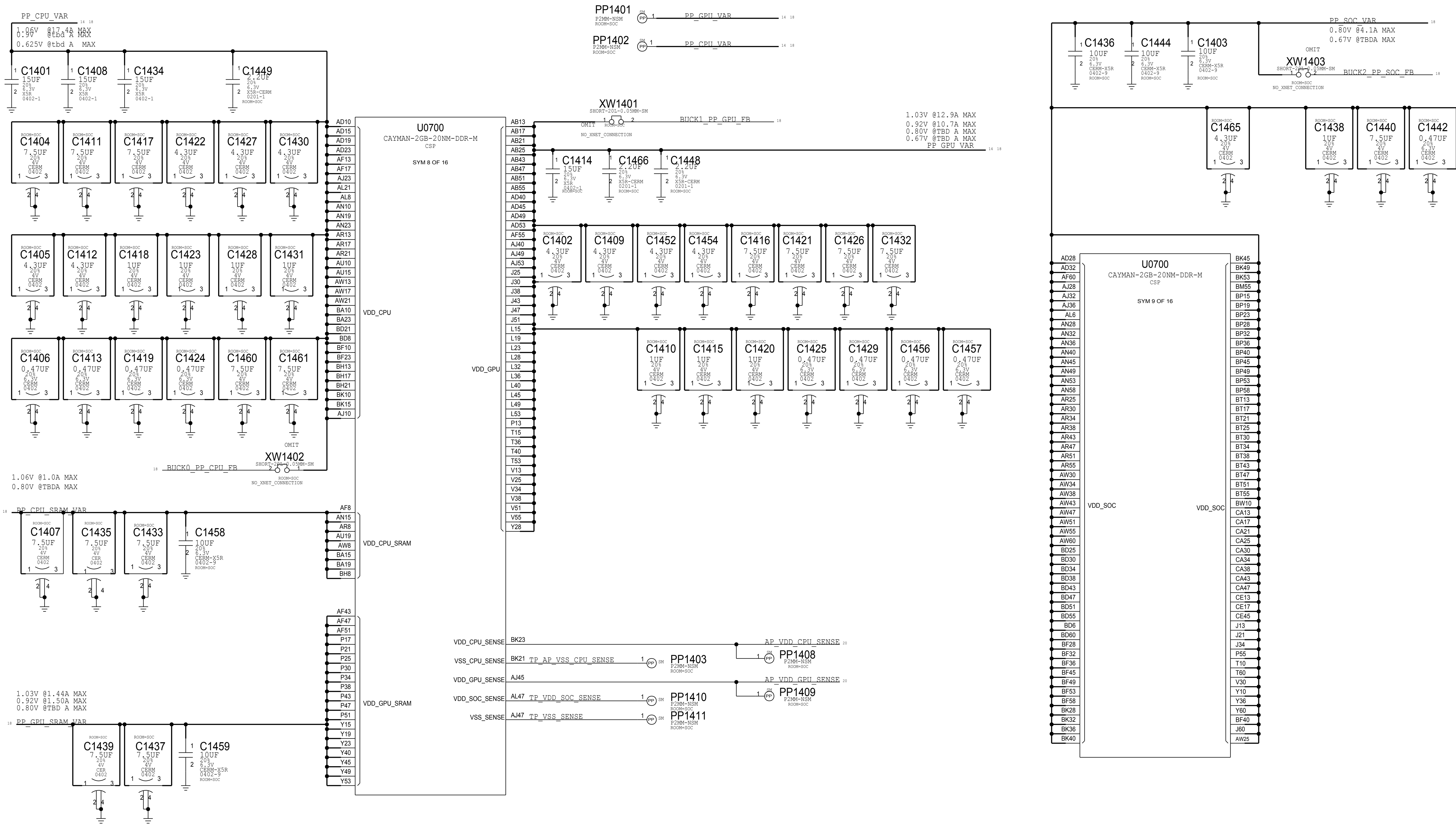
B

A

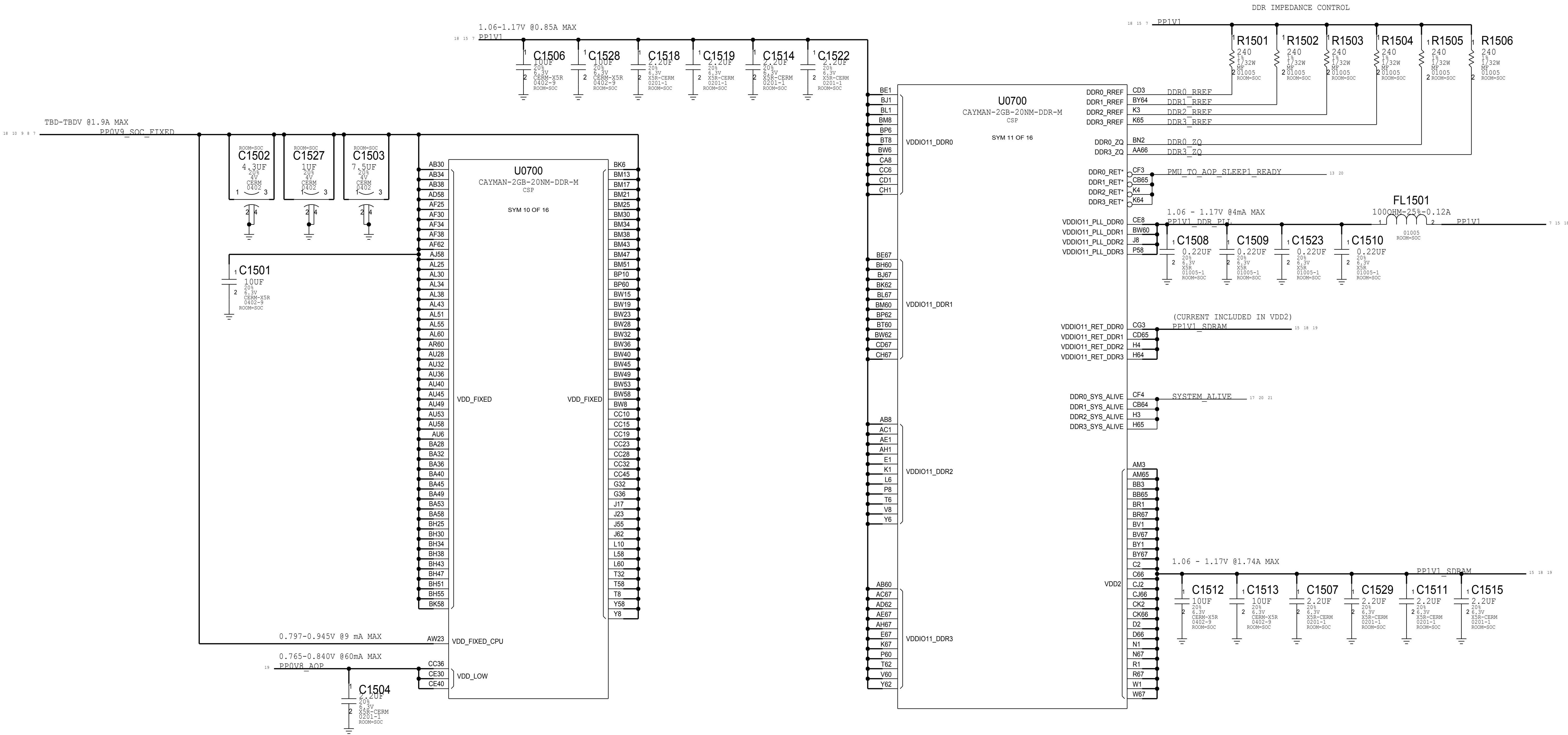
A



SOC - CPU, GPU & SOC RAILS



SOC - POWER SUPPLIES



SOC - POWER SUPPLIES

D

C

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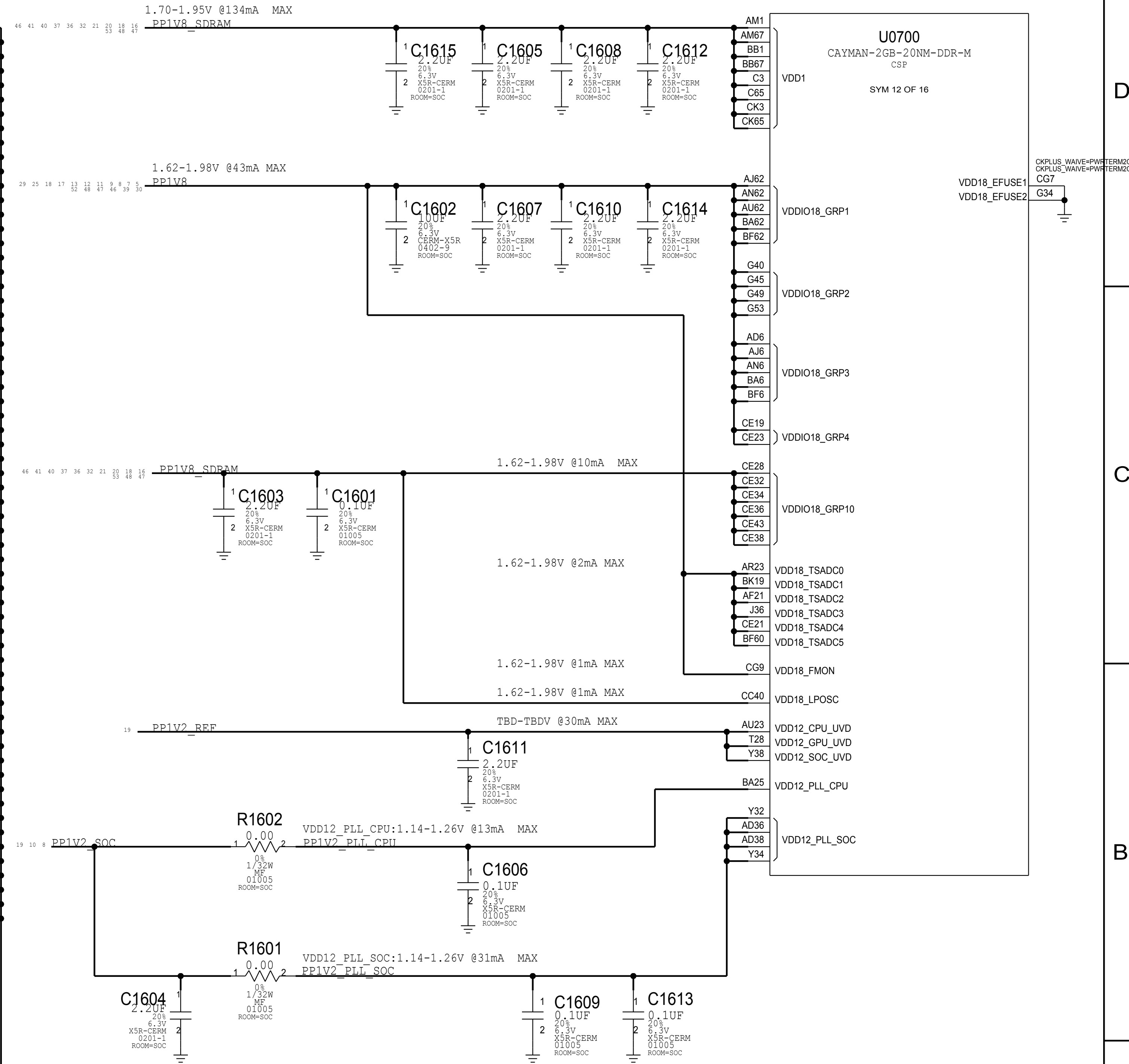
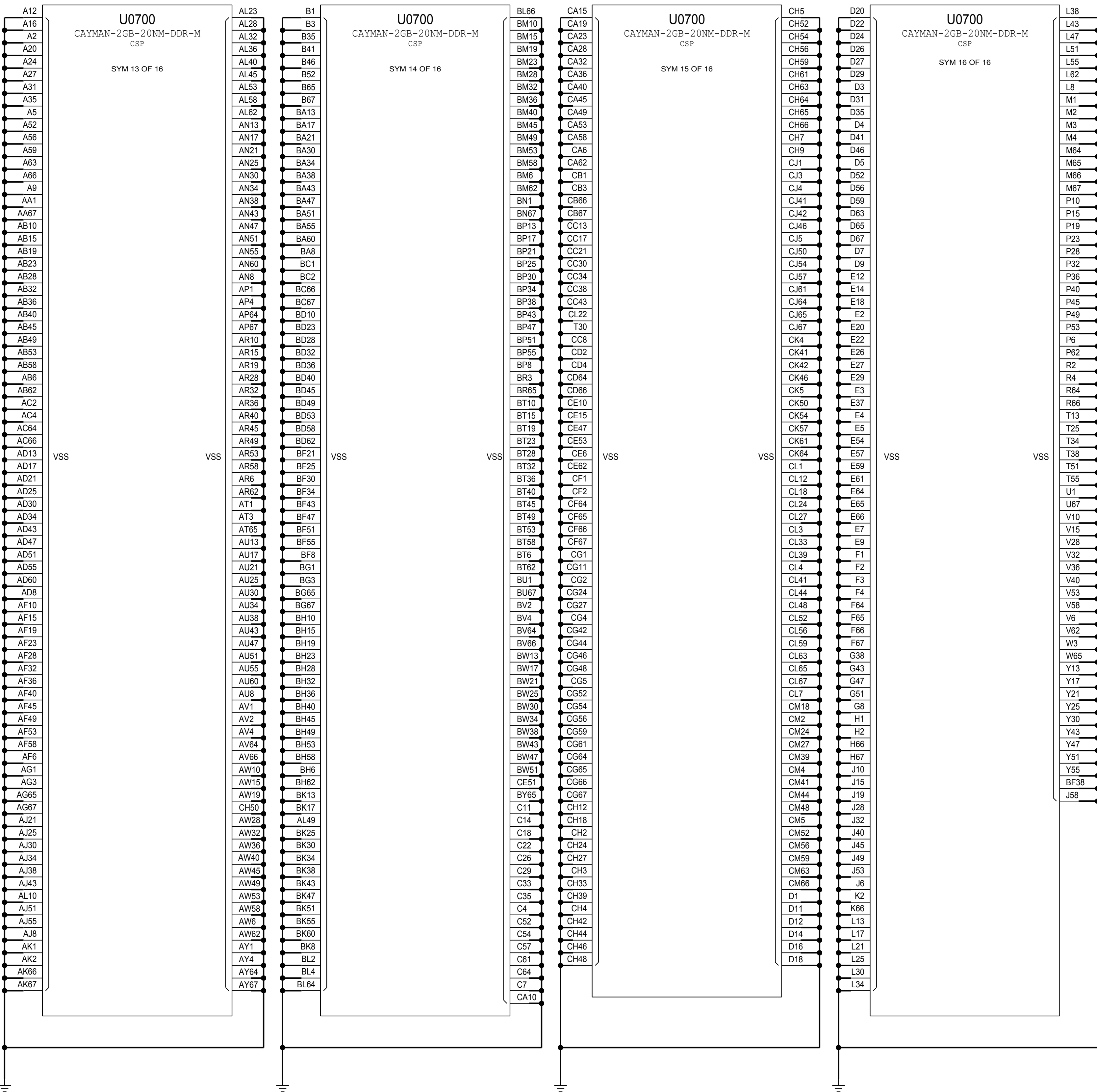
A

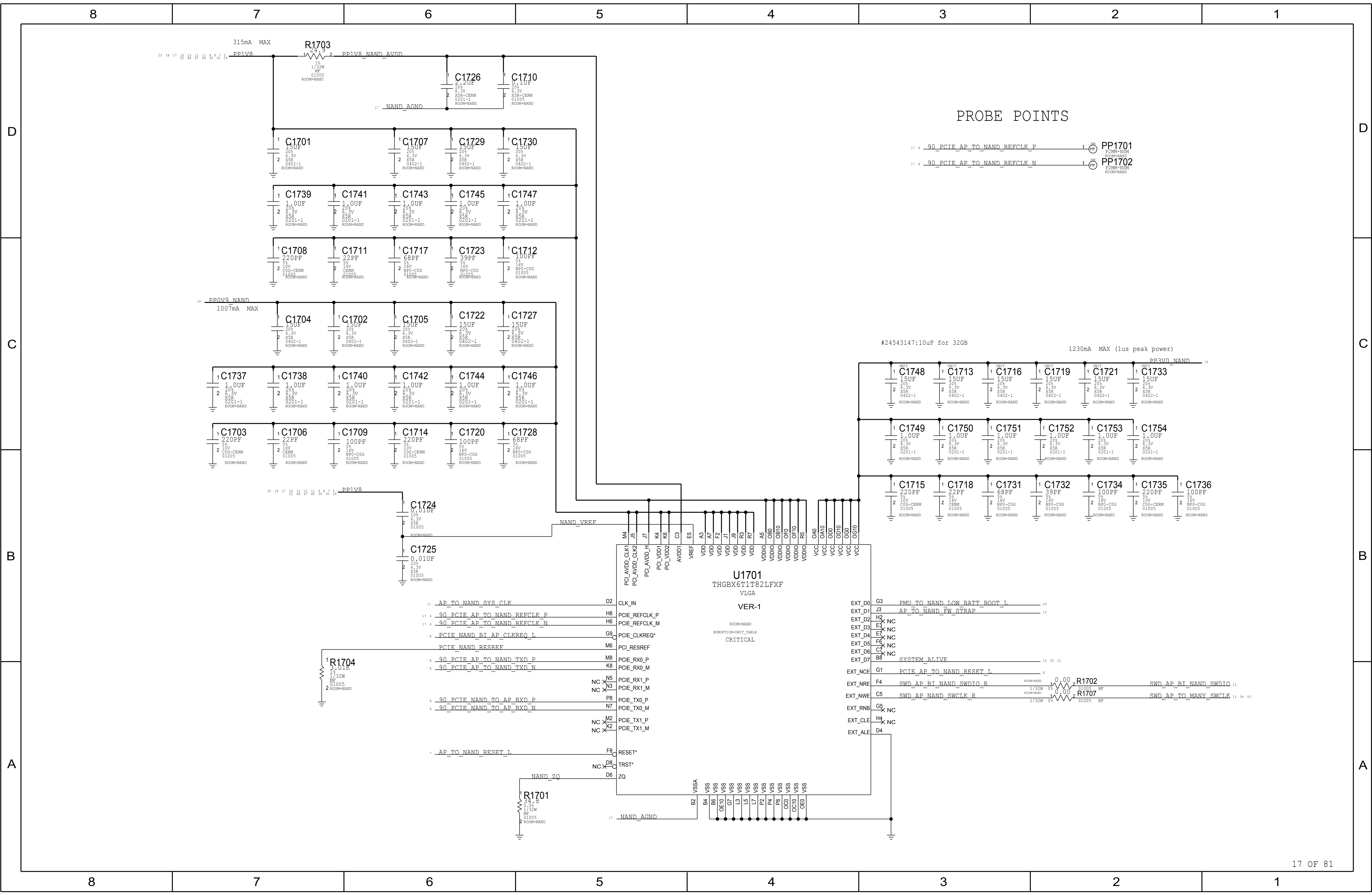
D

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D

C

B

A

D

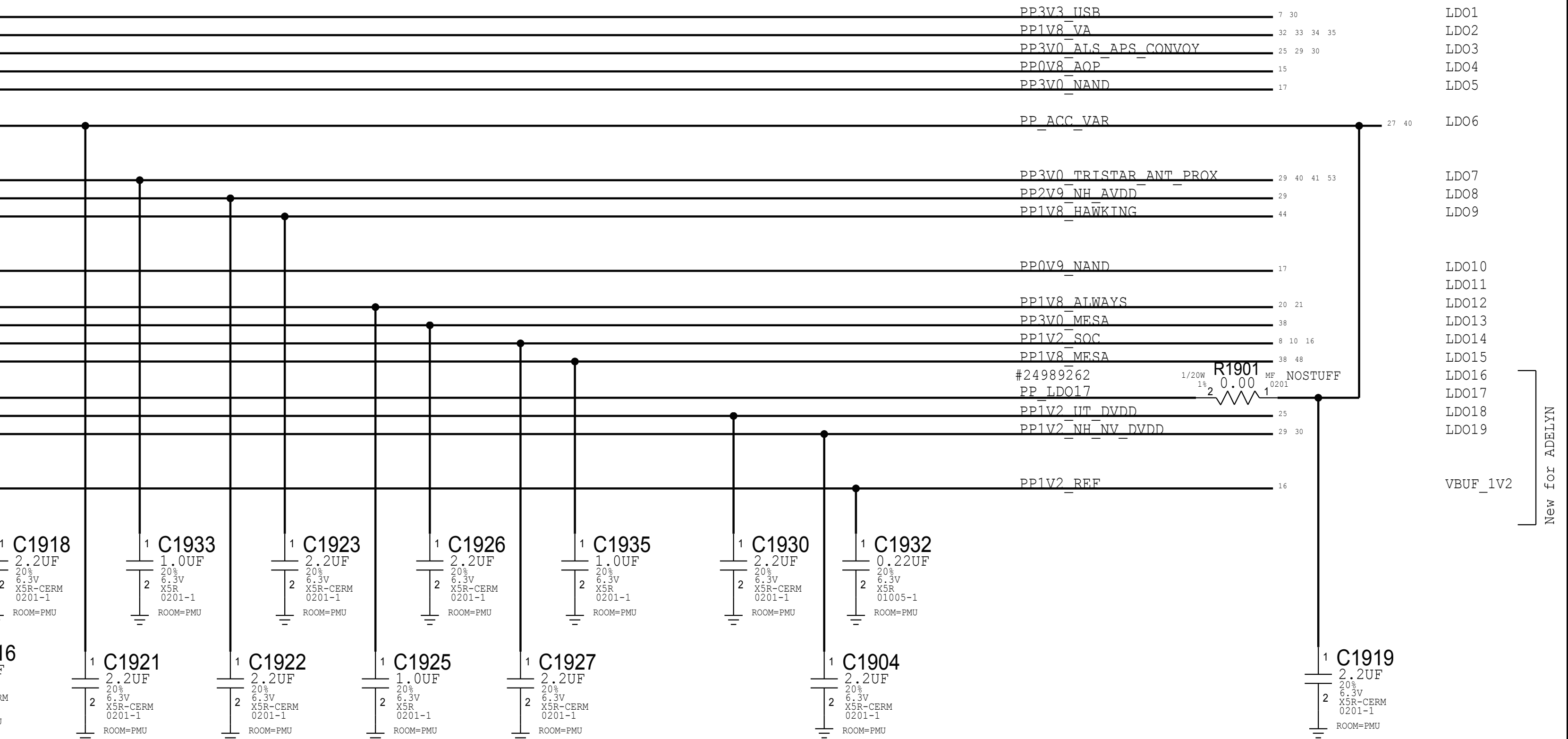
C

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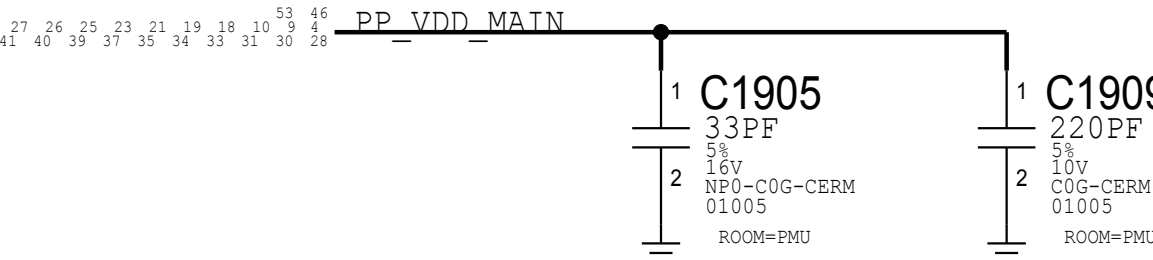
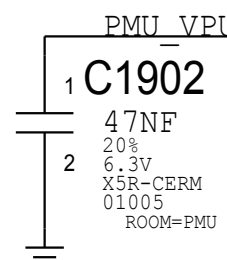
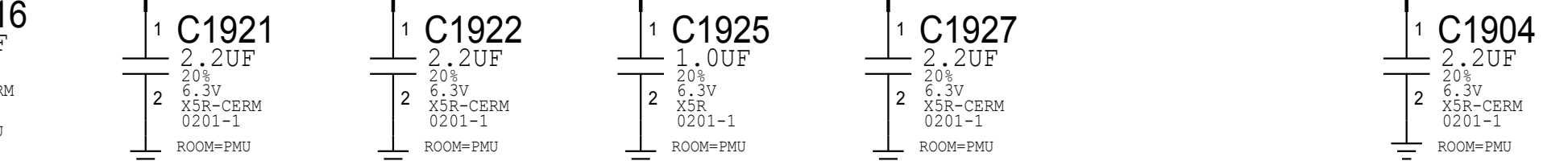
LDO#	ADJ.RANGE, LOW	ADJ.RANGE, HI	ACCURACY	MAX.CURRENT
LD01 (Ca)	1.2-2.475V	2.4-3.675V	+/-1.4%	50mA
LD02 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LD03 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LD04 (D)	0.7-1.2V		+/-2.5%	60mA
LD05 (F)	2.5-3.6V(tbc)		+/-75mV	1000mA
LD06 (Cb)	1.2-2.475V	2.4-3.675V	+/-2.5%	250mA (500/100mA in bypass)
LD07 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LD08 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LD09 (Cb)	1.2-2.475V	2.4-3.675V	+/-25mV	250mA
LD010 (Ga)	0.7-1.2V		+/-4.5%	1150mA

LDO#	ADJ.RANGE, LOW	ADJ.RANGE, HI	ACCURACY	MAX.CURRENT
LD011 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LD012 (E)	1.8V		+/-5%	10mA
LD013 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LD014 (Gb)	0.7-1.4V		+/-3.0%	400mA
LD015 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LD016 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LD017 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LD018 (Gb)	0.7-1.4V		+/-3.0%	400mA
LD019 (Gb)	0.7-1.4V		+/-3.0%	400mA
LDO_RTC	2.5V		+/-2.0%	10mA
BUF_1V2	1.2V		+/-5.0%	10mA



#24989262:OTP-AO LD017 default off,50mA Iout_max

VPUMP: 10nF min. @4.6V



PMU_VSS_RTC
U15 = PMU XTAL GND

D

C

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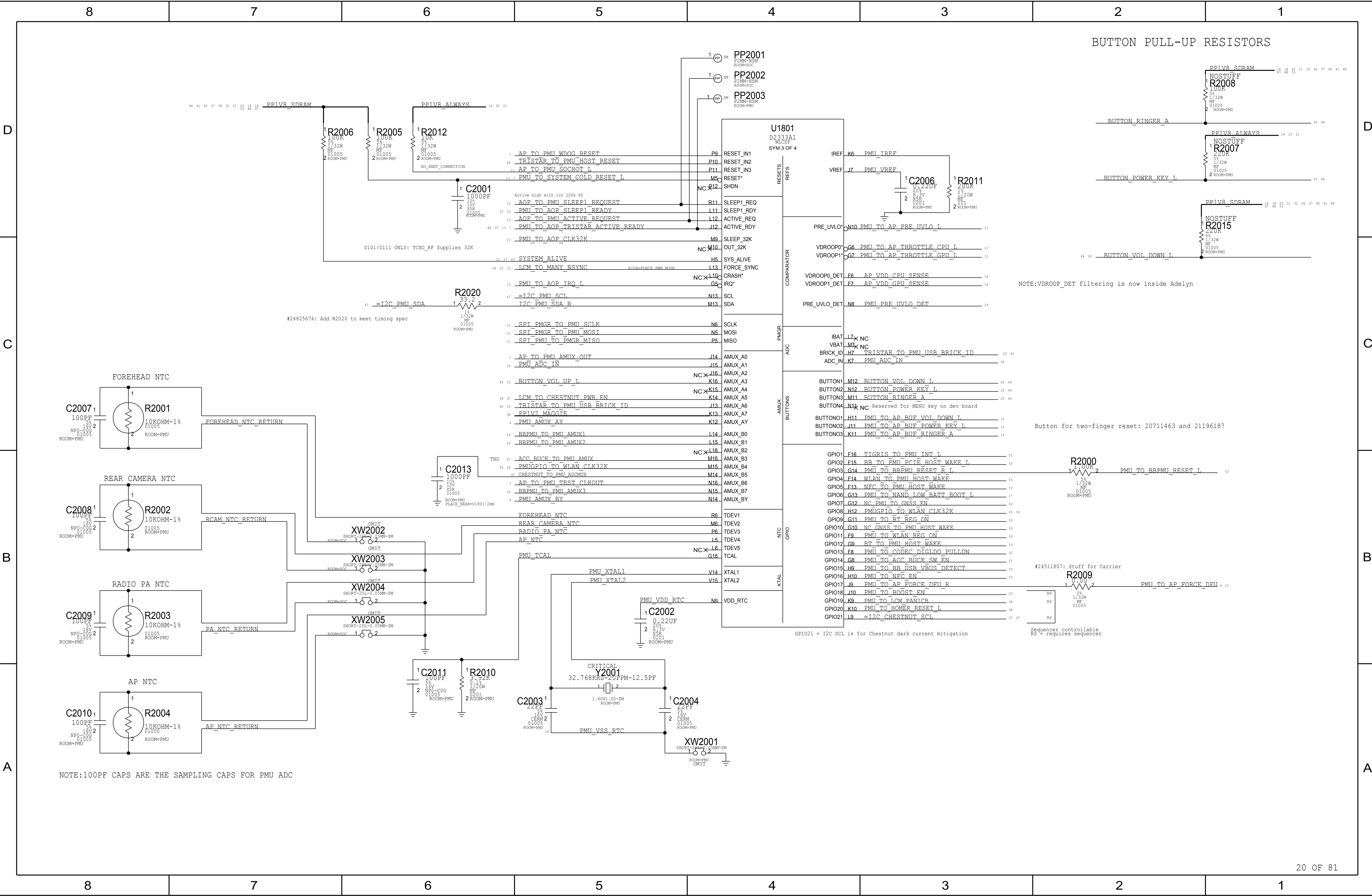
A

D

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A



BUTTON PULL-UP RESISTORS

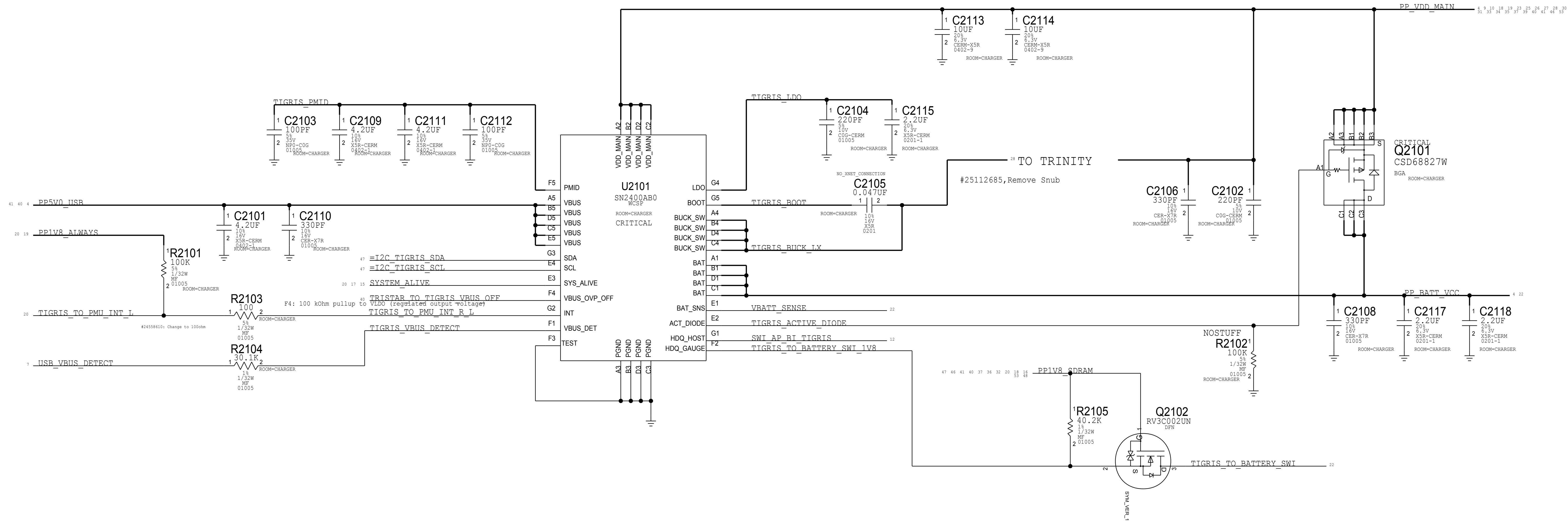
NOTE:VDROOP_DET filtering is now inside Adelyn

Button for two-finger reset: 20711463 and 21196187

#24511807: Stuff for Carrier

Sequencer controllable
RS= requires sequencer

TIGRIS CHARGER



D

C

B

A

D

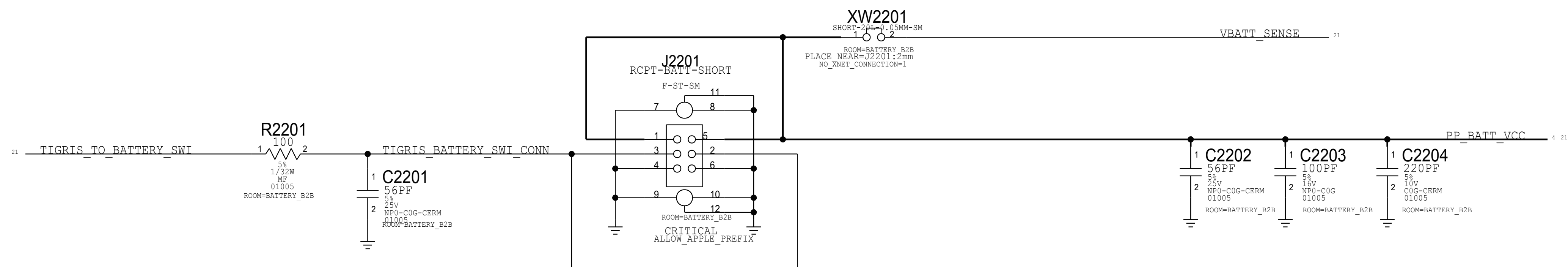
C

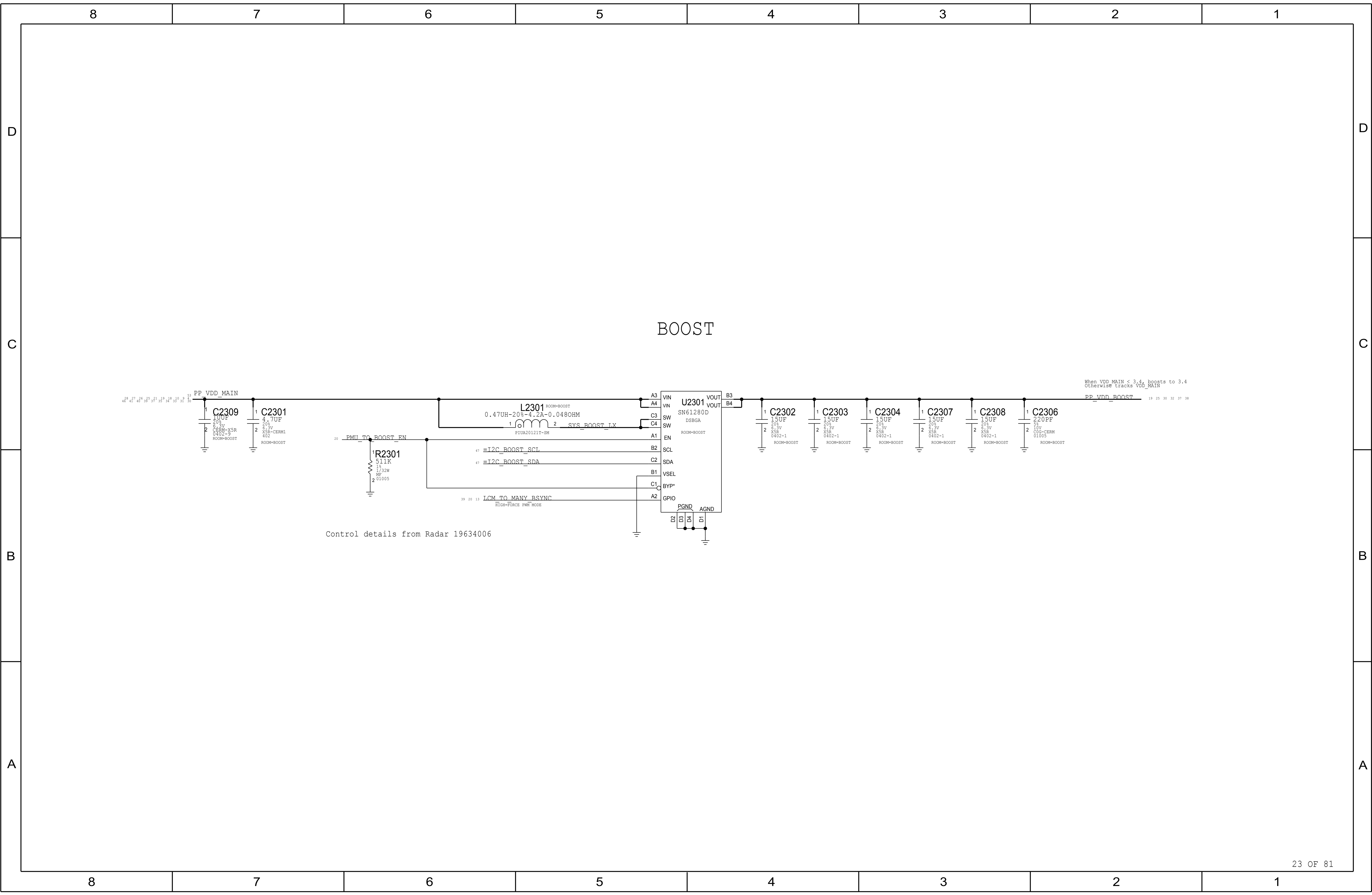
B

A

BATTERY CONNECTOR

THIS ONE ON MLB ---> 516S00172 (matches d10 mlb MCO rev 27)





D

C

B

A

D

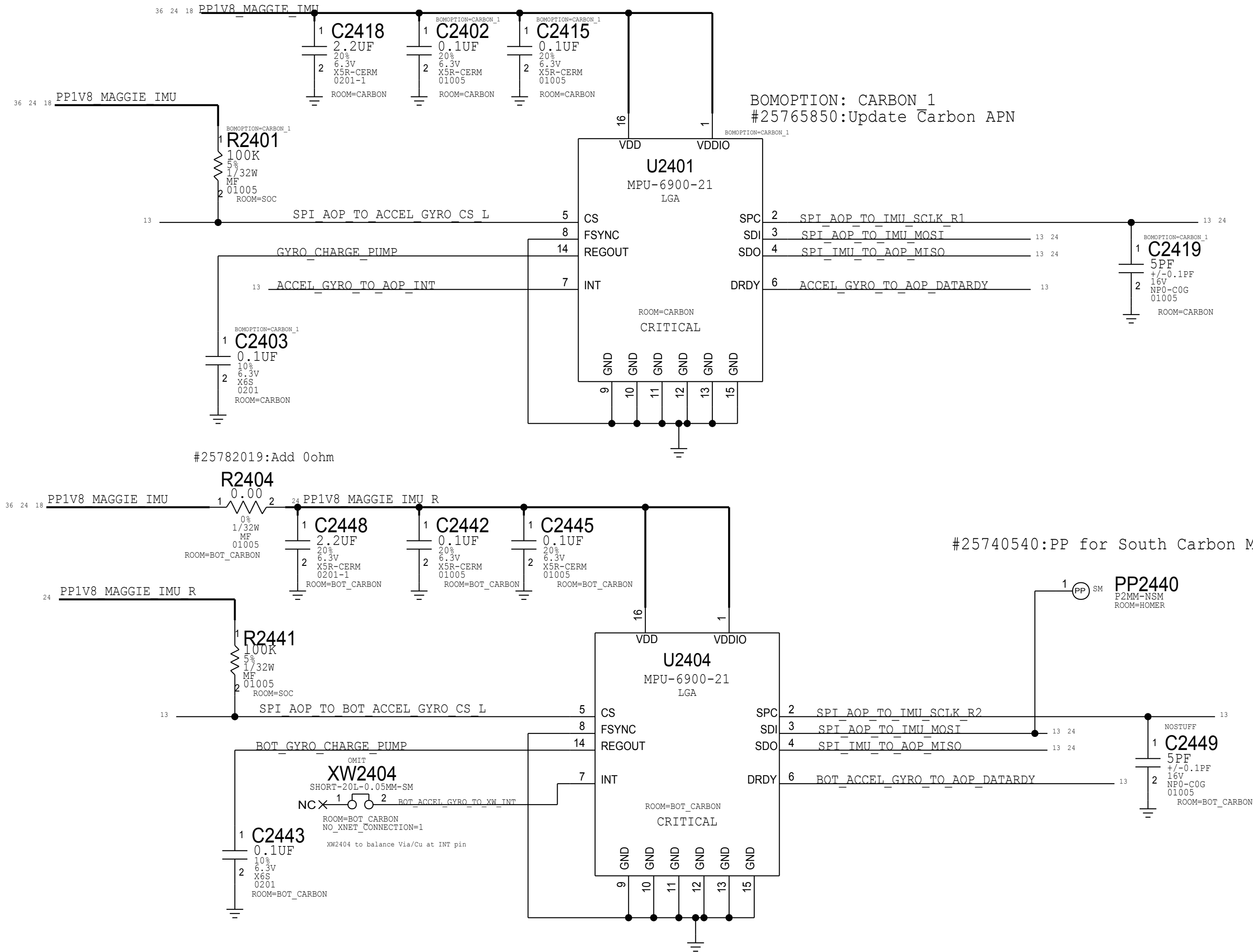
C

B

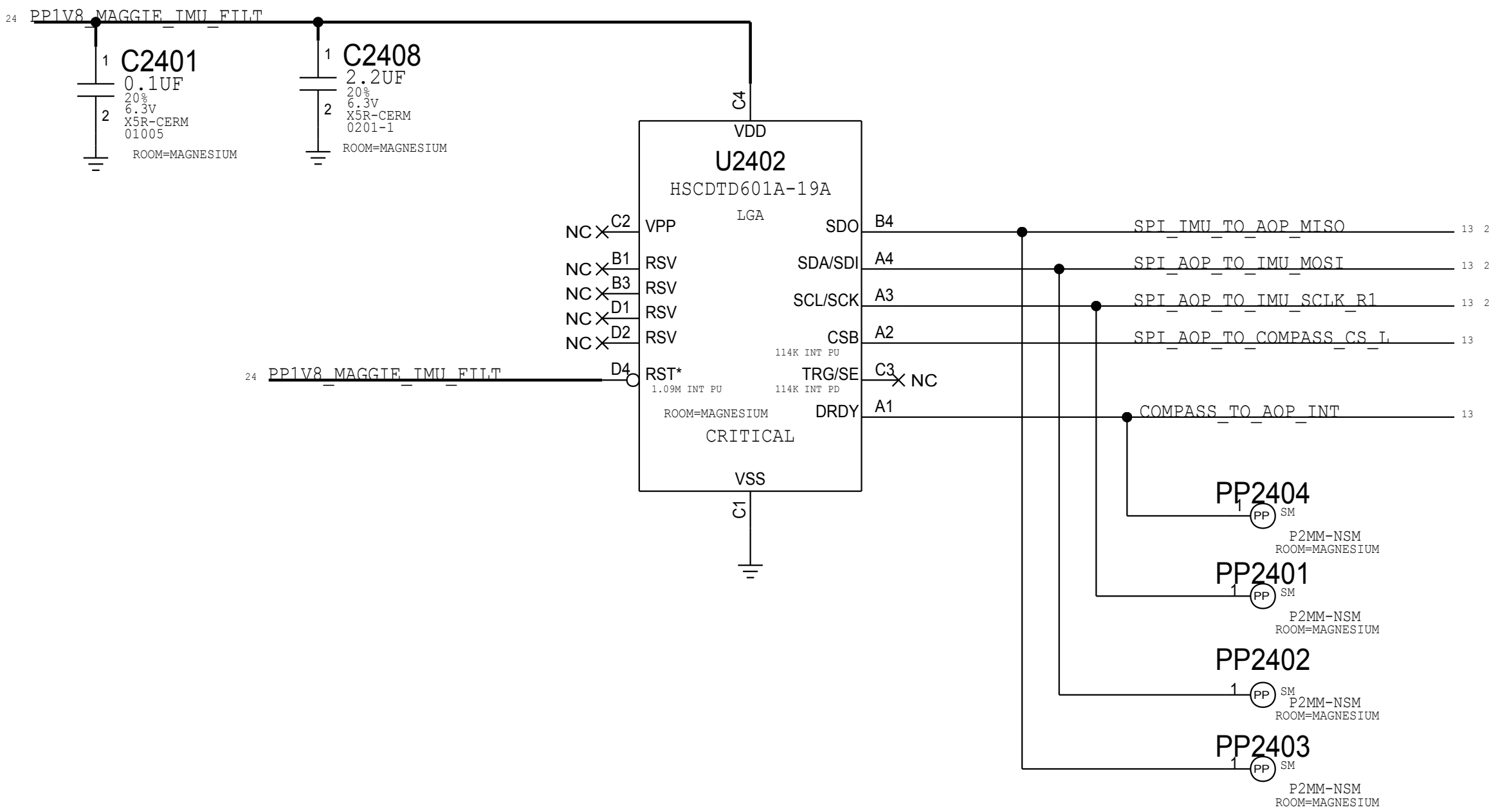
A

CARBON - ACCEL & GYRO

INVENSENSE, MPU-6800: C2403=0.1uF



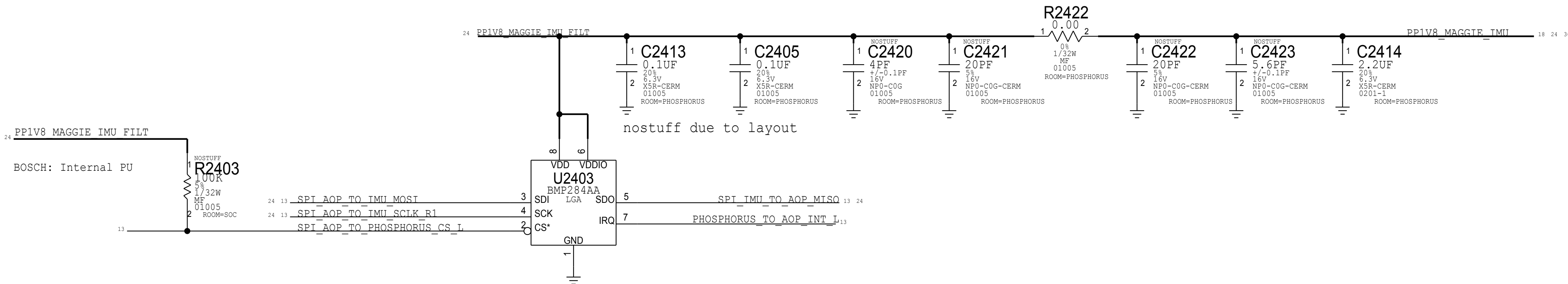
MAGNESIUM - COMPASS



PHOSPHORUS

#24593845

BOSCH (APN:338S00188): nostuff C2420/C2421/C2422/C2423 and R2403 PU
ST (APN:338S00230): stuff C2420-C2423, C2420R2422 with 155S00017, stuff R2403 PU
C2420=4pF(131S0253),C2405=3pF(131S0251) per #25691124



D

C



D



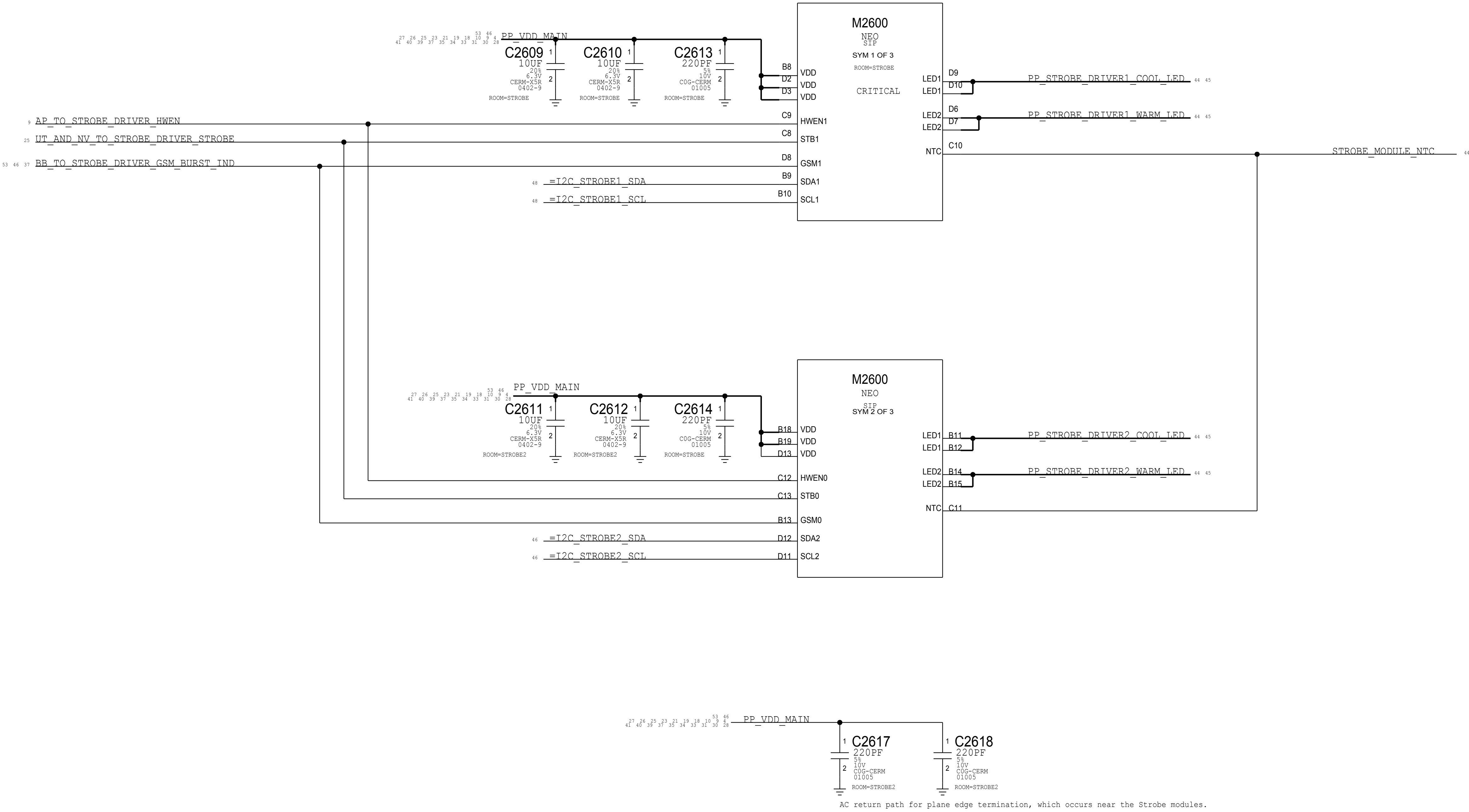
B

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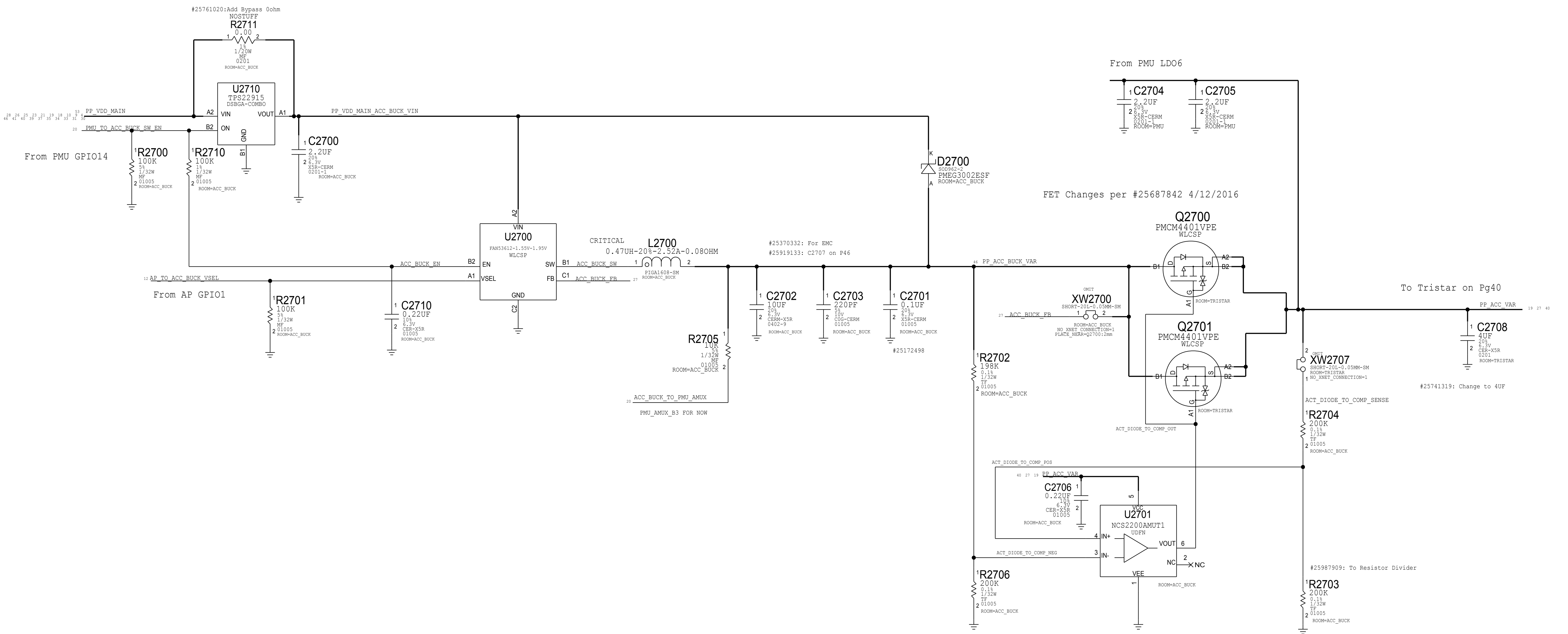


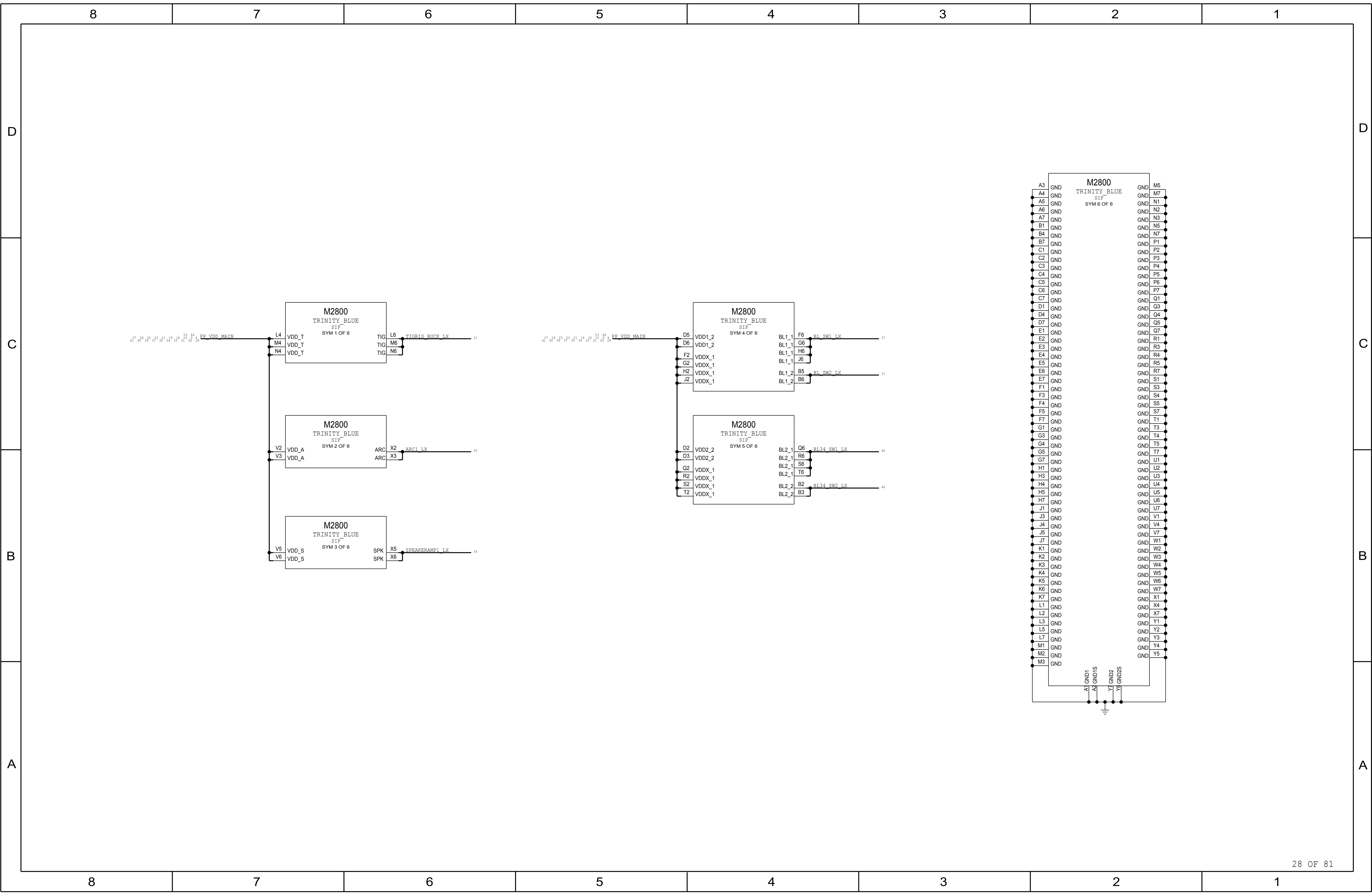
STROBE DRIVERS INSIDE NEO SIP MODULE

D10/sip_neo

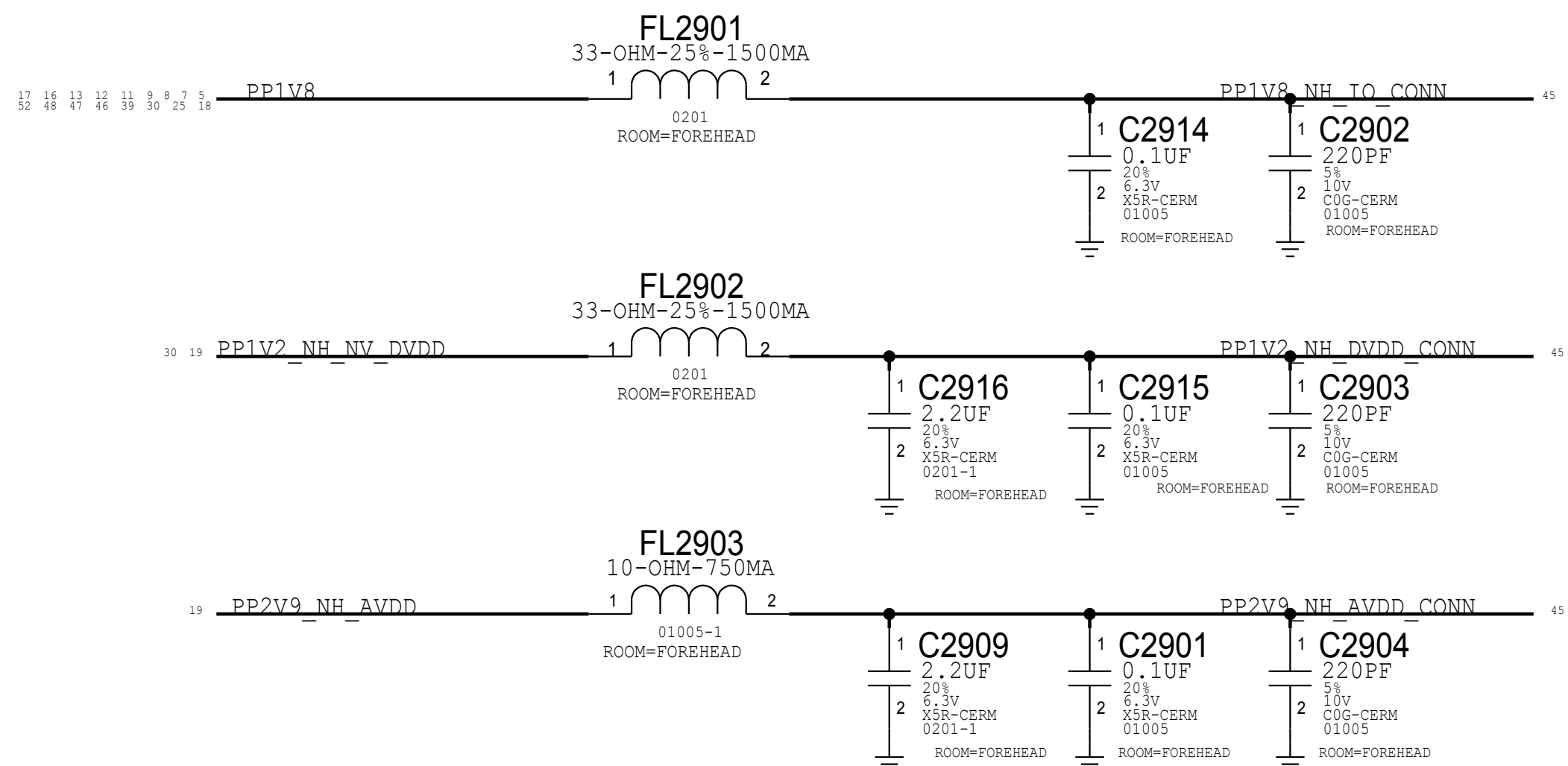


ACCESSORY BUCK

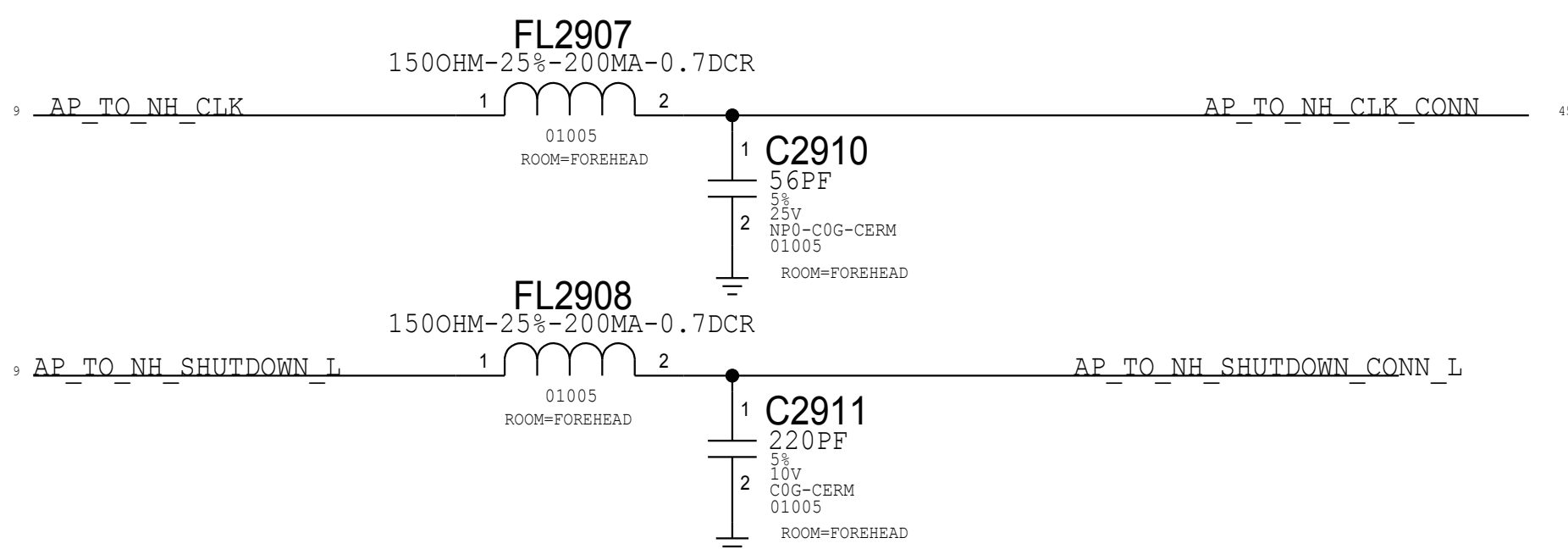




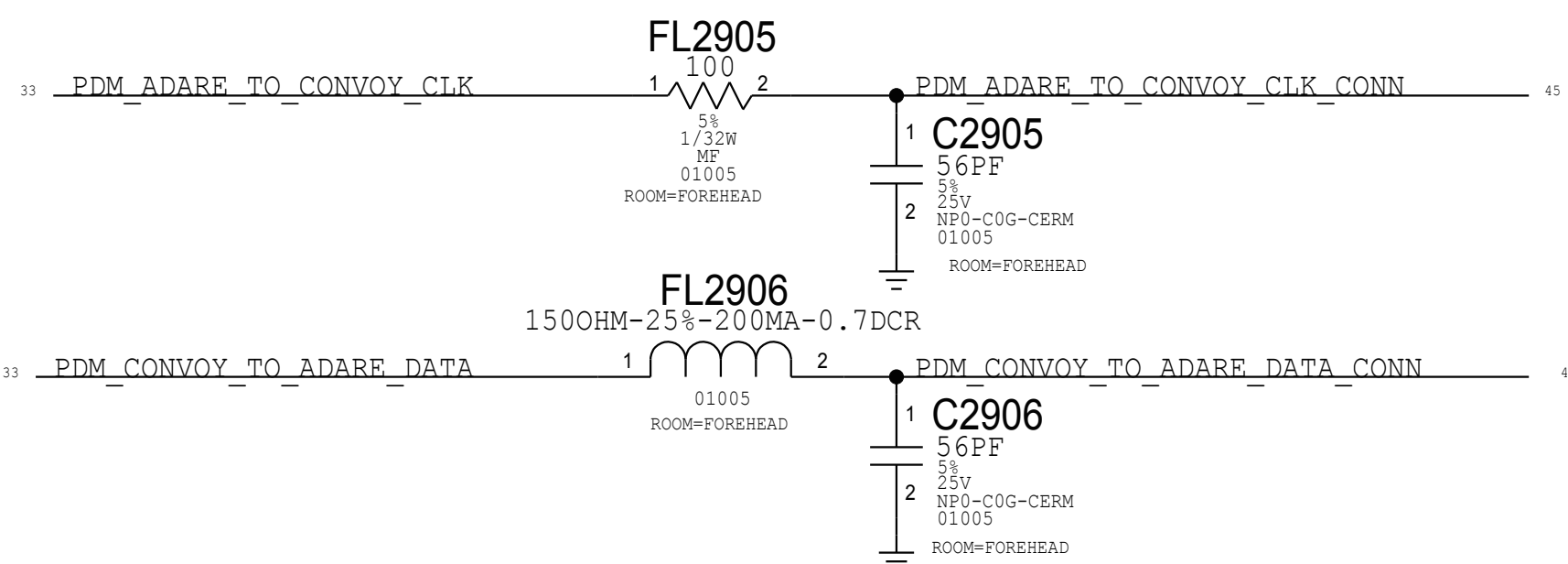
NEW HAMPSHIRE POWER



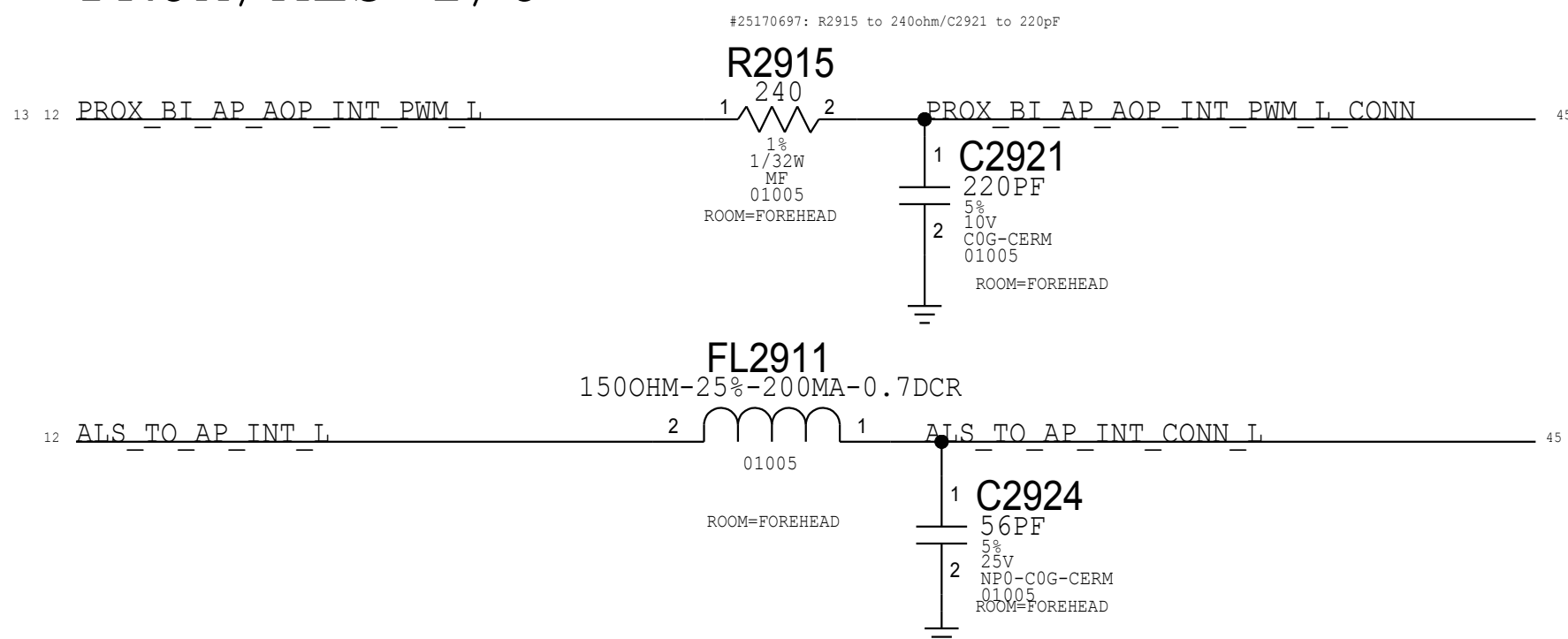
NEW HAMPSHIRE I/O



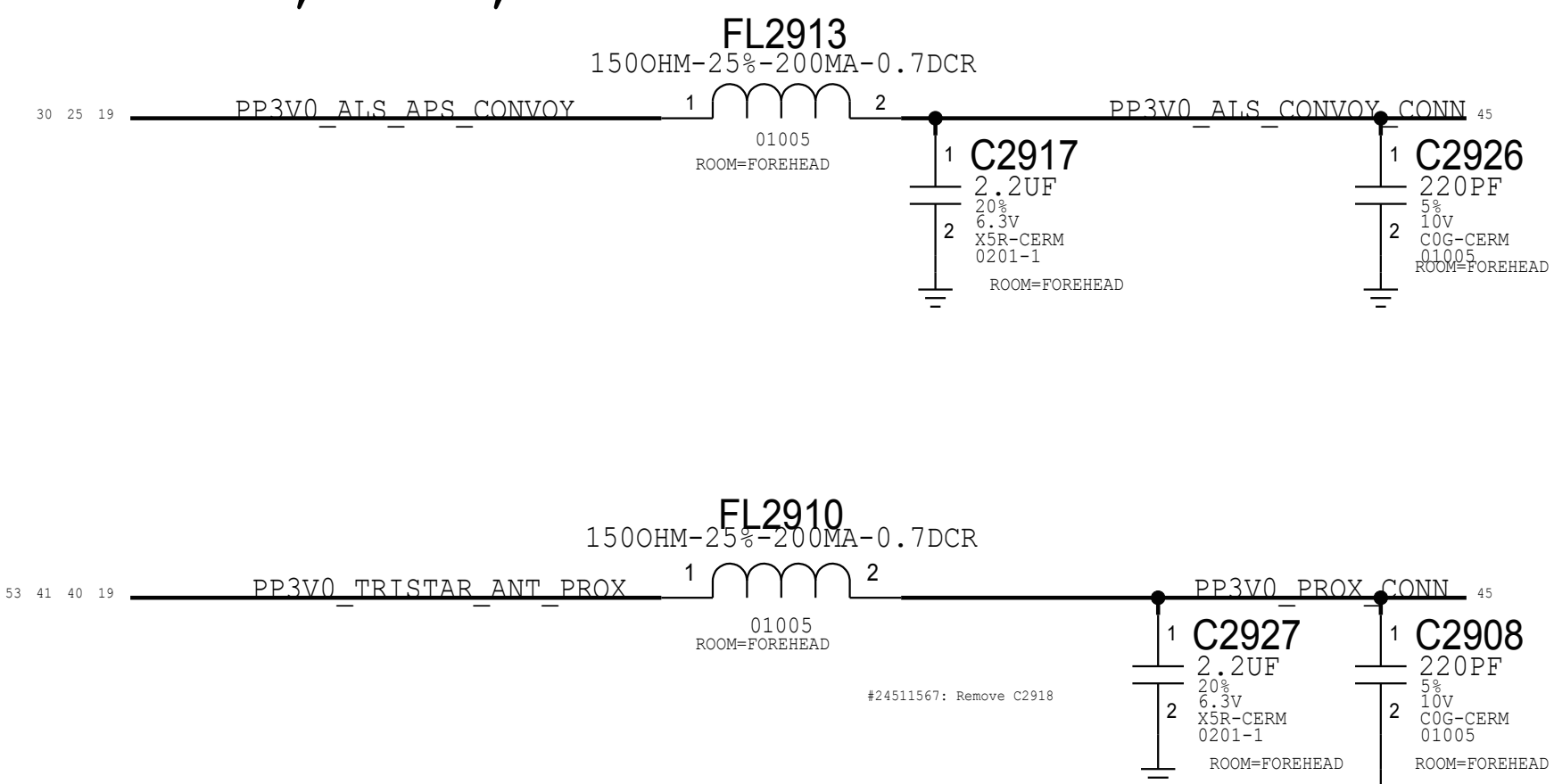
CONVOY I/O



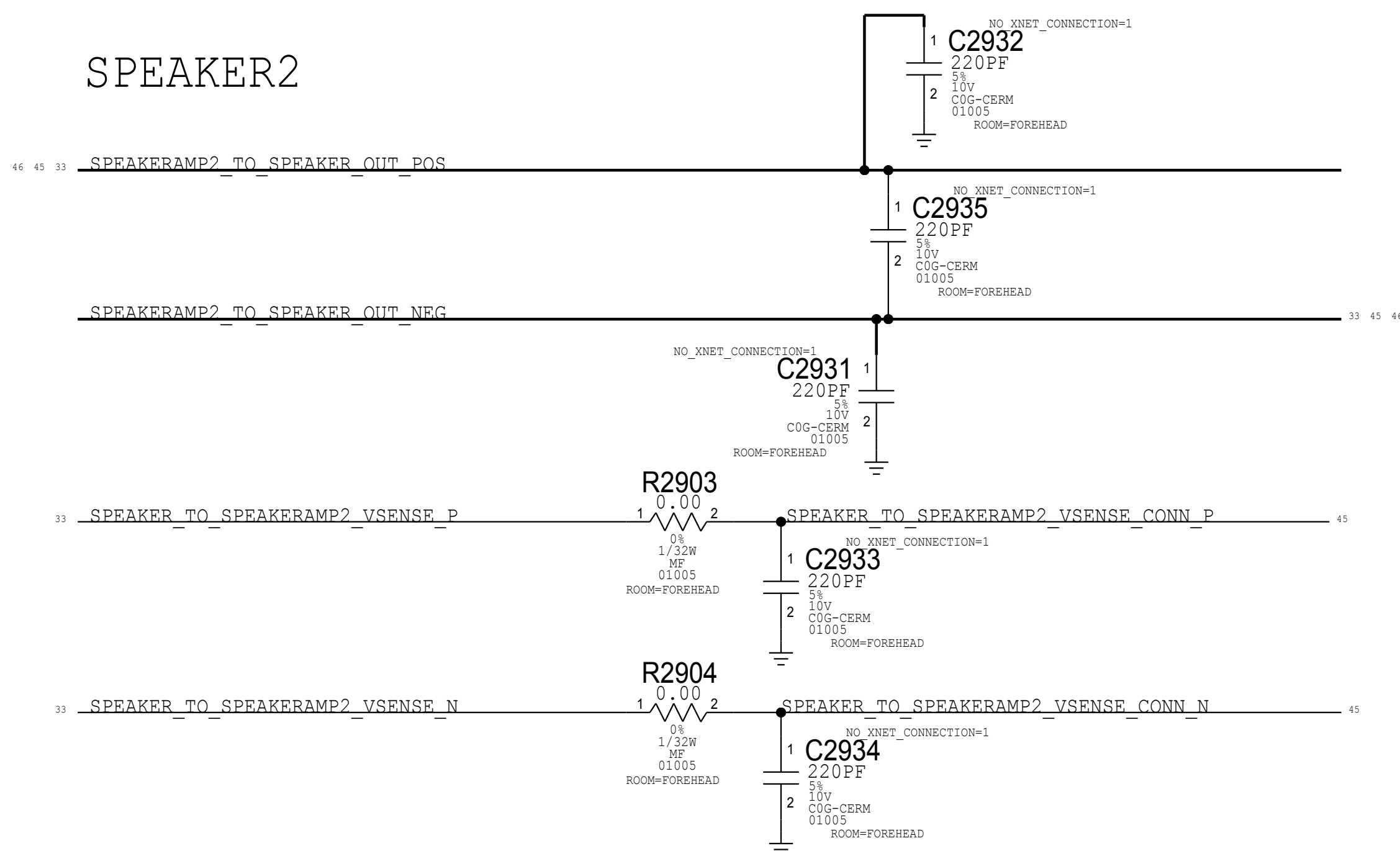
PROX/ALS I/O



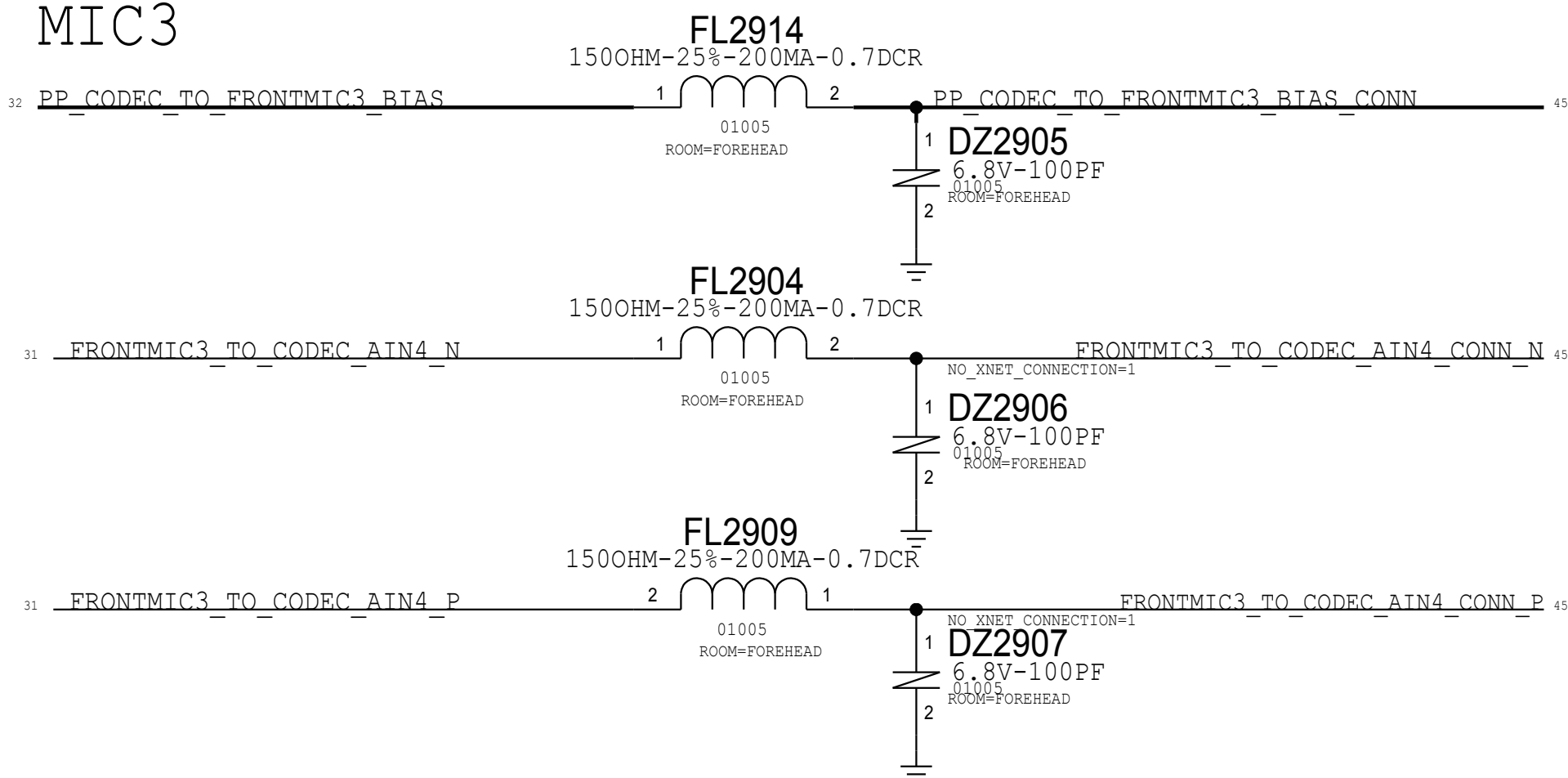
PROX, ALS, & CONVOY POWER



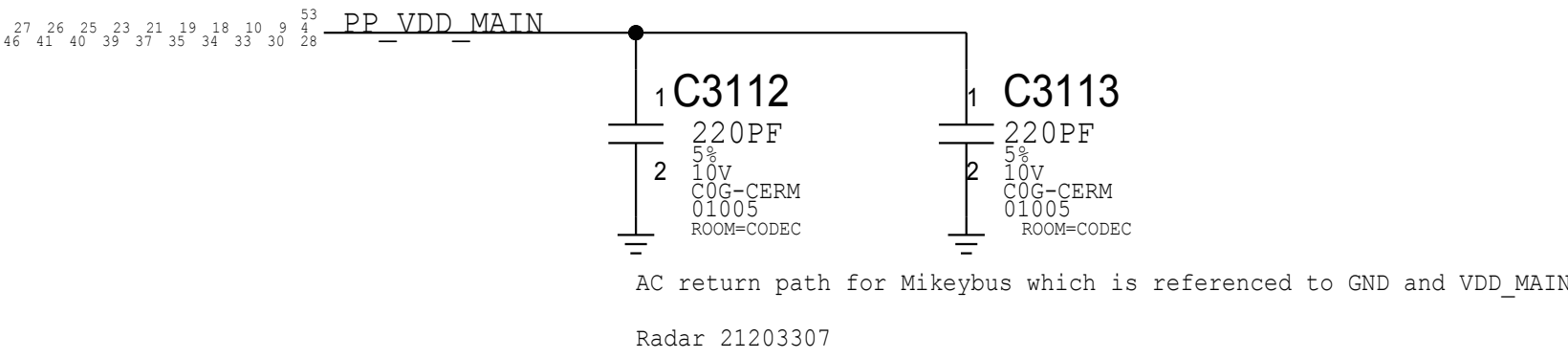
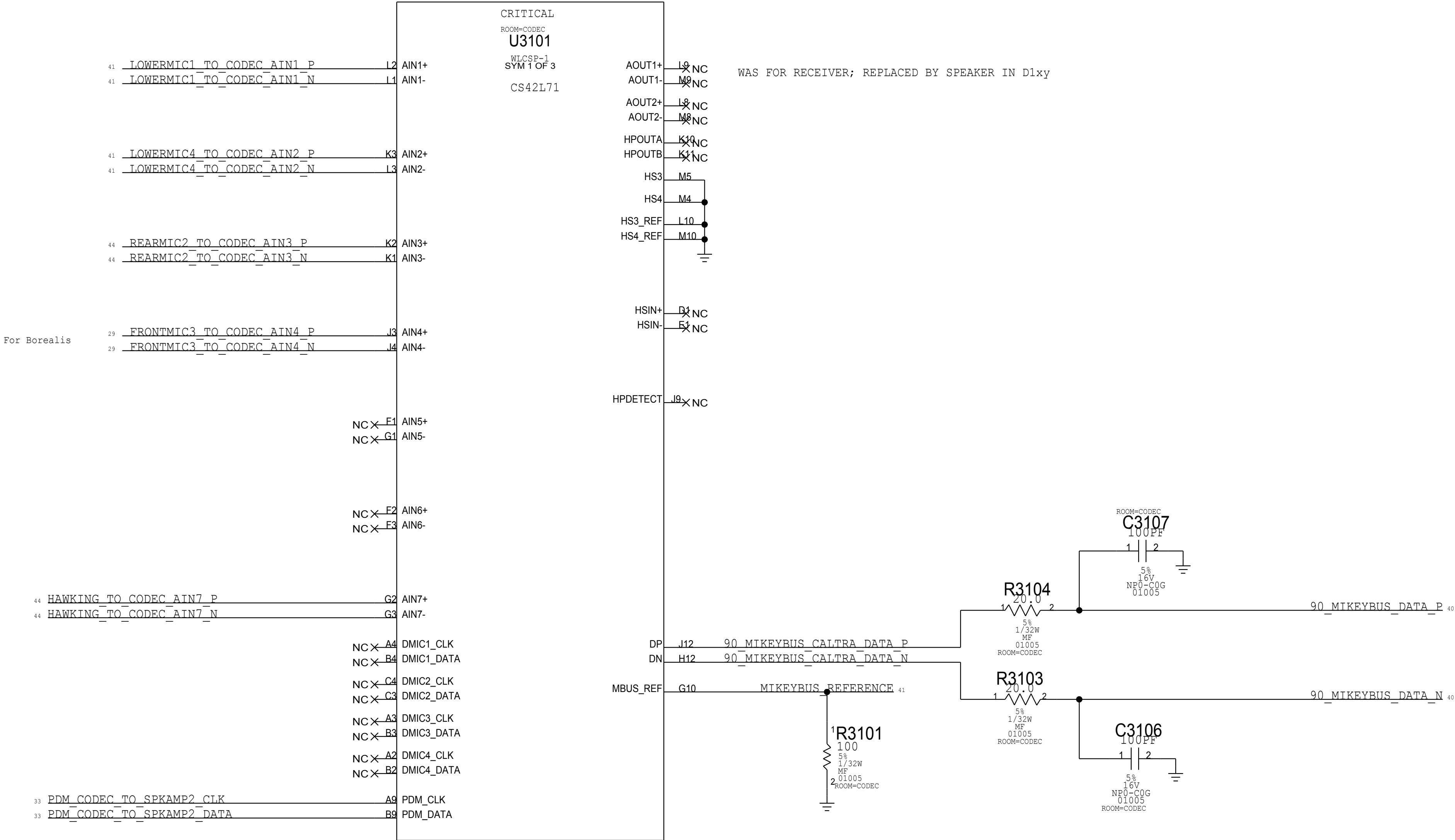
SPEAKER2



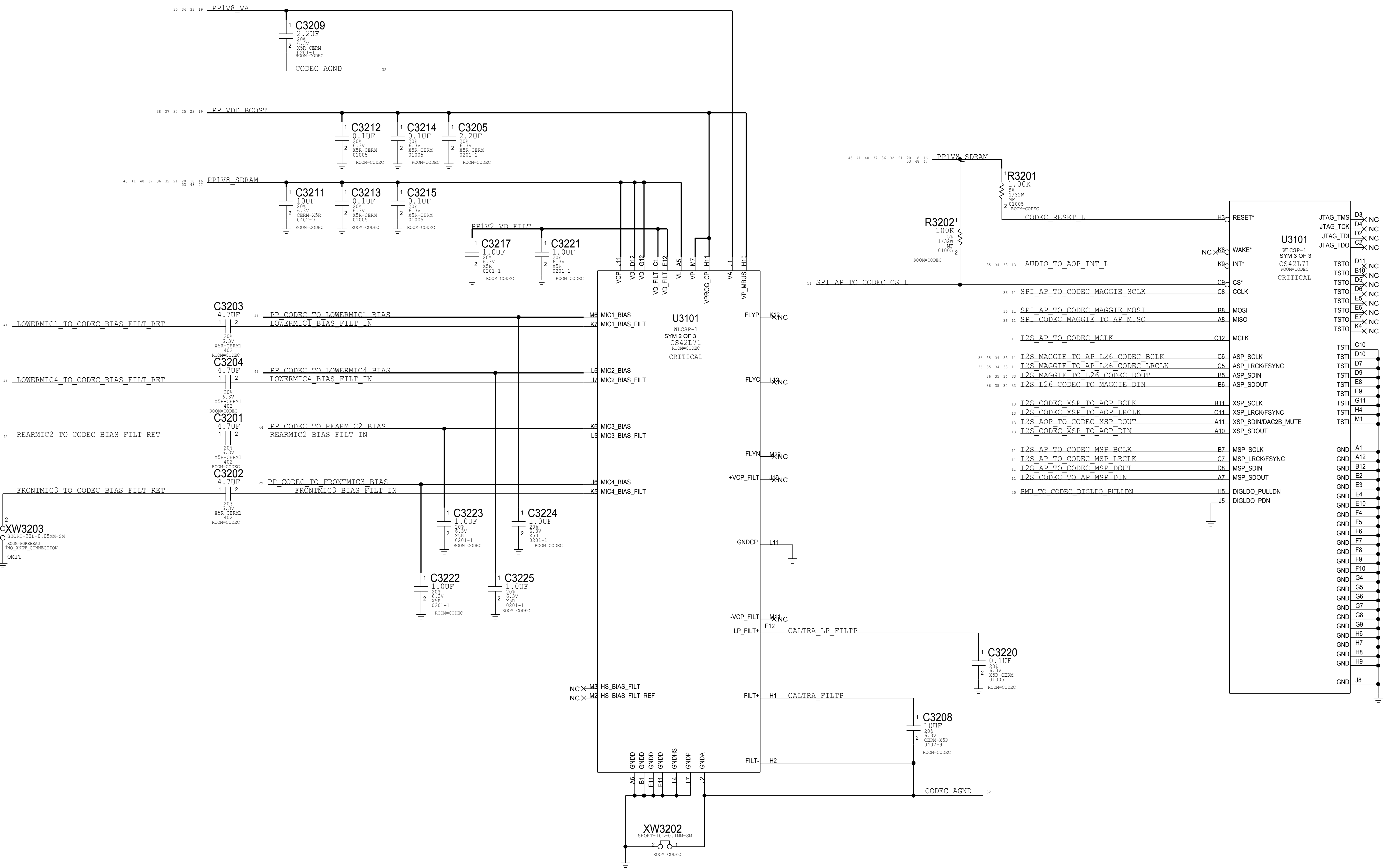
MIC3



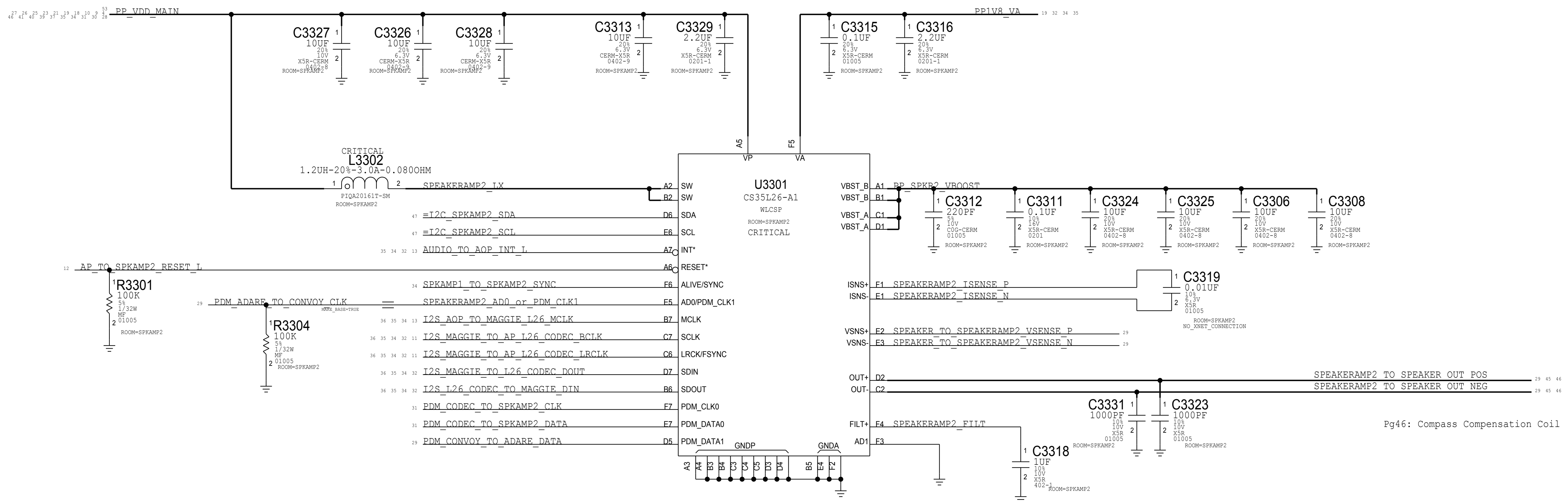
CALTRA AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



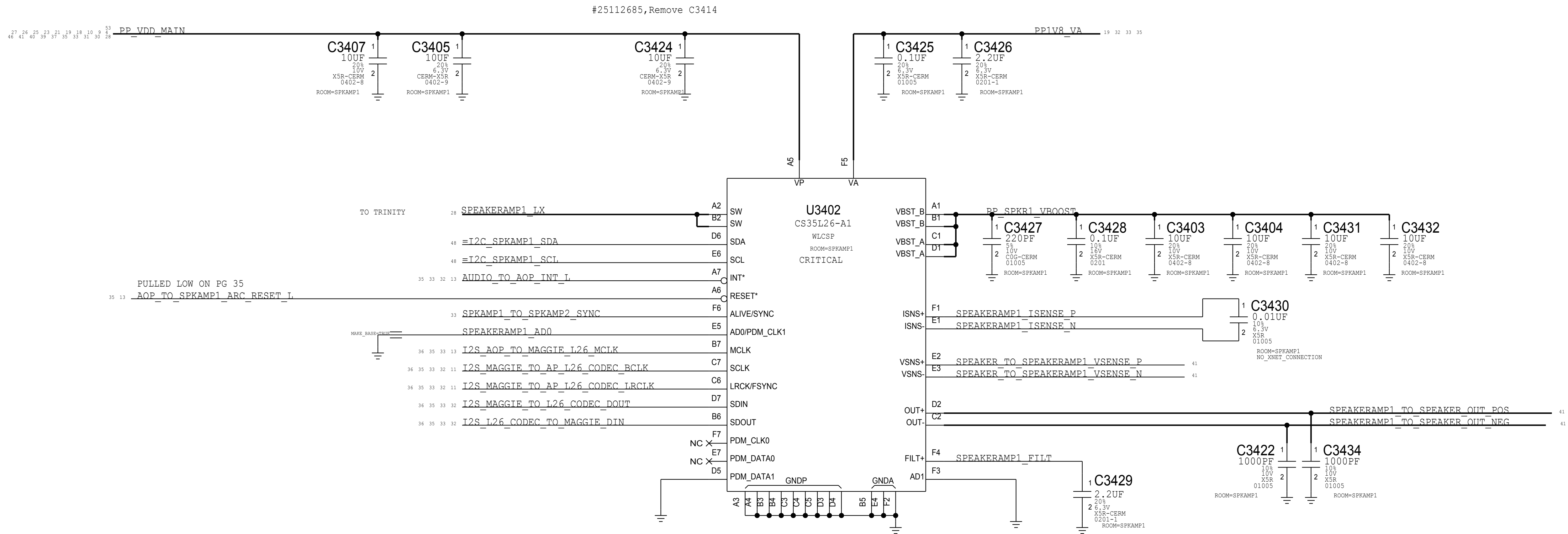
CALTRA AUDIO CODEC (POWER & I/O)



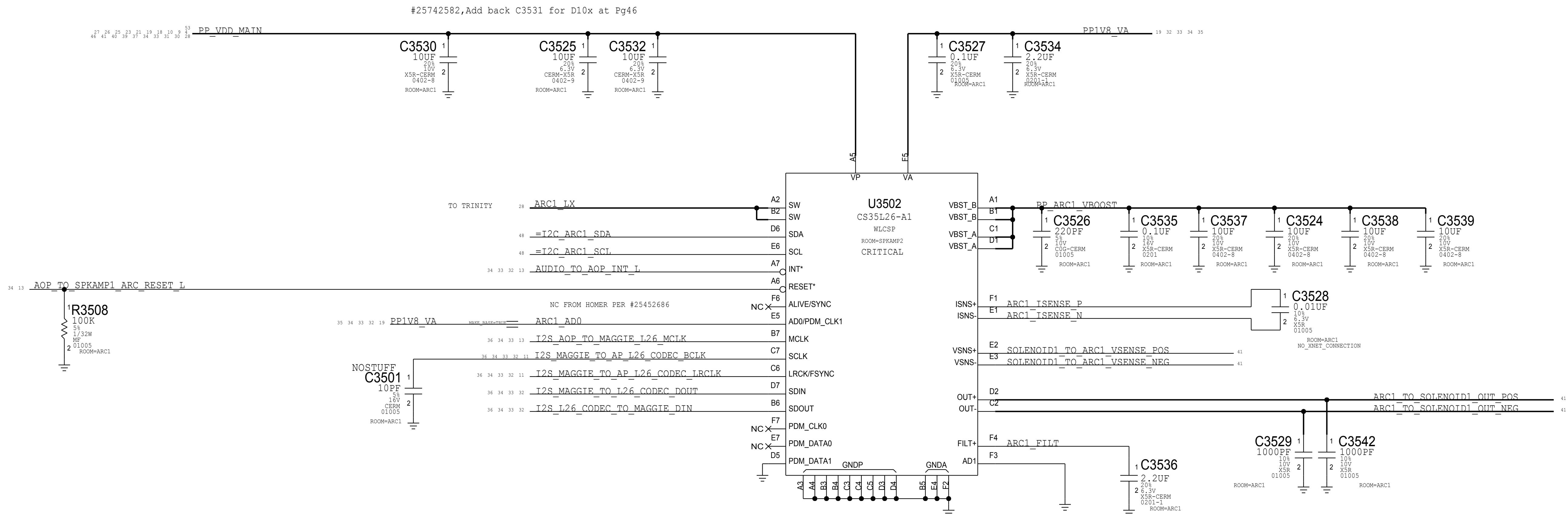
SPEAKER AMPLIFIER 2 (North)



SPEAKER AMPLIFIER 1 (South)



ARC DRIVER

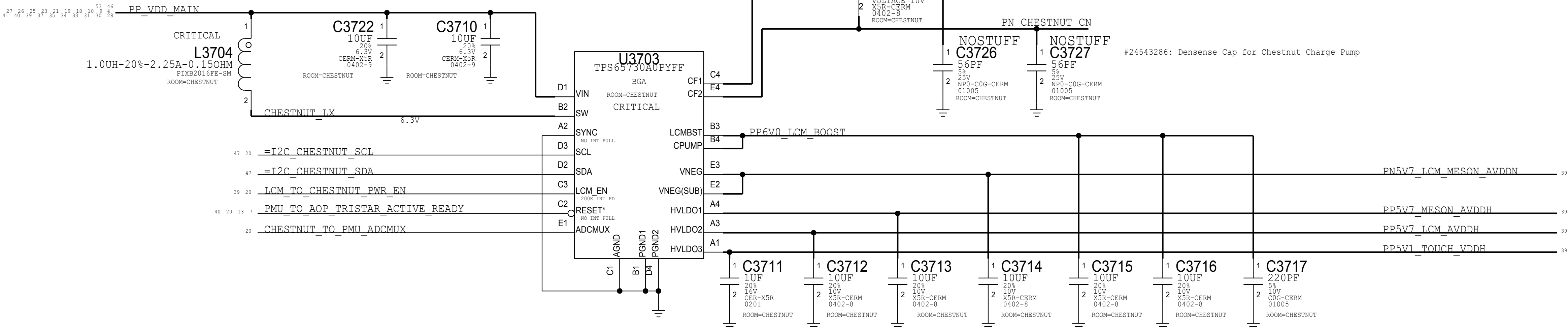




DISPLAY & TOUCH - POWER SUPPLIES

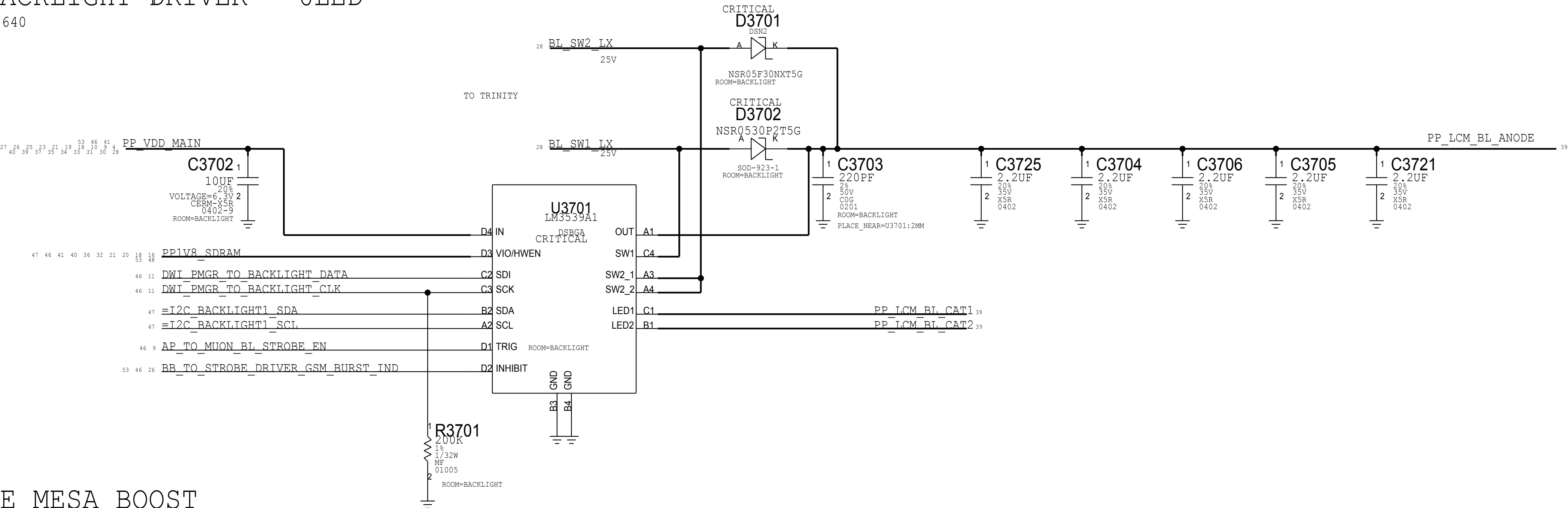
CHESTNUT DISPLAY PMU

APN:338S1172



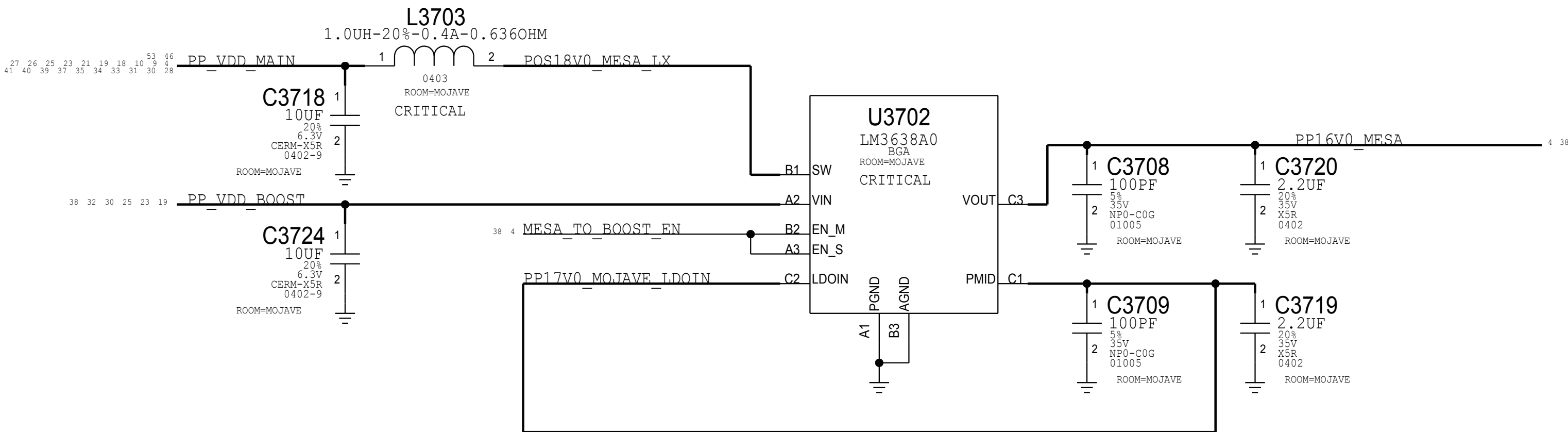
LED BACKLIGHT DRIVER - 6LED

APN:353s00640

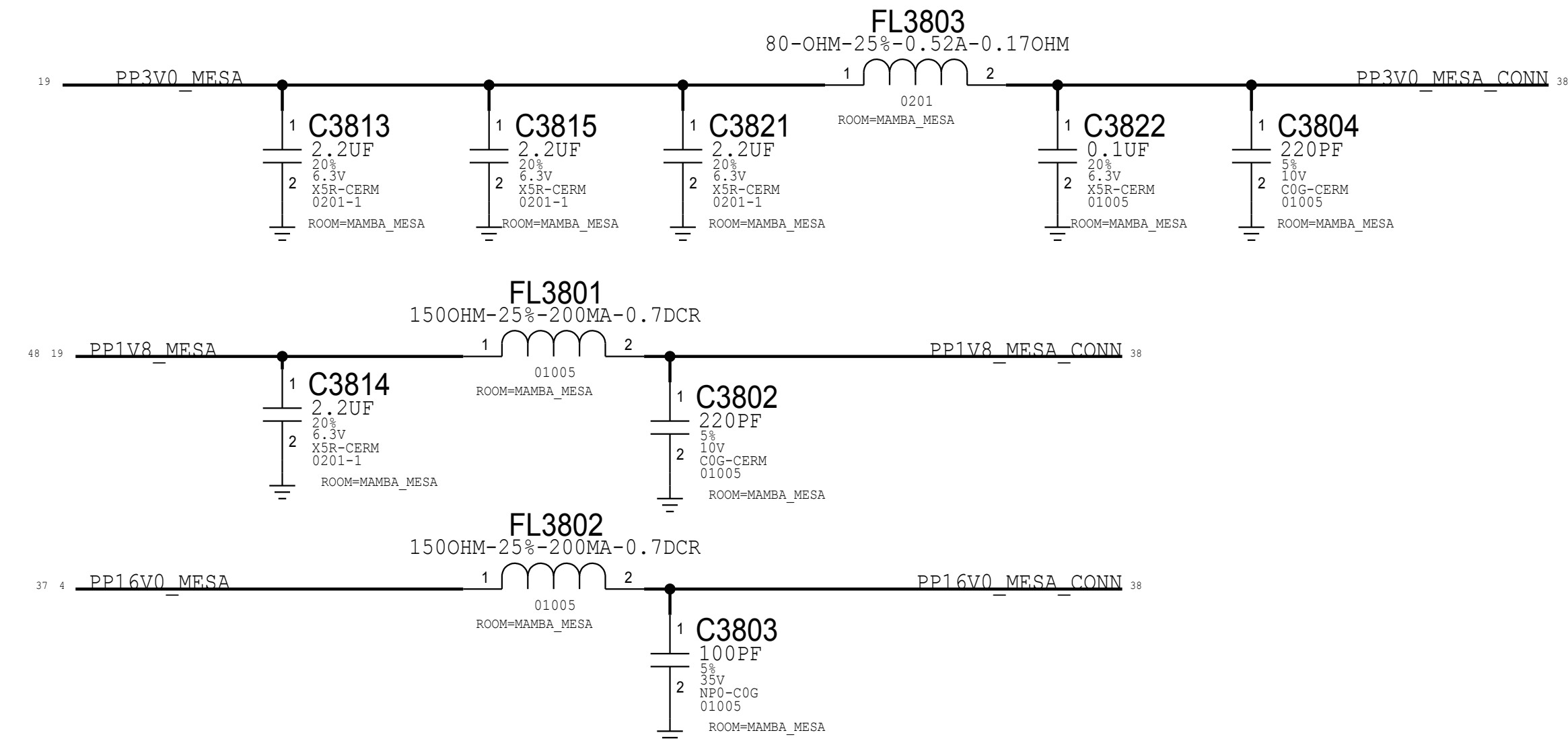


MOJAVE MESA BOOST

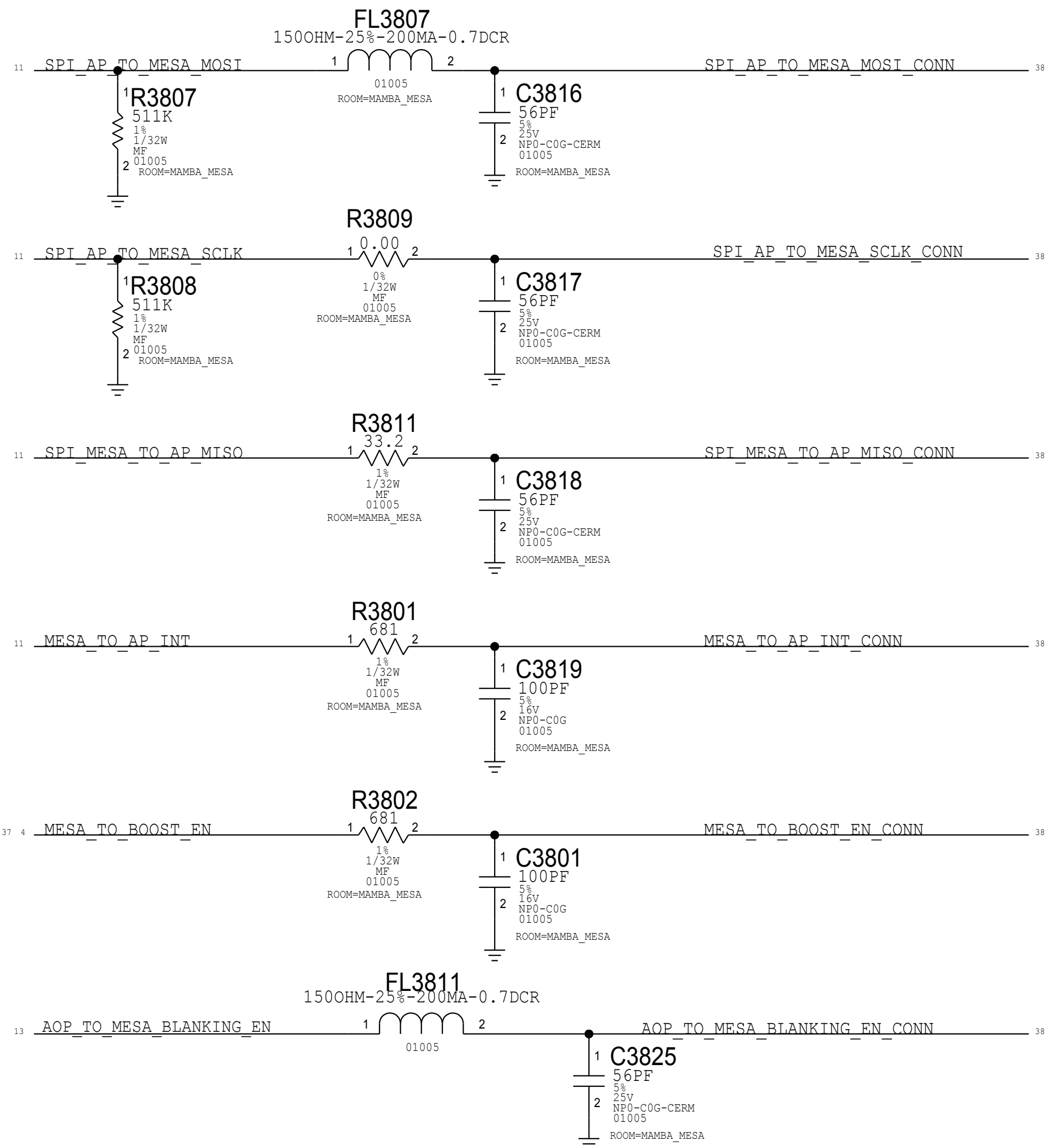
APN:353S00671



MESA POWER

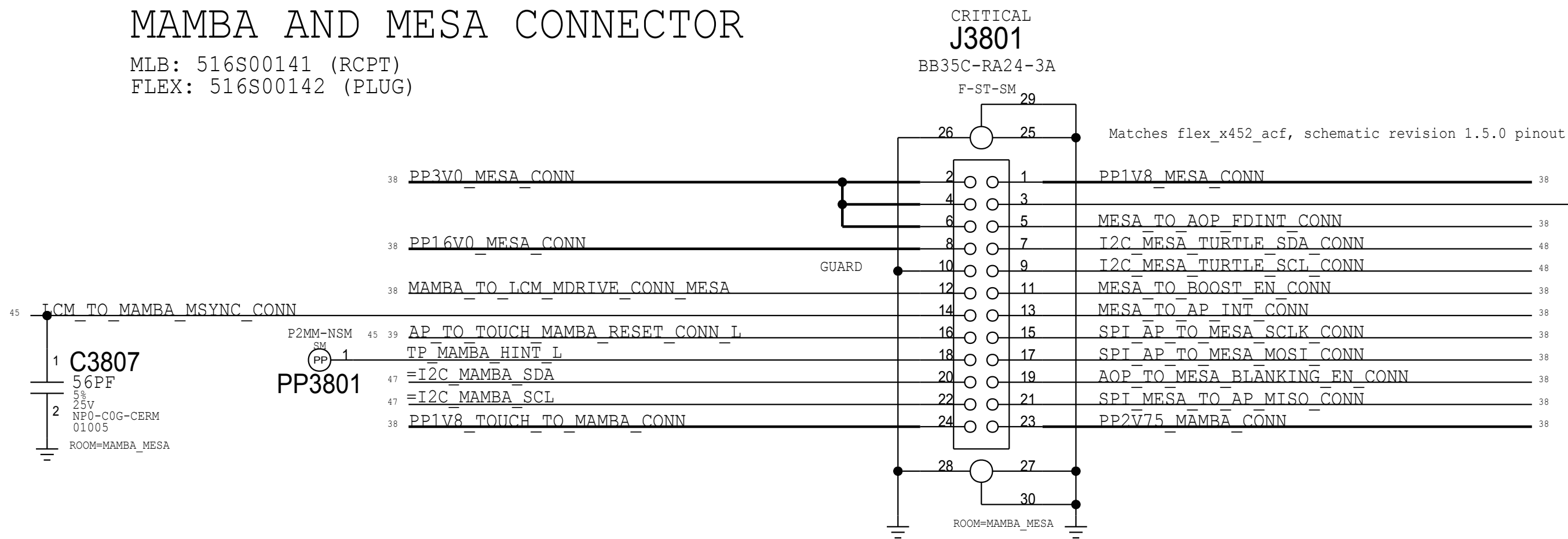


MESA DIGITAL I/O



MAMBA AND MESA CONNECTOR

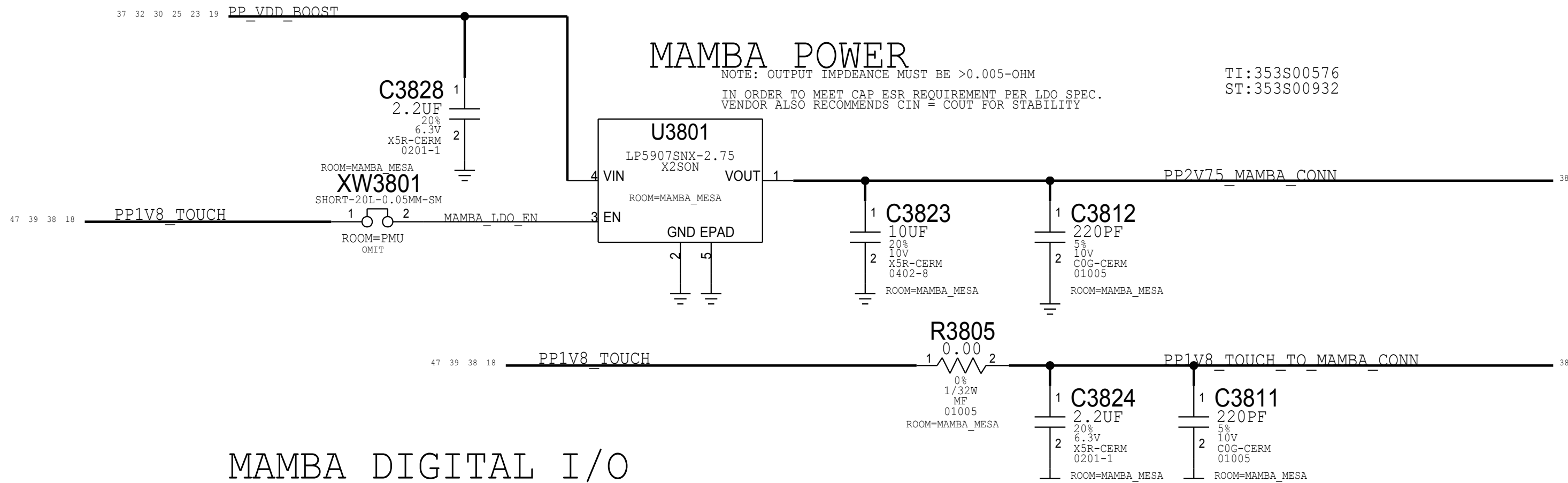
MLB: 516S00141 (RCPT)
FLEX: 516S00142 (PLUG)



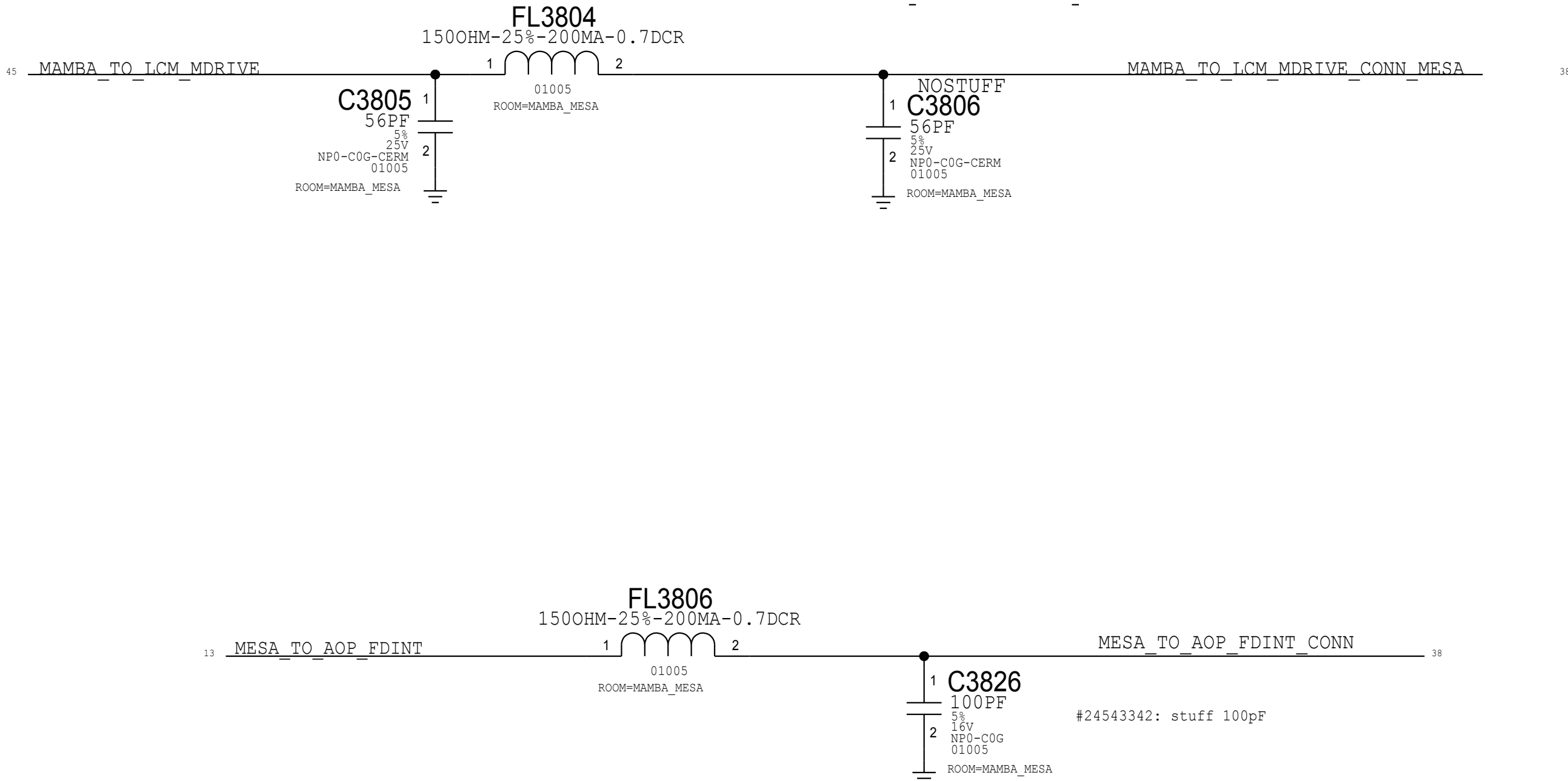
MAMBA POWER

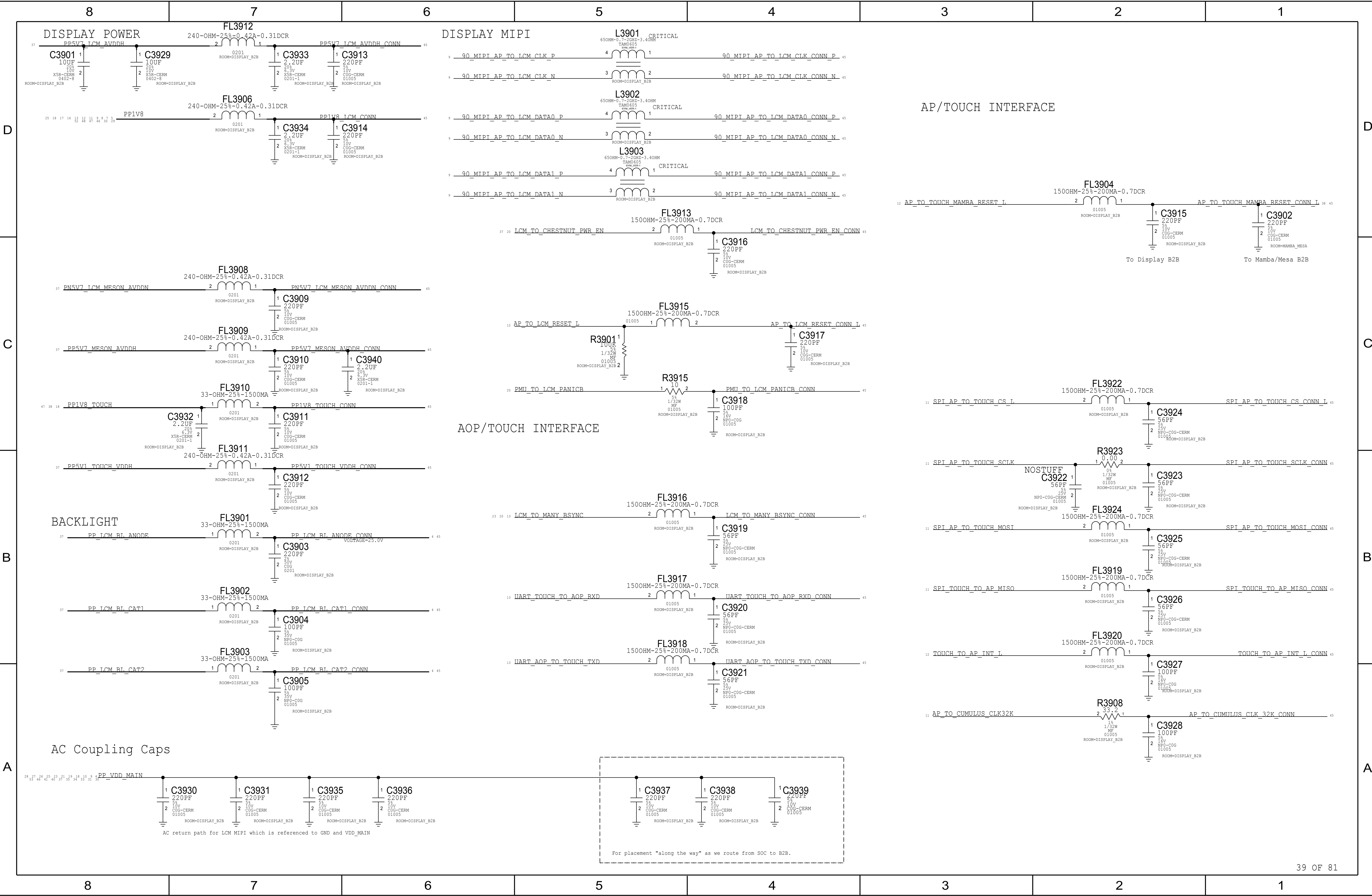
NOTE: OUTPUT IMPEDANCE MUST BE >0.005-OHM
IN ORDER TO MEET CAP ESR REQUIREMENT PER LDO SPEC.
VENDOR ALSO RECOMMENDS CIN = COUT FOR STABILITY.

TI:353800576
ST:353800932



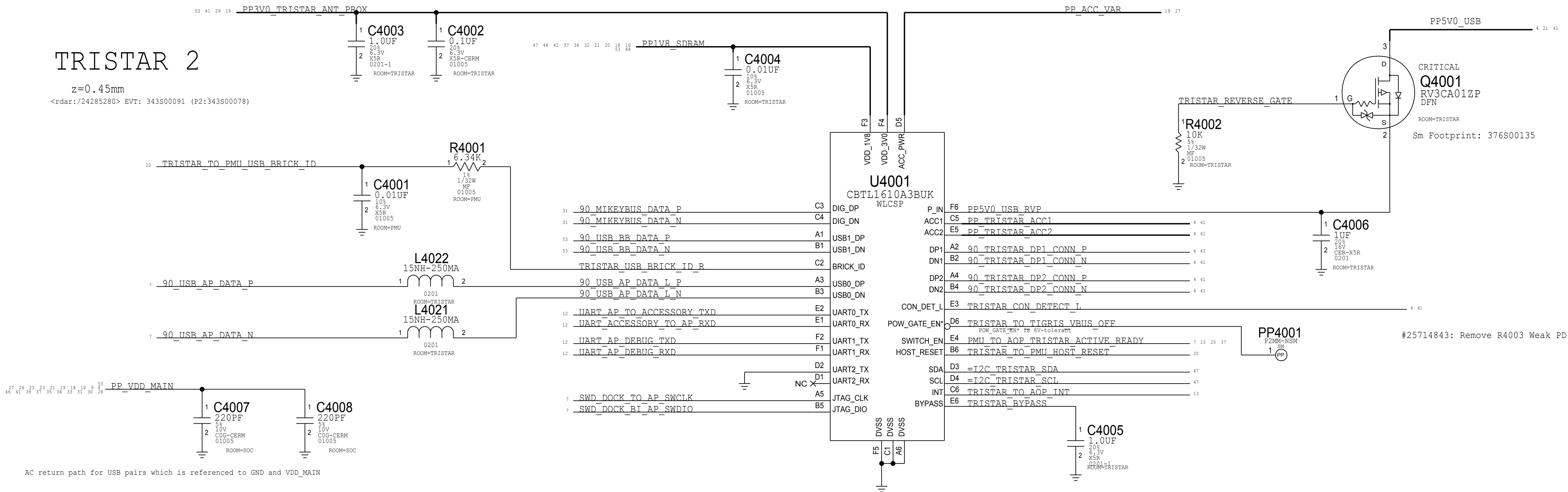
MAMBA DIGITAL I/O





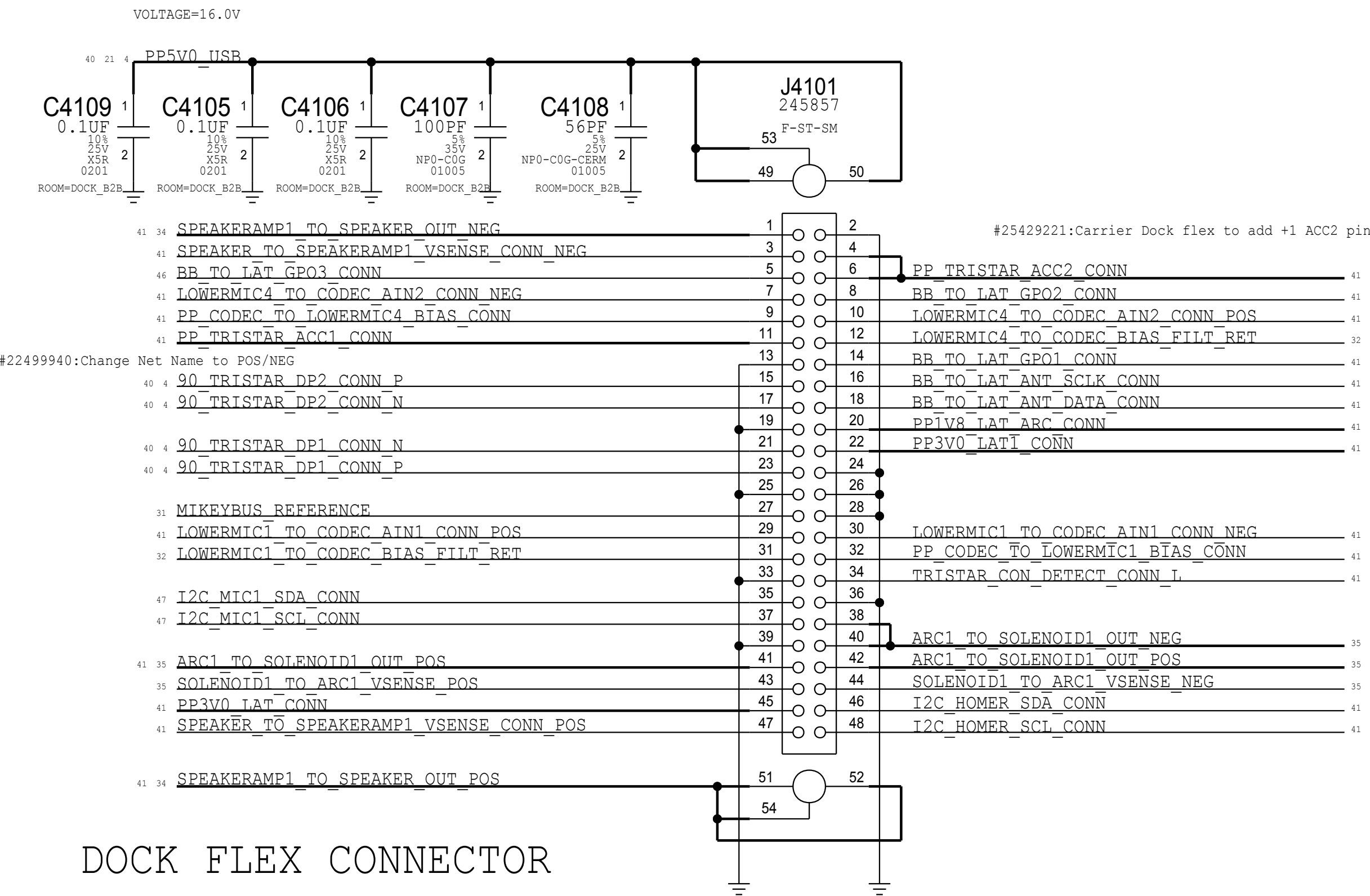
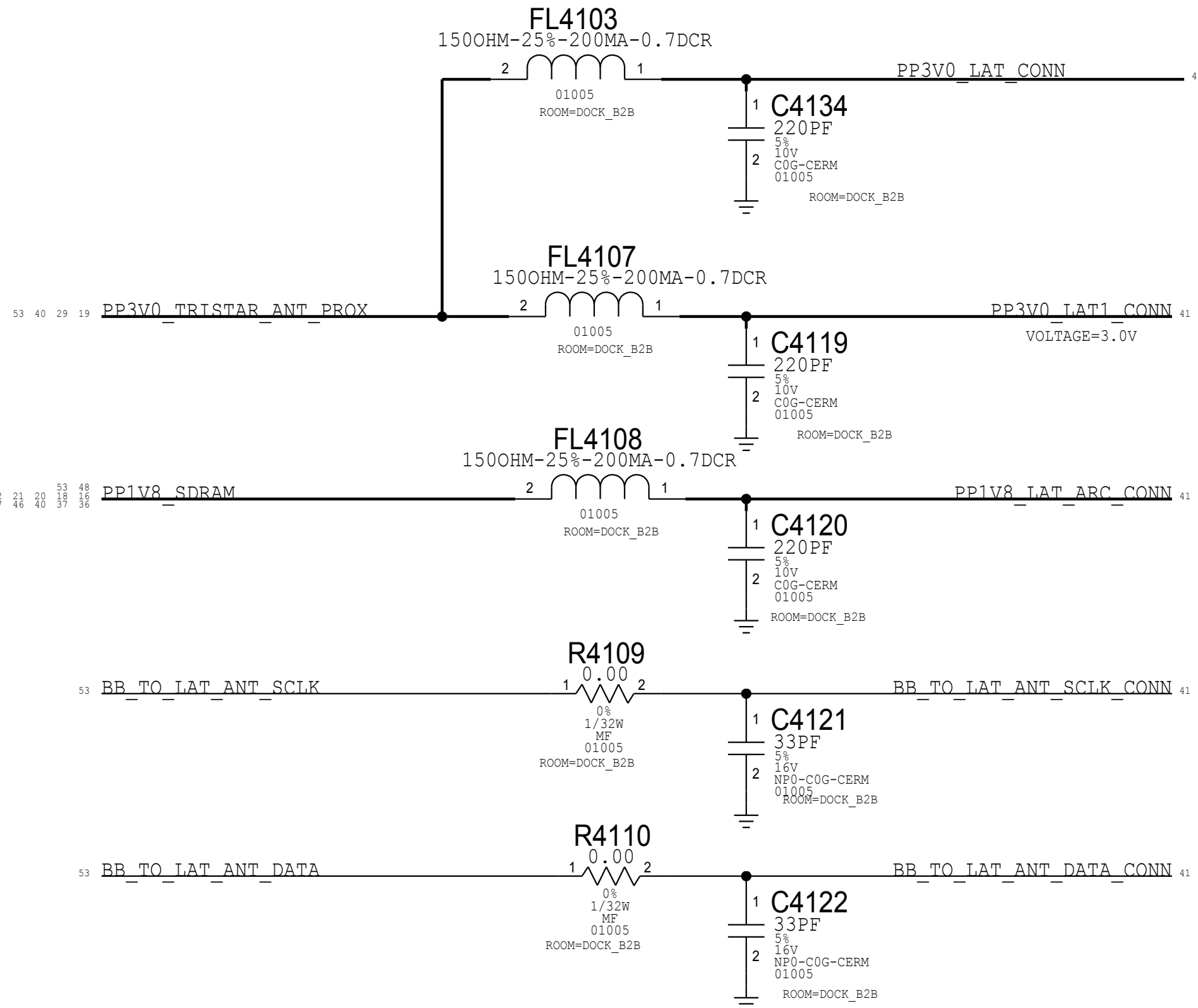
TRISTAR 2

z=0.45mm
<rdar://24285280> EVT: 343S00091 (P2:343S00078)

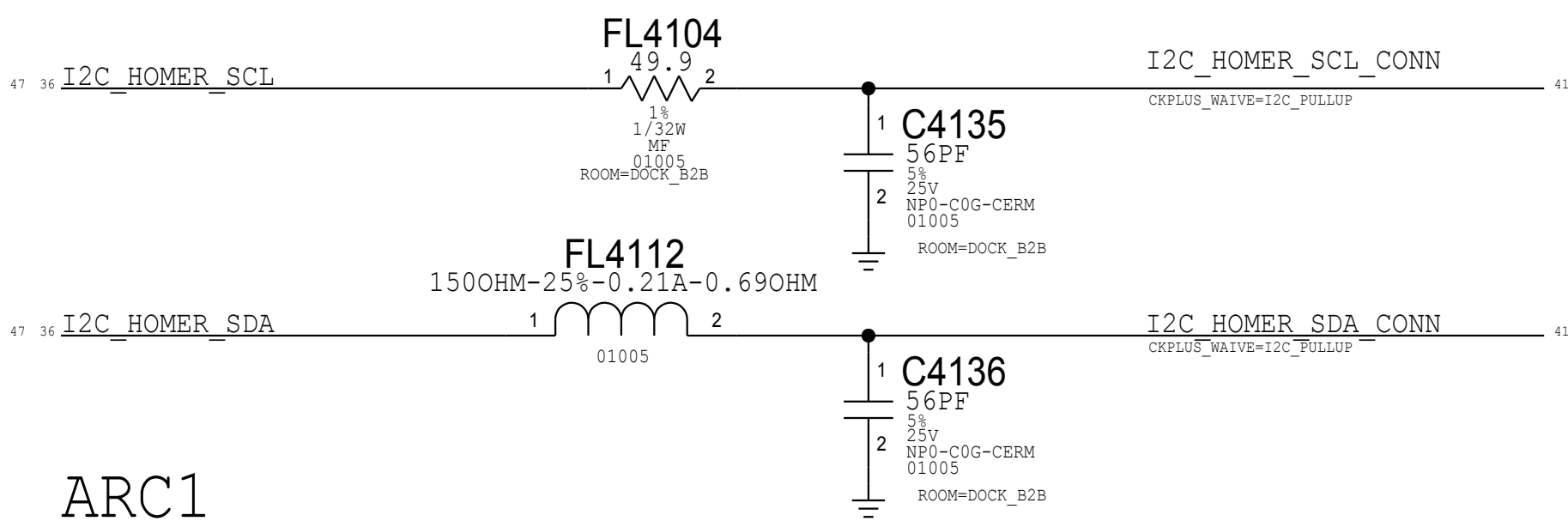


ANTENNA

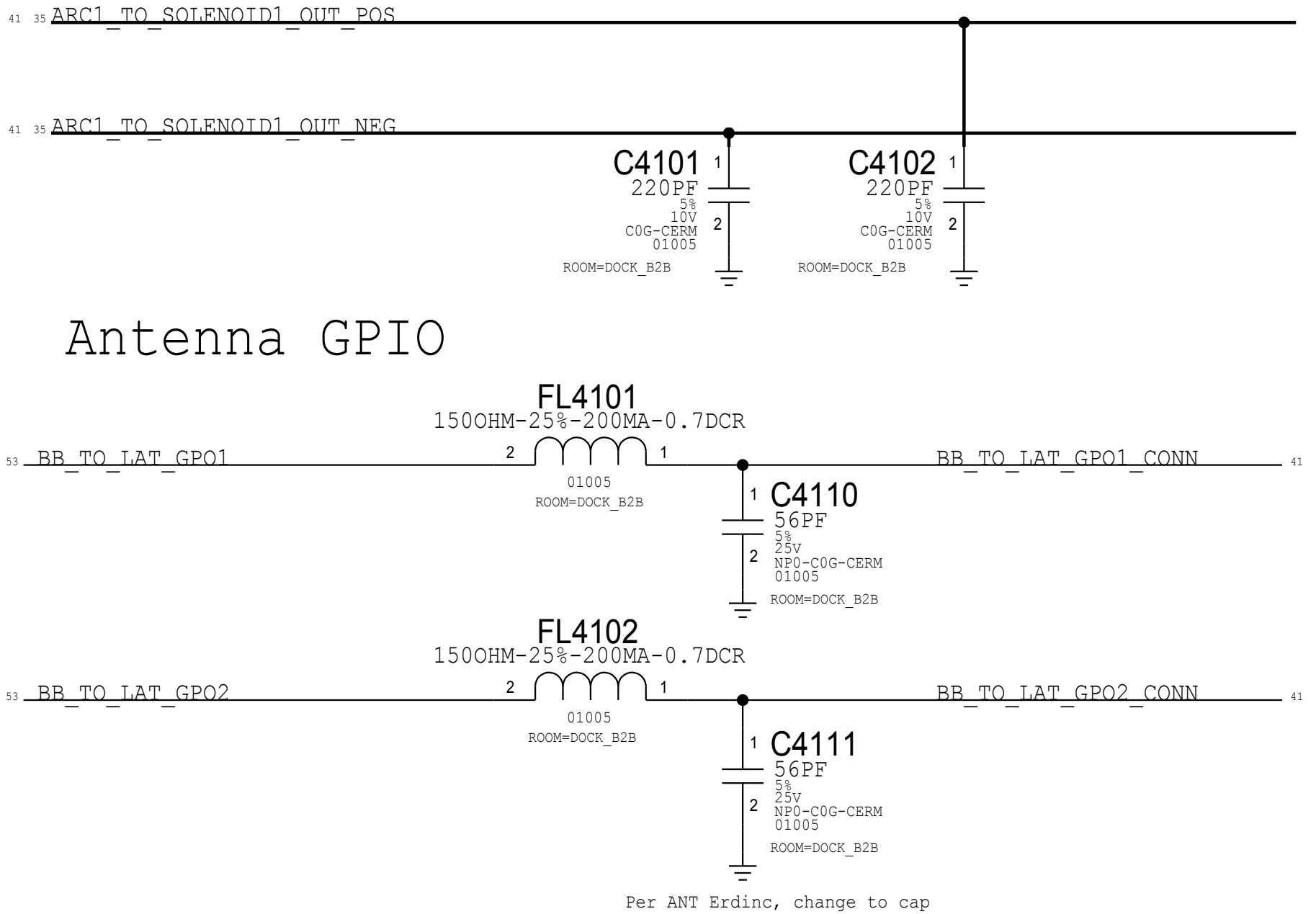
Please loop in Matt Mow (Antenna Team)
when changing these components!



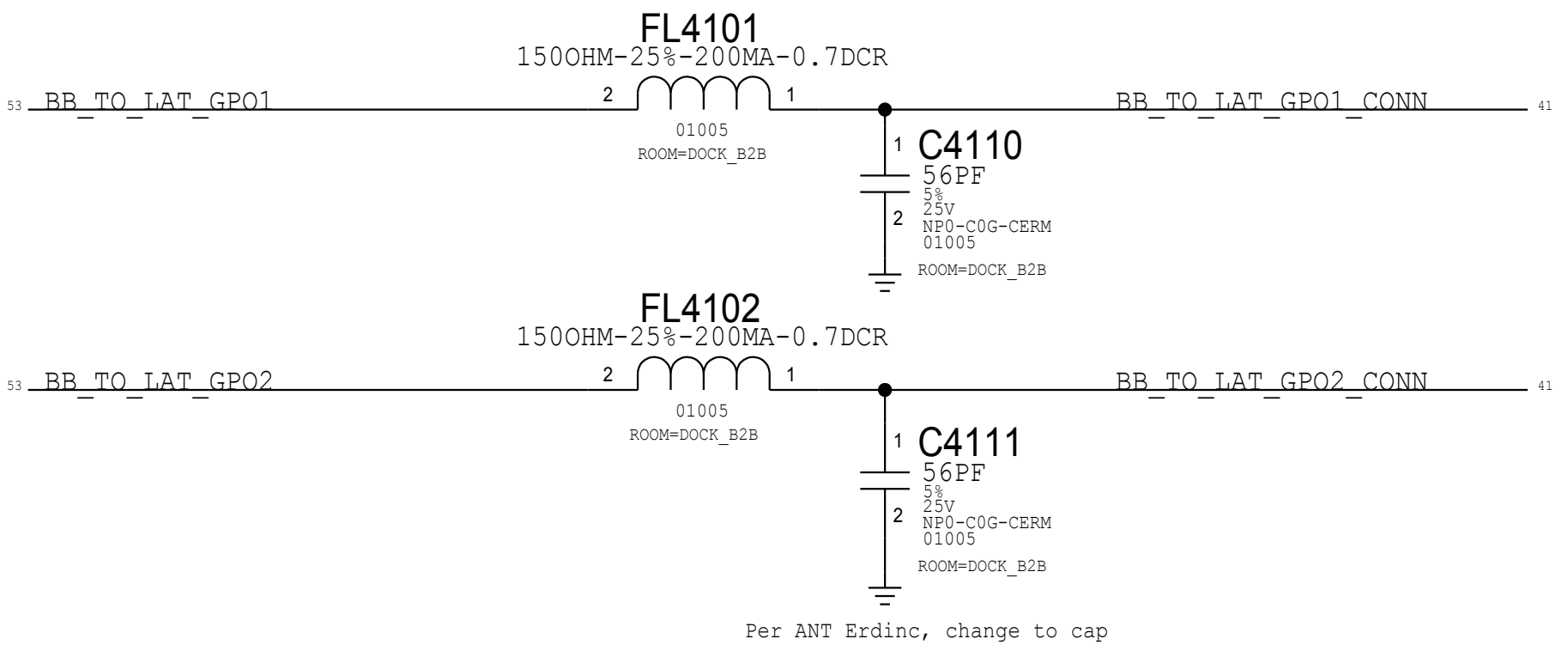
ARC CONTROL



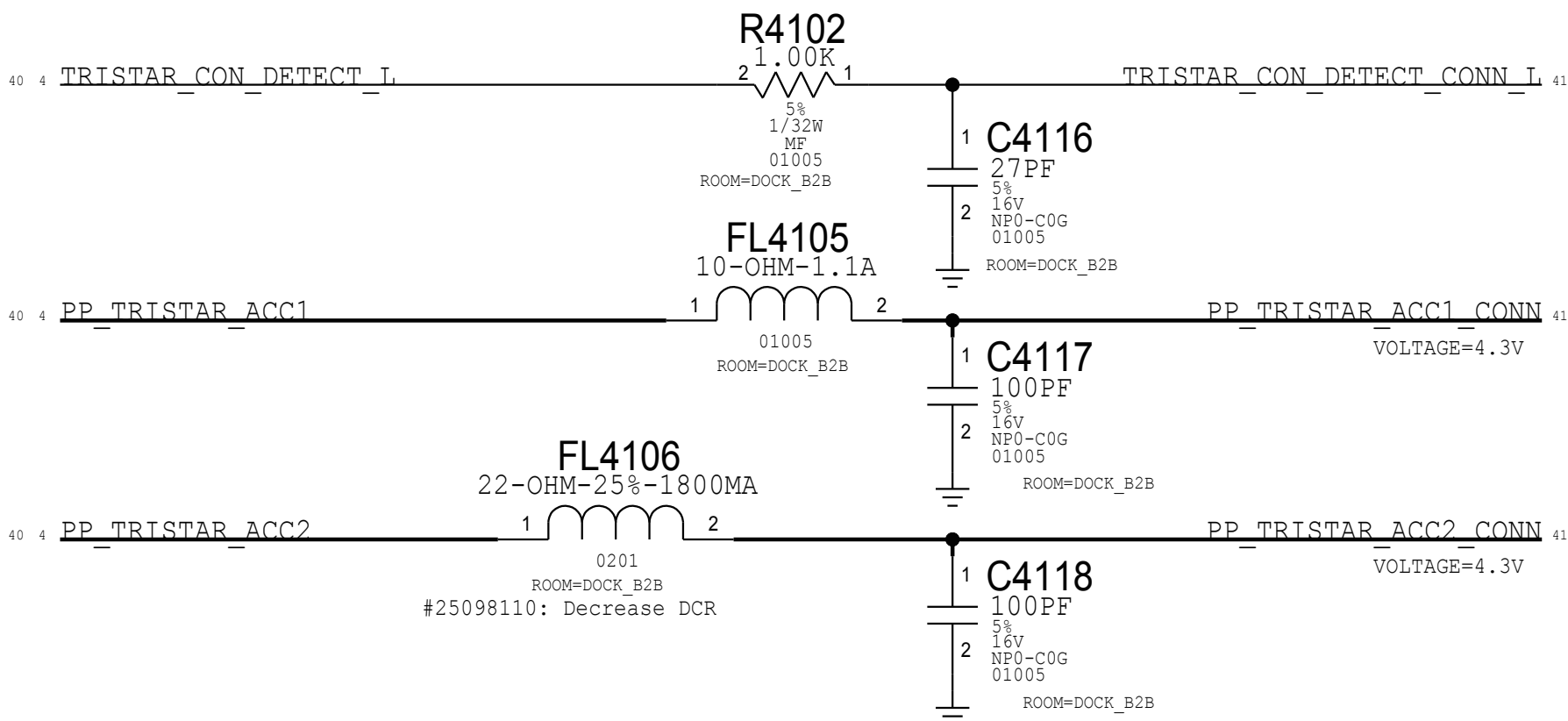
ARC1



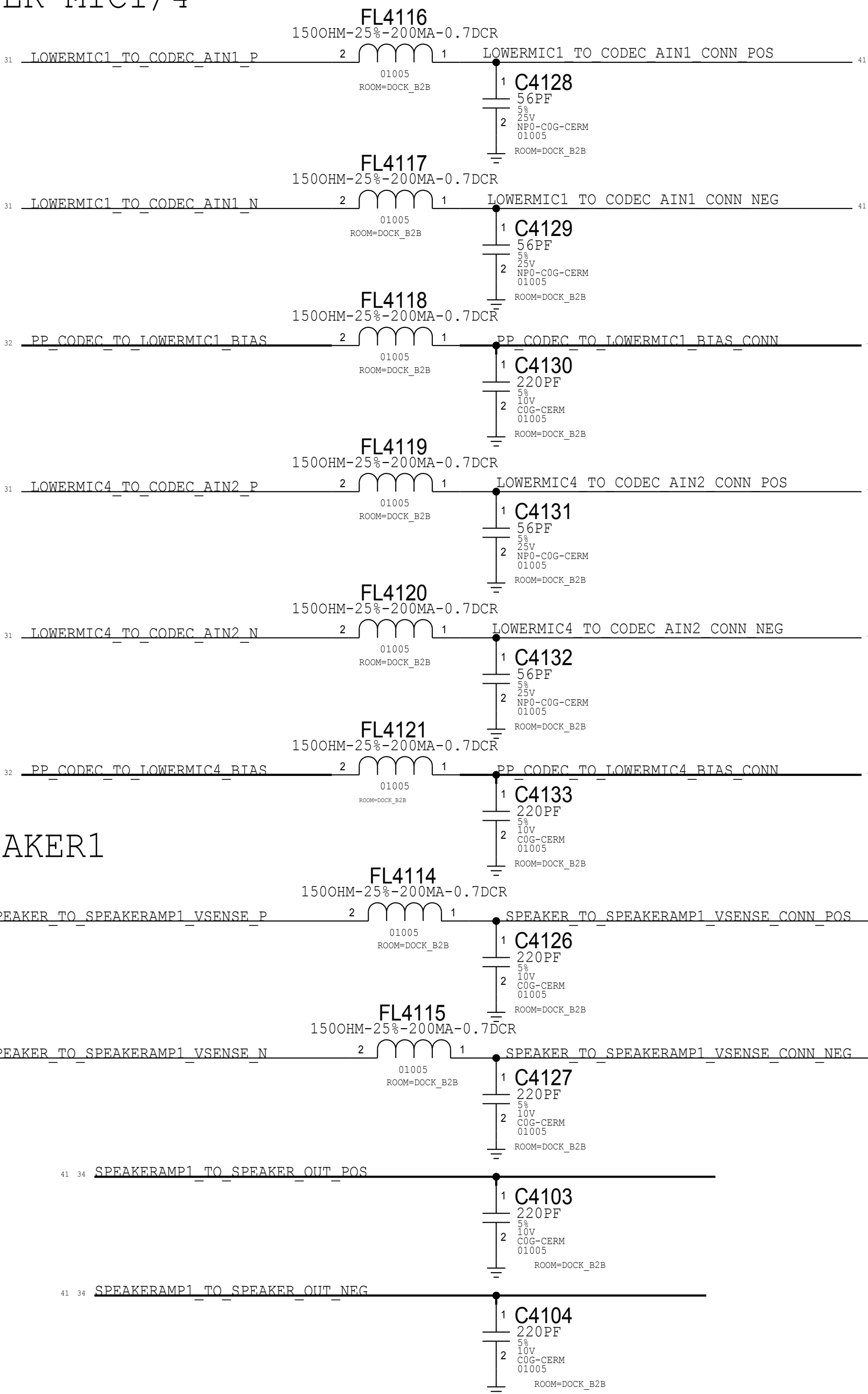
Antenna GPIO



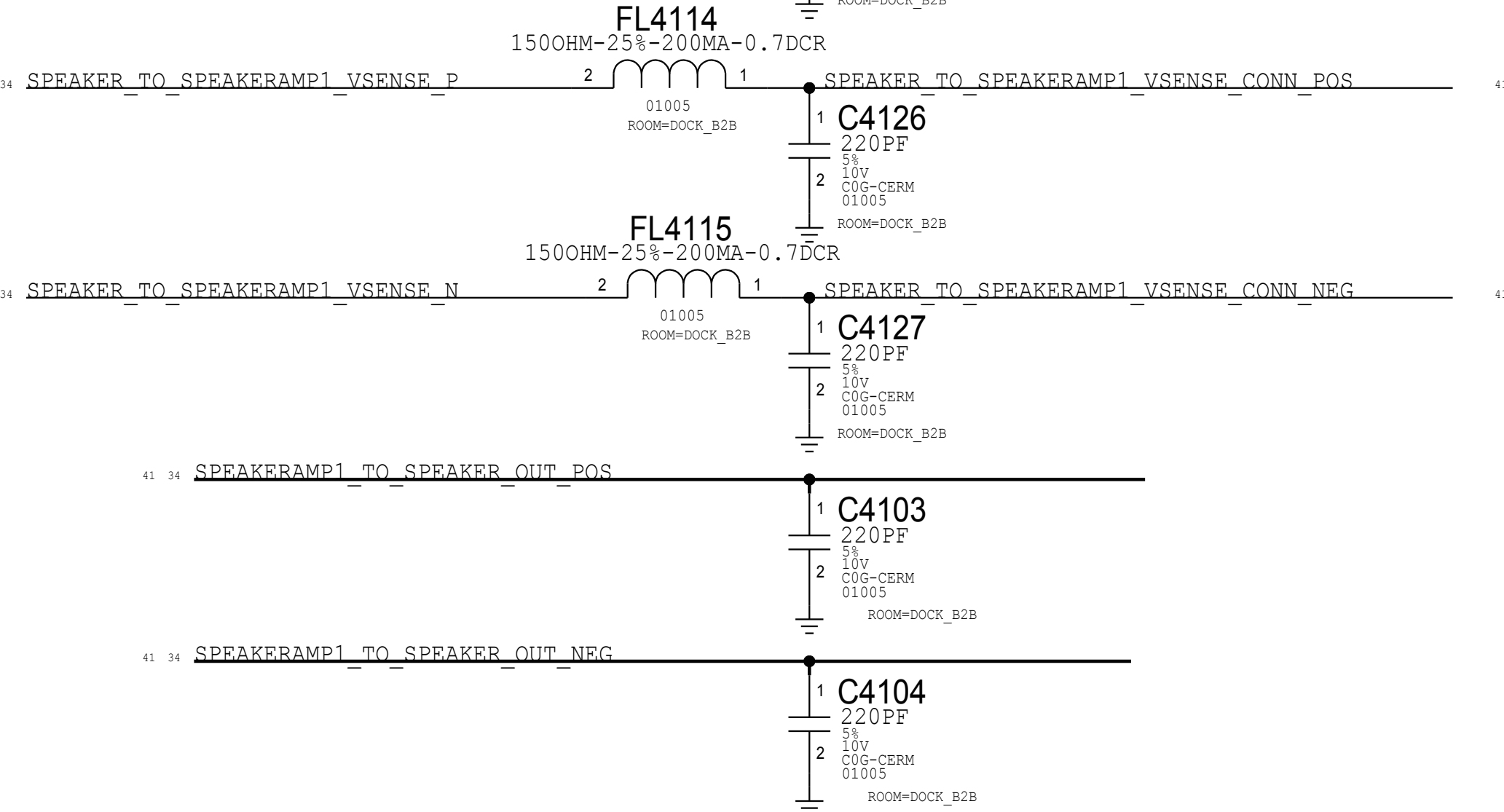
TRISTAR



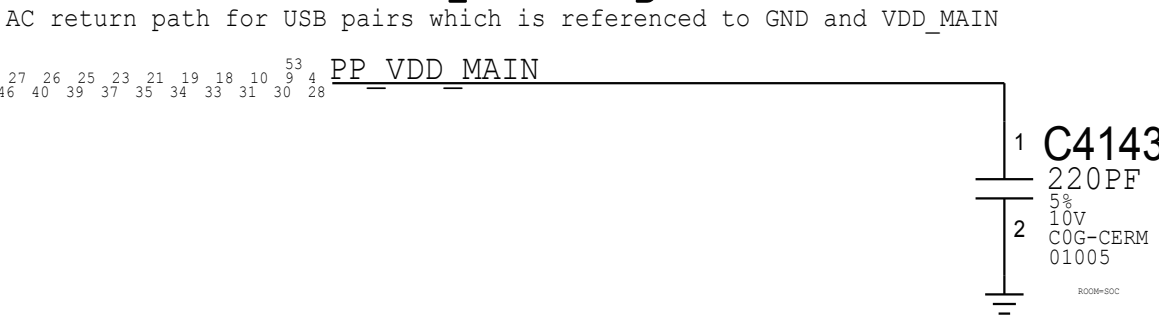
LOWER MIC1/4



SPEAKER1



USB AC Coupling



DOCK FLEX CONNECTOR

D

D

C

C

B

B

A

A

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43 OF 81

D

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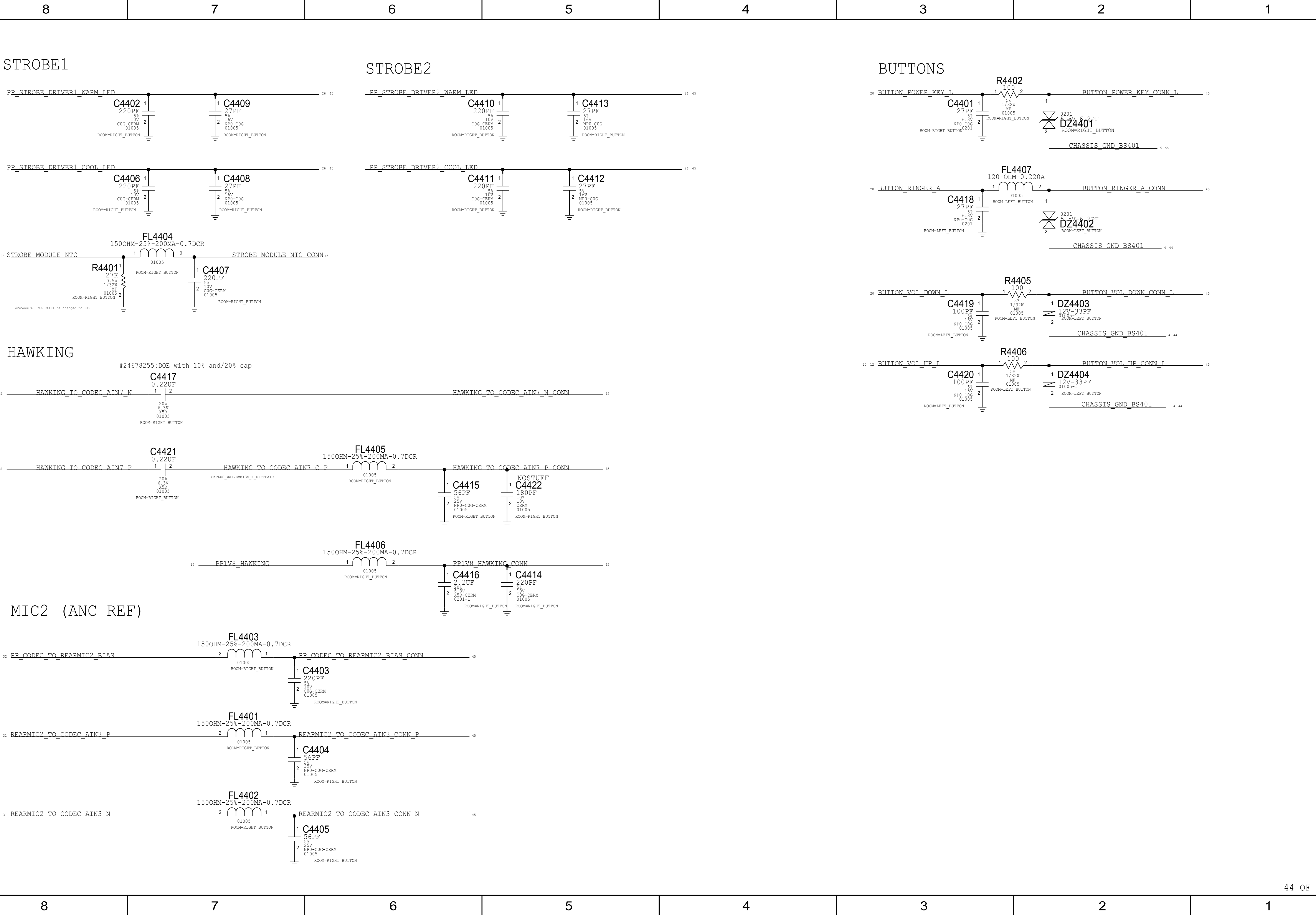
A

D

C

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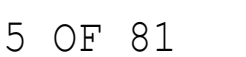
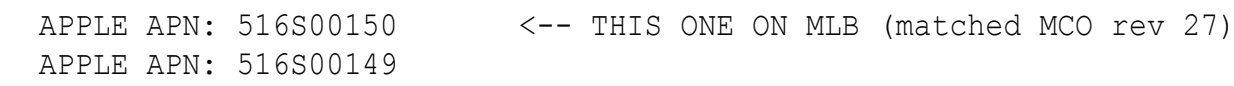
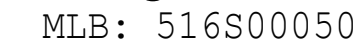
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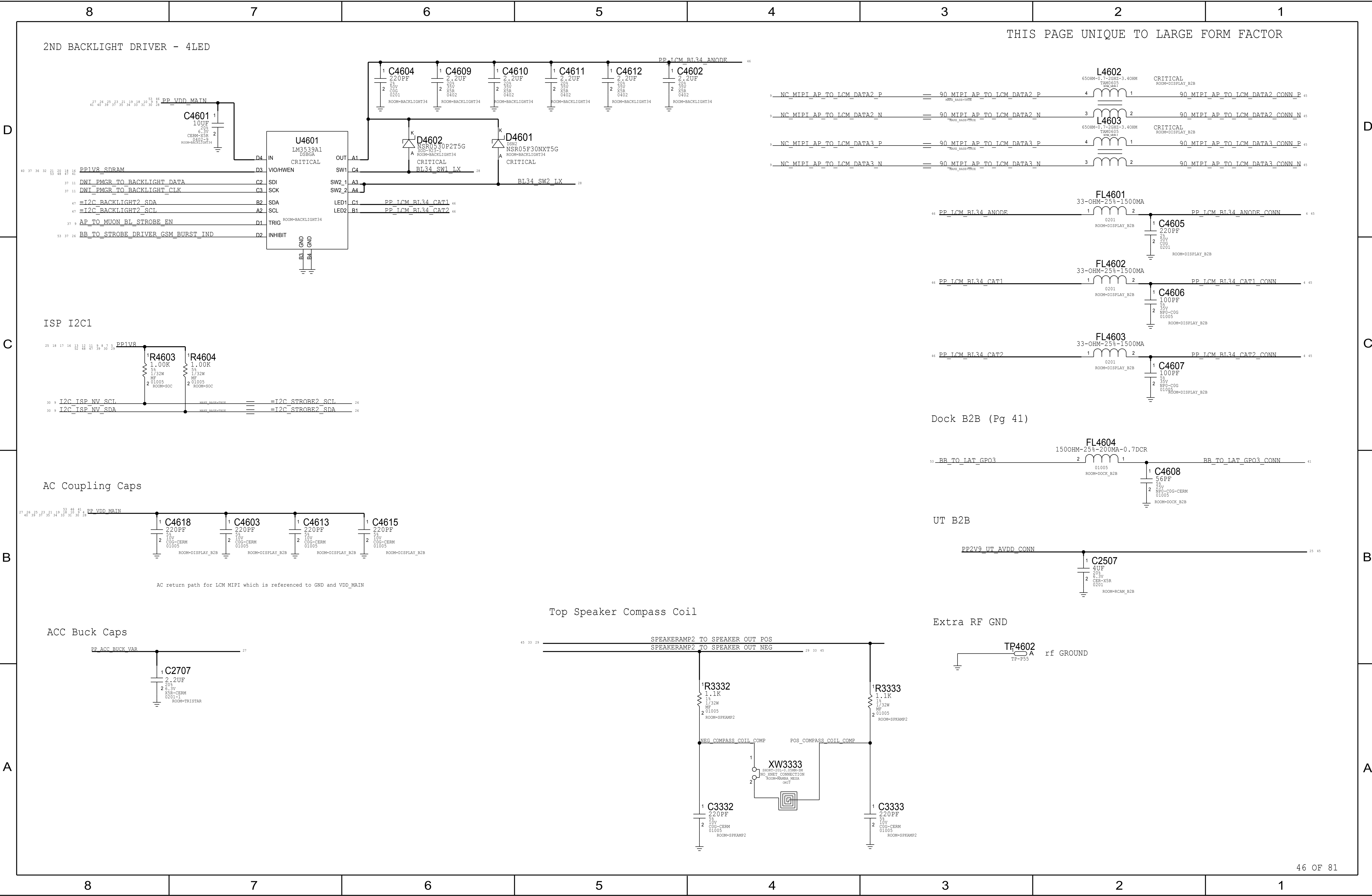



```
516S00152 RCPT (USED ON MLB)          <-- THIS ONE
516S00151 PLUG
```



MLB: 516S00146 (MCO REV 27)

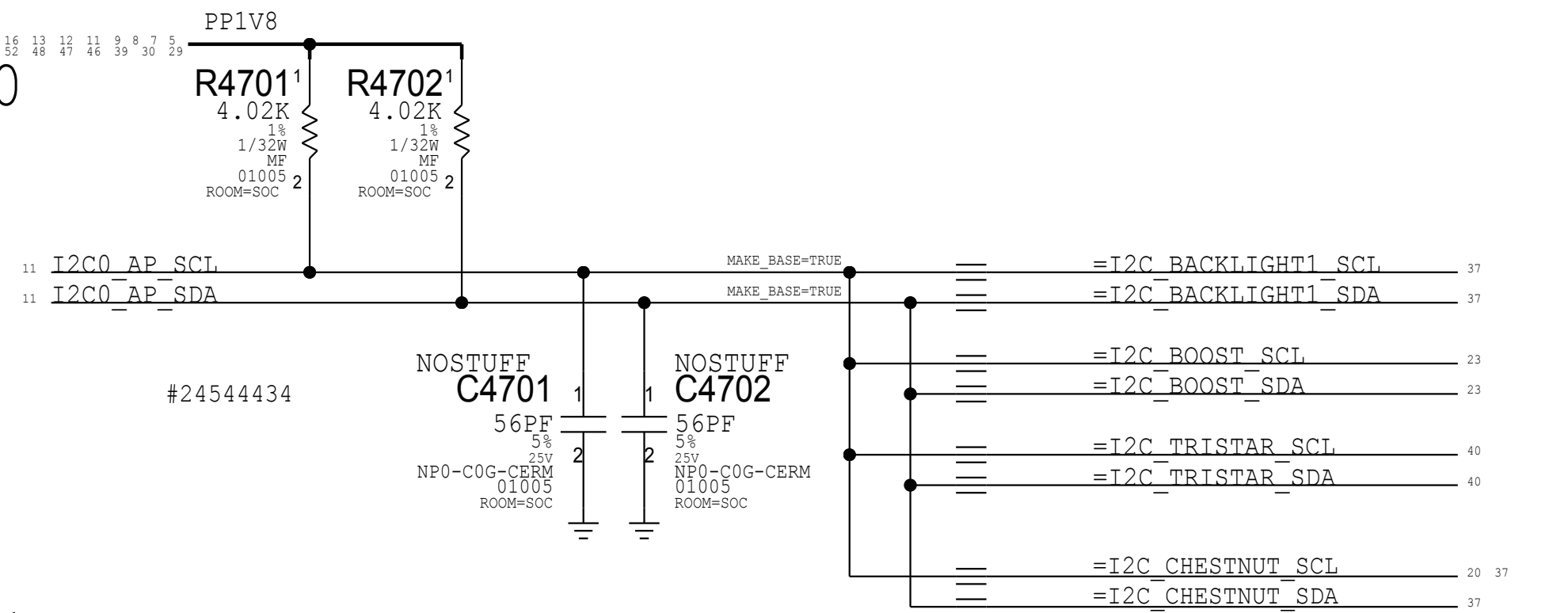




46 OF 81

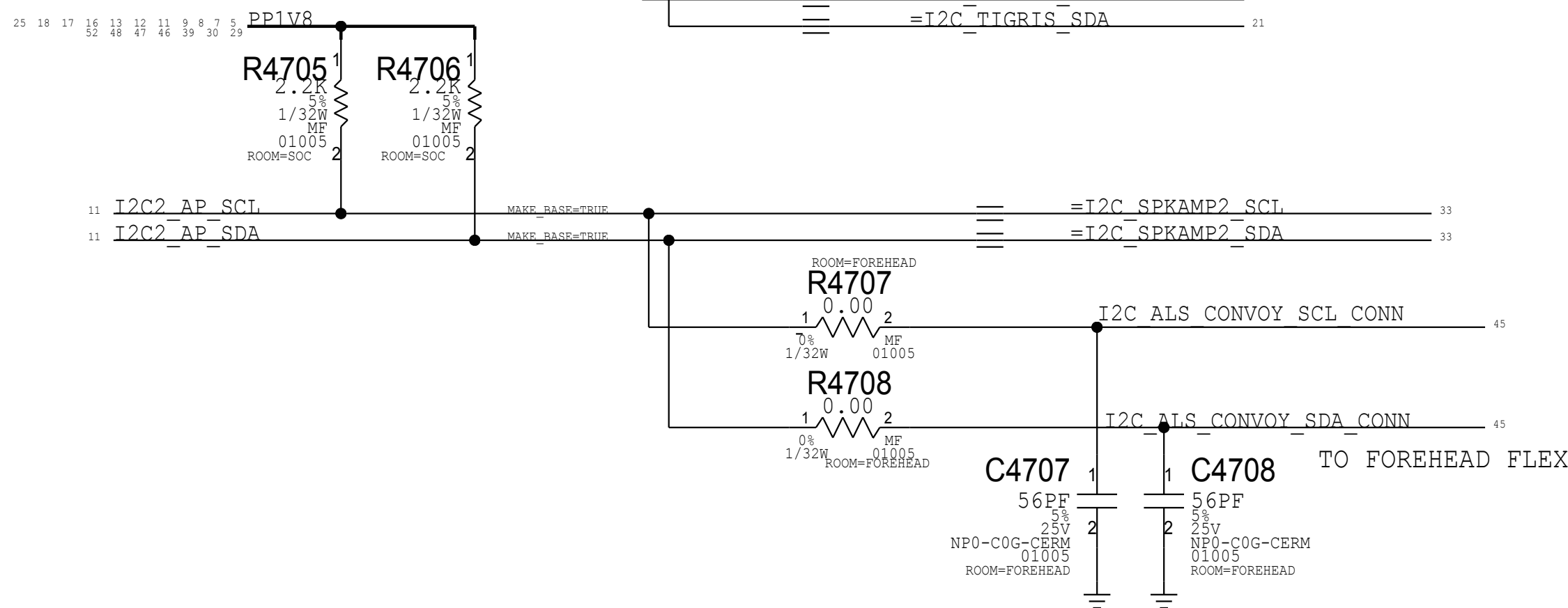
AP

I2C0

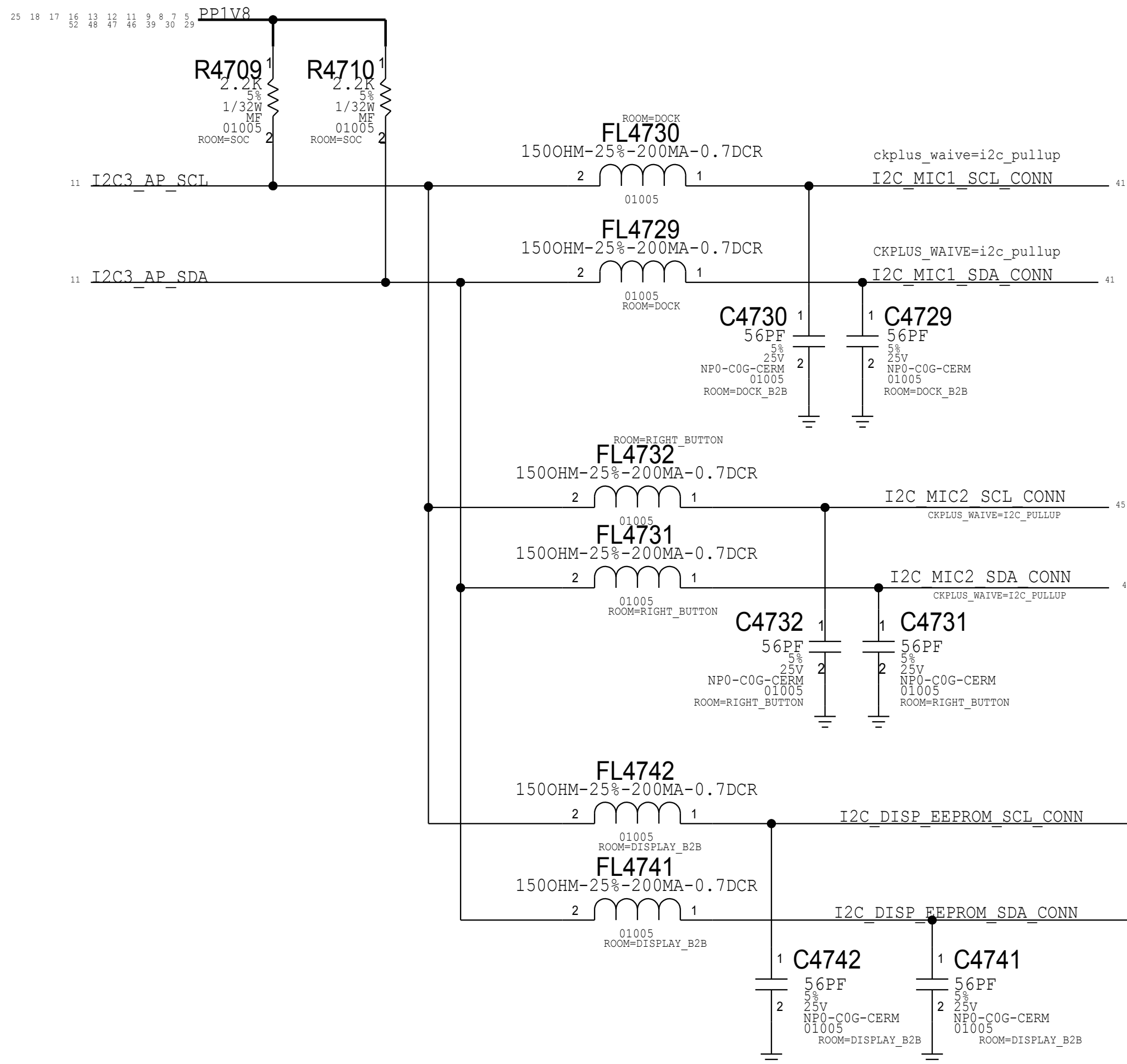


I2C1

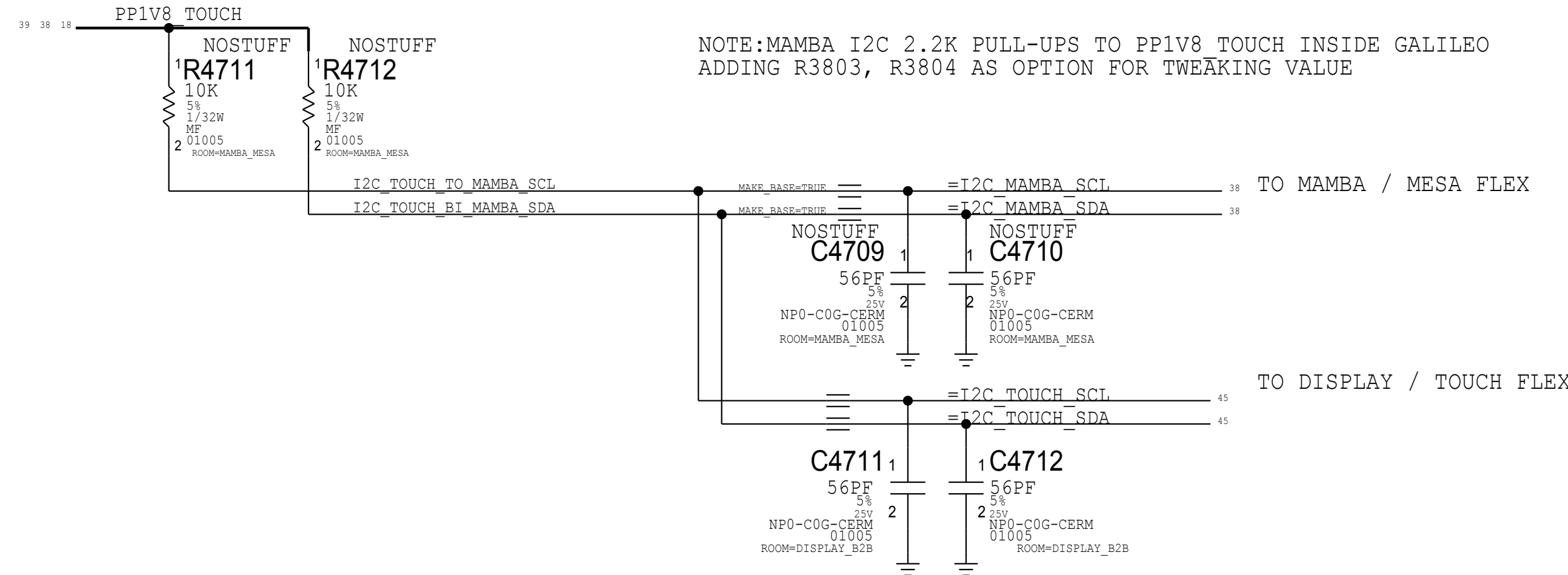
I2C2



I2C3



TOUCH



HOMER

I2C5

NOTE:MAMBA I2C 2.2K PULL-UPS TO PPIV8 TOUCH INSIDE GALILEO
ADDING R3803, R3804 AS OPTION FOR TWEAKING VALUE

AOP

I2C

ISP

I2C0

#24544699: Support 1MHz

#24550735: ISP I2C0 PU

#24958320: Intentional R4815 Change
Reduce undershoot when Prox Driving

Intentional R4815 Change

I2C1
See page 46

I2C2

TO MAMBA/MESA FLEX

8		7		6		5		4		3		2		1			
D																D	

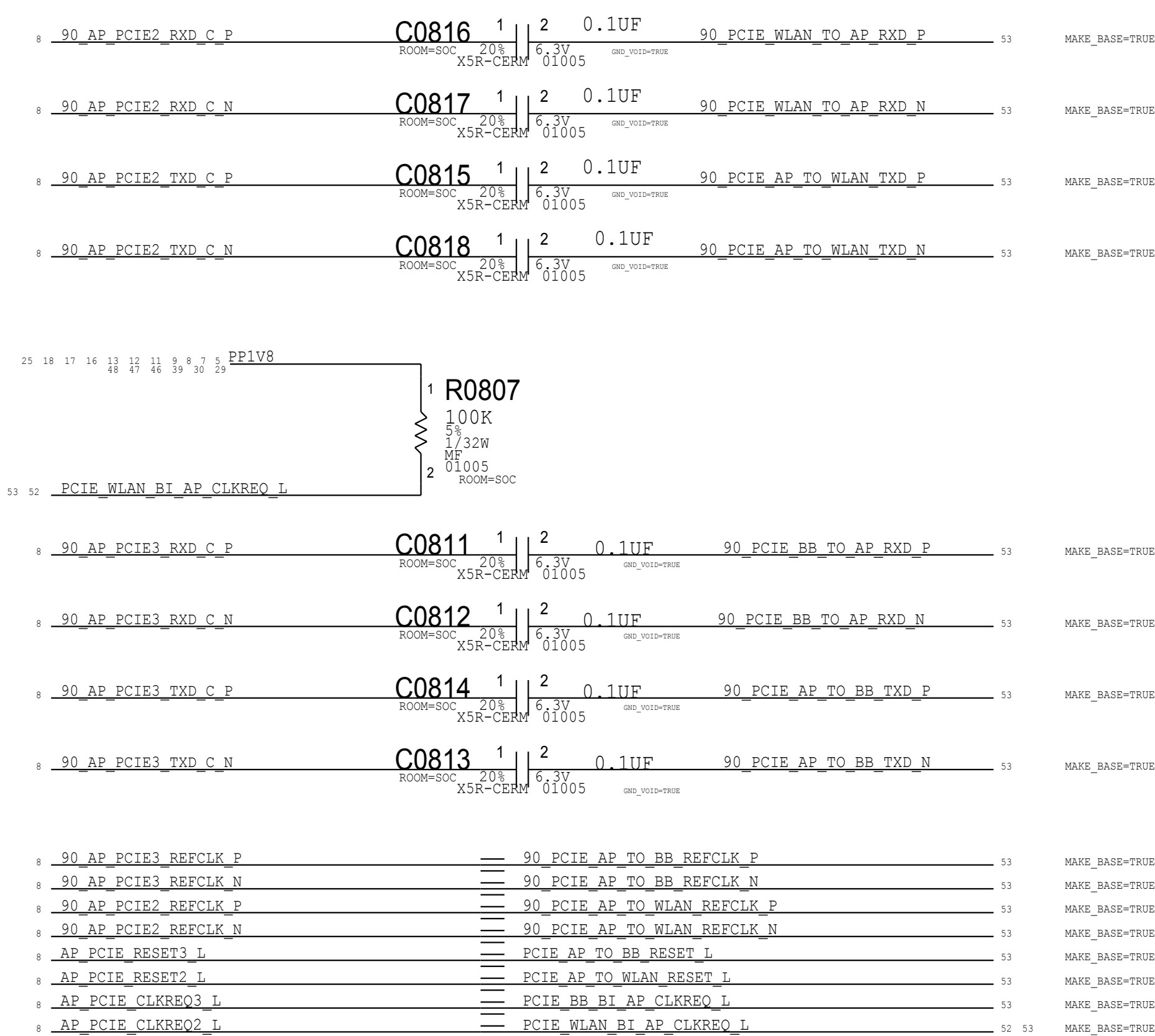
D1x I2C Table

Bus	Device	Binary	7-bit Address	8-bit Address	Max Speed
ISP I2C0 <i>1MHz</i>	SALT LAKE (PRIMARY)	0010000X	0x10	0x20	1 MHz
	SALT LAKE (SECONDARY)	0110000X	0x30	0x60	1 MHz
	GRUNBERG (STANDARD)	0011100X	0x1C	0x38	1 MHz
	GRUNBERG (SPECIAL ABSOLUTE R/W)	0011101X	0x1D	0x3A	1 MHz
	GRUNBERG (SPECIAL DELTA READ)	0011110X	0x1E	0x3C	1 MHz
	STROBE 1	1100011X	0x63	0xC6	1 MHz
ISP I2C1 <i>1MHz</i>	LAS VEGAS (PRIMARY)	0100000X	0x20	0x40	1 MHz
	LAS VEGAS (SECONDARY)	0110000X	0x30	0x60	1 MHz
	EDWIN	1100011X	0x63	0xC6	400 kHz
	STROBE 2	0000110X	0x06	0x0C	1 MHz
ISP I2C2 <i>400kHz</i>	CONCORD	0010000X	0x10	0x20	1 MHz
AOP I2C <i>1MHz</i>	ARC DRIVER 1	100000YX	0x41	0x82	1 MHz
	PROX	1011000X	0x58	0xB0	1 MHz
	SPKAMP1 (L26)	100000YX	0x40	0x80	1 MHz
	TURTLE	0101100X	0x2c	0x58	400 kHz
	MESA	1100001X	0x61	0xC2	400 kHz
Touch I2C <i>600kHz</i>	MAMBA	1100000X	0x60	0xC0	1 MHz
	MESON	1000000X	0x40	0x80	510 kHz
EEPROM I2C	EEPROM	1010001X	0x51	0xA2	400 kHz
AP I2C0 <i>400kHz</i>	BOOST	1110101X	0x75	0xEA	3.4 MHz
	TRISTAR	0011010X	0x1A	0x34	400 kHz
	BACKLIGHT 1	1100010X	0x62	0xC4	1 MHz
	CHESTNUT	0100111X	0x27	0x4E	400 kHz
AP I2C1 <i>400kHz</i>	ADELYN	1110100X	0x74	0xE8	400 kHz
	TIGRIS	1110101X	0x75	0xEA	400 kHz
	BACKLIGHT 2	1100010X	0x62	0xC4	1 MHz
AP I2C2 <i>400kHz</i>	ALS	0101001X	0x29	0x52	400 kHz
	CONVOY	0100001X	0x21	0x42	3.4 MHz
		0100010X	0x22	0x44	400 kHz
	SPKAMP 2 (L26)	10000ZYX	0x40	0x80	400 MHz
AP I2C3 <i>100kHz</i>	DOCK FLEX (MIC1)	1010100X	0x54	0xA8	1 MHz
	COMBINED BUTTON FLEX (MIC2)	1010100X	0x56	0xAC	1 MHz
	DISP EEPROM	1010001X	0x51	0xA2	400kHz
Homer I2C <i>1MHz</i>	SCHRODINGER (STANDARD)	0001110X	0xE	0x1C	1 MHz
	SCHRODINGER (CAL REG'S)	0001111X	0xF	0x1D	1 MHz
	EEPROM (STANDARD)	1010000X	0x50	0xA0	
	EEPROM (CONFIG)	1010000X	0x50	0xA0	

8								7								6								5								4								3								2								1							

This page controls elements different accross all MLBs

PCIE Lanes



Wifi/BT

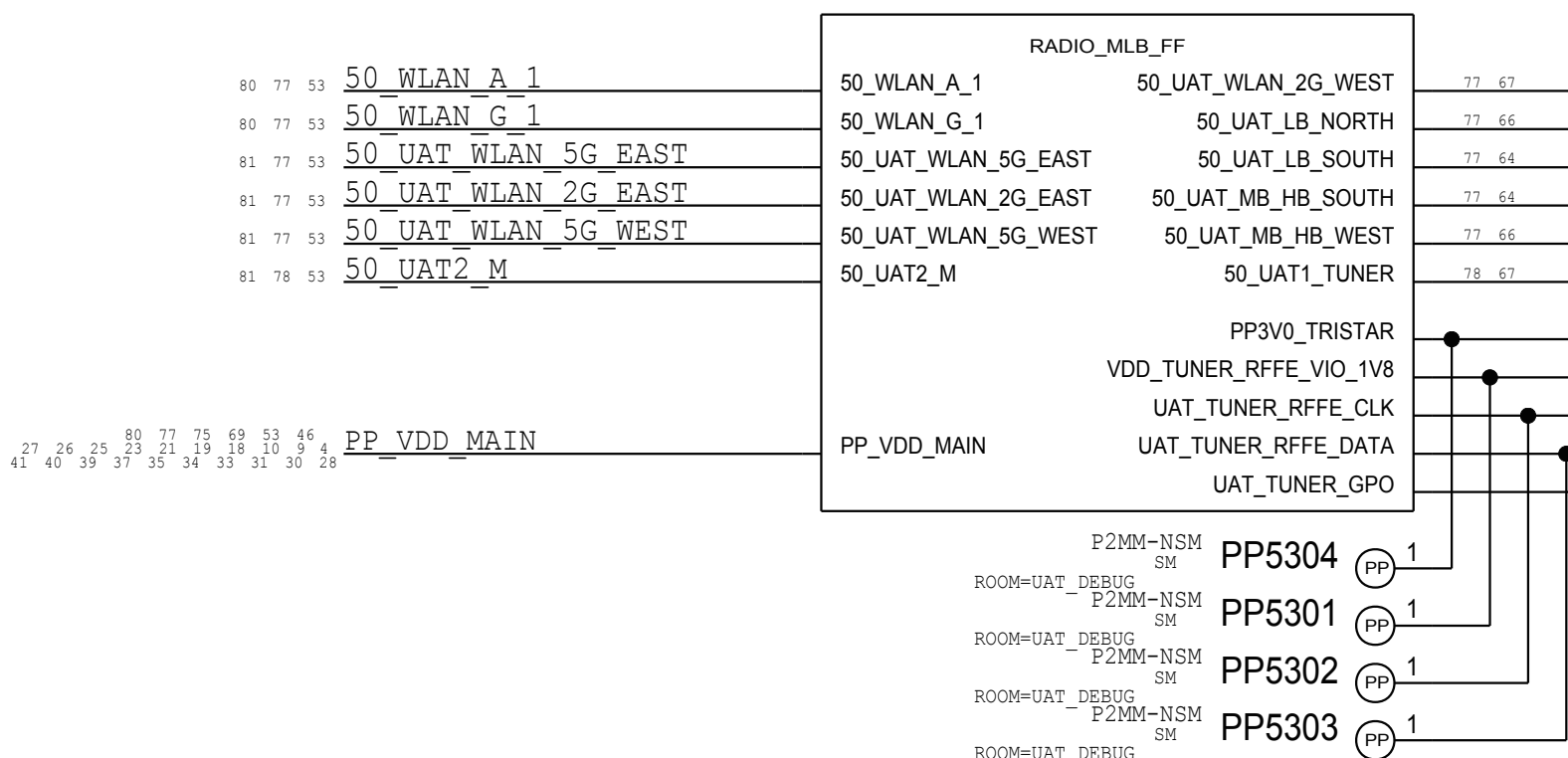
PP_VDD_MAIN	PP_VDD_MAIN
PP1V8_SDRAM	PP1V8_SDRAM
PMUGPIO_TO_WLAN_CLK32K	PMU_TO_WLAN_32K
PMU_TO_WLAN_REG_ON	PMU_TO_WLAN_REG_ON
PMU_TO_BT_REG_ON	PMU_TO_BT_REG_ON
BT_TO_PMU_HOST_WAKE	BT_TO_PMU_HOST_WAKE
AP_TO_BT_WAKE	AP_TO_BT_WAKE
90_PCIE_AP_TO_WLAN_REFCLK_P	100_PCIE_AP_TO_WLAN_REFCLK_P
90_PCIE_AP_TO_WLAN_REFCLK_N	100_PCIE_AP_TO_WLAN_REFCLK_N
90_PCIE_AP_TO_WLAN_TXD_P	100_PCIE_AP_TO_WLAN_TX_P
90_PCIE_AP_TO_WLAN_TXD_N	100_PCIE_AP_TO_WLAN_TX_N
90_PCIE_WLAN_TO_AP_RXD_P	100_PCIE_WLAN_TO_AP_RX_P
90_PCIE_WLAN_TO_AP_RXD_N	100_PCIE_WLAN_TO_AP_RX_N
PCIE_AP_TO_WLAN_RESET_L	PCIE_AP_TO_WLAN_PERST_L
PCIE_WLAN_BI_AP_CLKREQ_L	PCIE_AP_BI_WLAN_CLKREQ_L
WLAN_TO_PMU_HOST_WAKE	PCIE_WLAN_TO_PMU_WAKE
AP_TO_WLAN_DEVICE_WAKE	AP_TO_WLAN_DEV_WAKE
UART_AP_TO_WLAN_TXD	UART_AP_TO_WLAN_TXD
UART_WLAN_TO_AP_RXD	UART_WLAN_TO_AP_RXD
UART_AP_TO_WLAN_RTS_L	UART_AP_TO_WLAN_RTS_L
UART_WLAN_TO_AP_CTS_L	UART_WLAN_TO_AP_CTS_L
UART_AP_TO_BT_TXD	UART_AP_TO_BT_TXD
UART_BT_TO_AP_RXD	UART_BT_TO_AP_RXD
UART_AP_TO_BT_RTS_L	UART_AP_TO_BT_RTS_L
UART_BT_TO_AP_CTS_L	UART_BT_TO_AP_CTS_L
AOP_TO_WLAN_CONTEXT_A	AOP_TO_WLAN_CONTEXT_A
AOP_TO_WLAN_CONTEXT_B	AOP_TO_WLAN_CONTEXT_B
I2S_AP_TO_BT_BCLK	I2S_AP_TO_BT_BCLK
I2S_AP_TO_BT_LRCLK	I2S_AP_TO_BT_LRCLK
I2S_BT_TO_AP_DIN	I2S_BT_TO_AP_DIN
I2S_AP_TO_BT_DOUT	I2S_AP_TO_BT_DOUT
UART_BB_TO_WLAN_COEX	UART_BB_TO_WLAN_COEX
UART_WLAN_TO_BB_COEX	UART_WLAN_TO_BB_COEX
50_UAT_WLAN_5G_WEST	50_UAT_WLAN_5G_WEST
50_UAT_WLAN_2G_EAST	50_UAT_WLAN_2G_EAST
50_UAT_WLAN_5G_EAST	50_UAT_WLAN_5G_EAST
50_WLAN_G_1	50_WLAN_G_1
50_WLAN_A_1	50_WLAN_A_1
50_UAT2_M	50_UAT2_M

WIFI_MLB

NFC

PP_VDD_MAIN	PP_VDD_MAIN
PP1V8_SDRAM	PP1V8_SDRAM
PMU_TO_NFC_EN	PMU_TO_NFC_EN
BB_TO_NFC_CLK	BB_TO_NFC_CLK
NFC_TO_BB_CLK_REQ	NFC_TO_BB_CLK_REQ
AP_TO_NFC_FW_DWLD_REQ	AP_TO_NFC_FW_DWLD_REQ
AP_TO_NFC_DEV_WAKE	AP_TO_NFC_DEV_WAKE
NFC_TO_PMU_HOST_WAKE	NFC_TO_PMU_HOST_WAKE
UART_AP_TO_NFC_TXD	UART_AP_TO_NFC_TXD
UART_NFC_TO_AP_RXD	UART_NFC_TO_AP_RXD
UART_AP_TO_NFC_RTS_L	UART_AP_TO_NFC_RTS_L
UART_NFC_TO_AP_CTS_L	UART_NFC_TO_AP_CTS_L
SIM1_SWP	NFC_SWP
NC_SE2_READY	SE2_READY
NC_SE2_PWR_REQ	SE2_PWR_REQ
NC_SE2_PRESENT	SE2_PRESENT
NC_NFC_SWP_MUX	NFC_SWP_MUX
NC_ICEFALL_LDO_ENABLE	ICEFALL_LDO_ENABLE

FF SPECIFIC



Cellular

Aliases used to correct name for Karoo radio system

PP_VDD_MAIN	PP_VDD_MAIN
PP1V8_SDRAM	PP1V8_SDRAM
BBPMU_TO_PMU_AMUX1	BBPMU_TO_PMU_AMUX1
BBPMU_TO_PMU_AMUX2	BBPMU_TO_PMU_AMUX2
BBPMU_TO_PMU_AMUX3	BBPMU_TO_PMU_AMUX3
AP_TO_BBPMU_SDWN_L	AP_TO_BBPMU_SDWN_L
PMU_TO_BBPMU_ON	PMU_TO_BBPMU_ON
AP_TO_BB_RESET_L	AP_TO_BB_RESET_L
BB_TO_AP_RESET_ACT_L	BB_TO_AP_RESET_ACT_L
BB_TO_AP_RESET_DETECT_L	BB_TO_AP_RESET_DETECT_L
BB_TO_STROBE_DRIVER_GSM_BURST_IND	BB_TO_STROBE_DRIVER_GSM_BURST_IND
AP_TO_BB_MESA_ON	AP_TO_BB_MESA_ON
AP_TO_BB_TIME_MARK	AP_TO_BB_TIME_MARK
AP_TO_BB_COREDUMP	AP_TO_BB_COREDUMP
AP_TO_BB_IPC_GPIO1	AP_TO_BB_IPC_GPIO1
AP_TO_BB_IPC_GPIO2	AP_TO_BB_IPC_GPIO2
90_PCIE_AP_TO_BB_REFCLK_P	100_PCIE_AP_TO_BB_REFCLK_P
90_PCIE_AP_TO_BB_REFCLK_N	100_PCIE_AP_TO_BB_REFCLK_N
90_PCIE_AP_TO_BB_TXD_P	100_PCIE_AP_TO_BB_TX_P
90_PCIE_AP_TO_BB_TXD_N	100_PCIE_AP_TO_BB_TX_N
90_PCIE_BB_TO_AP_RXD_P	100_PCIE_BB_TO_AP_RX_P
90_PCIE_BB_TO_AP_RXD_N	100_PCIE_BB_TO_AP_RX_N
PCIE_AP_TO_BB_RESET_L	PCIE_AP_TO_BB_PERST_L
PCIE_BB_BI_AP_CLKREQ_L	PCIE_AP_BI_BB_CLKREQ_L
BB_TO_PMU_PCIE_HOST_WAKE_L	PCIE_BB_TO_PMU_WAKE_L
AP_TO_BB_DEVICE_WAKE	AP_TO_BB_DEVICE_WAKE
UART_AP_TO_BB_TXD	UART_AP_TO_BB_TXD
UART_BB_TO_AP_RXD	UART_BB_TO_AP_RXD
UART_AOP_TO_BB_TXD	UART_AOP_TO_BB_TXD
UART_BB_TO_AOP_RXD	UART_BB_TO_AOP_RXD
I2S_BB_TO_AP_BCLK	I2S_AP_TO_BB_BCLK
I2S_BB_TO_AP_LRCLK	I2S_AP_TO_BB_LRCLK
I2S_AP_TO_BB_DOUT	I2S_AP_TO_BB_DOUT
I2S_BB_TO_AP_DIN	I2S_BB_TO_AP_DIN
UART_BB_TO_WLAN_COEX	UART_BB_TO_WLAN_COEX
UART_WLAN_TO_BB_COEX	UART_WLAN_TO_BB_COEX
SIM1_SWP	SIM1_SWP
NFC_TO_BB_CLK_REQ	SIM2_SWP
BB_TO_NFC_CLK	NFC_TO_BB_CLKREQ
BB_TO_NFC_CLK	BB_TO_NFC_CLK
NC_PMU_TO_GNSS_EN	PMU_TO_GNSS_EN
NC_AP_UART2_TXD	UART_AP_TO_GNSS_TXD
NC_AP_UART2_RXD	UART_GNSS_TO_AP_RXD
NC_AP_UART2_RTS_L	UART_AP_TO_GNSS_RTS_L
NC_AP_UART2_CTS_L	UART_GNSS_TO_AP_CTS_L
NC_GNSS_TO_PMU_HOST_WAKE	GNSS_TO_PMU_HOST_WAKE
NC_AP_TO_GNSS_TIME_MARK	AP_TO_GNSS_TIME_MARK
NC_AP_TO_GNSS_WAKE	AP_TO_GNSS_DEVICE_WAKE
SWD_AP_TO_MANY_SWCLK	SWD_AP_TO_BB_CLK
SWD_AOP_BI_BB_SWCLK	SWD_AP_BI_BB_IO
PMU_TO_BB_USB_VBUS_DETECT	USB_BB_VBUS
90_USB_BB_DATA_P	90_USB_BB_P
90_USB_BB_DATA_N	90_USB_BB_N
BB_TO_LAT_ANT_SCLK	LAT_TUNER_RFFE_CLK
BB_TO_LAT_ANT_DATA	LAT_TUNER_RFFE_DATA
BB_TO_UAT_TUNER_GPO	UAT_TUNER_GPO
BB_TO_UAT_ANT_SCLK	UAT_TUNER_RFFE_CLK
BB_TO_UAT_ANT_DATA	UAT_TUNER_RFFE_DATA
BB_TO_LAT_GPO1	LAT_TUNER_GPO1
BB_TO_LAT_GPO2	LAT_TUNER_GPO2
BB_TO_LAT_GPO3	LAT_TUNER_GPO3

RADIO_MLB

BASEBAND

D

C

B

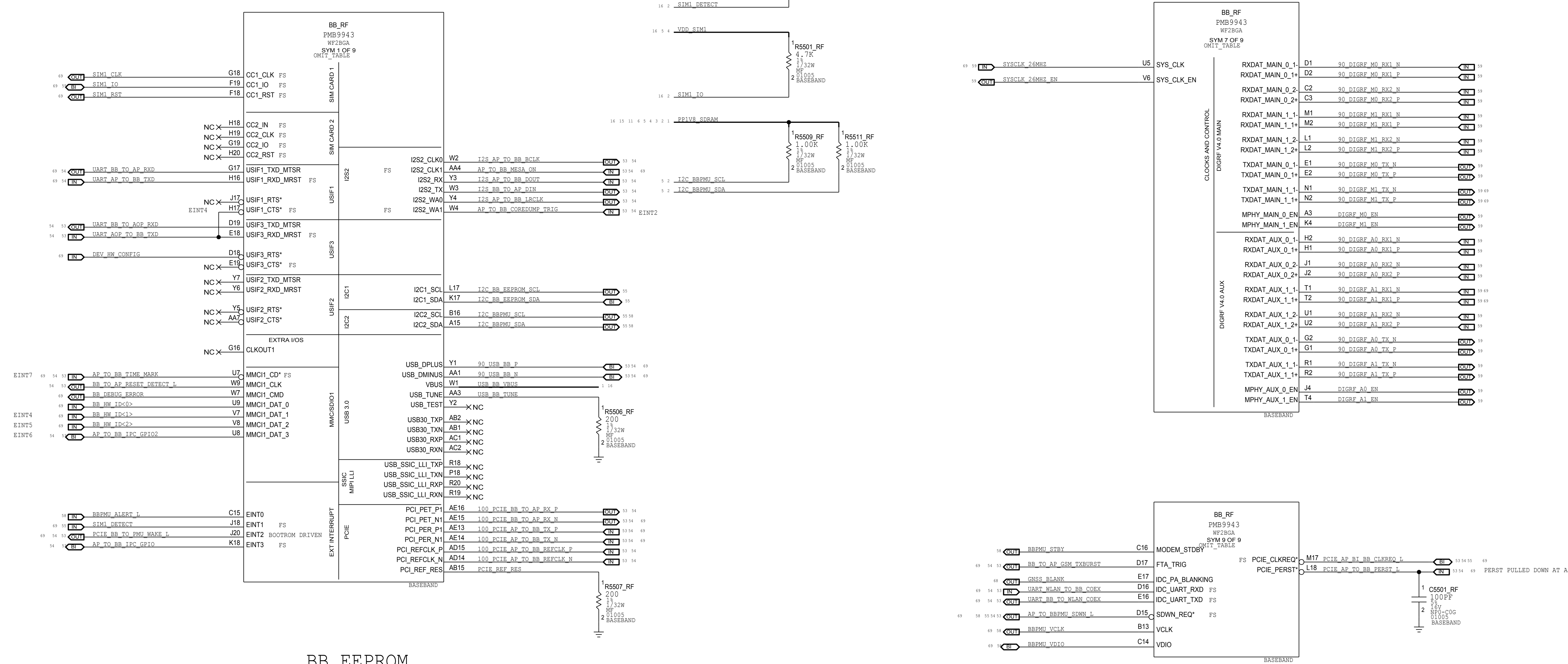
A

D

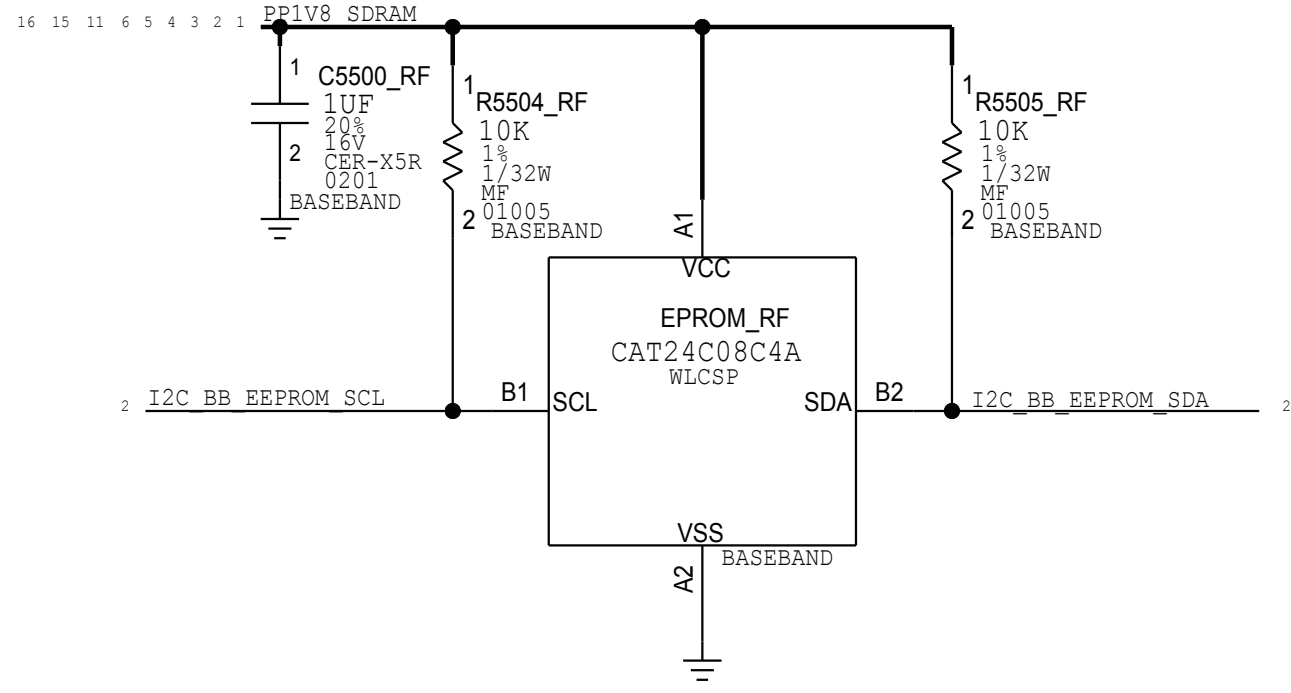
C

B

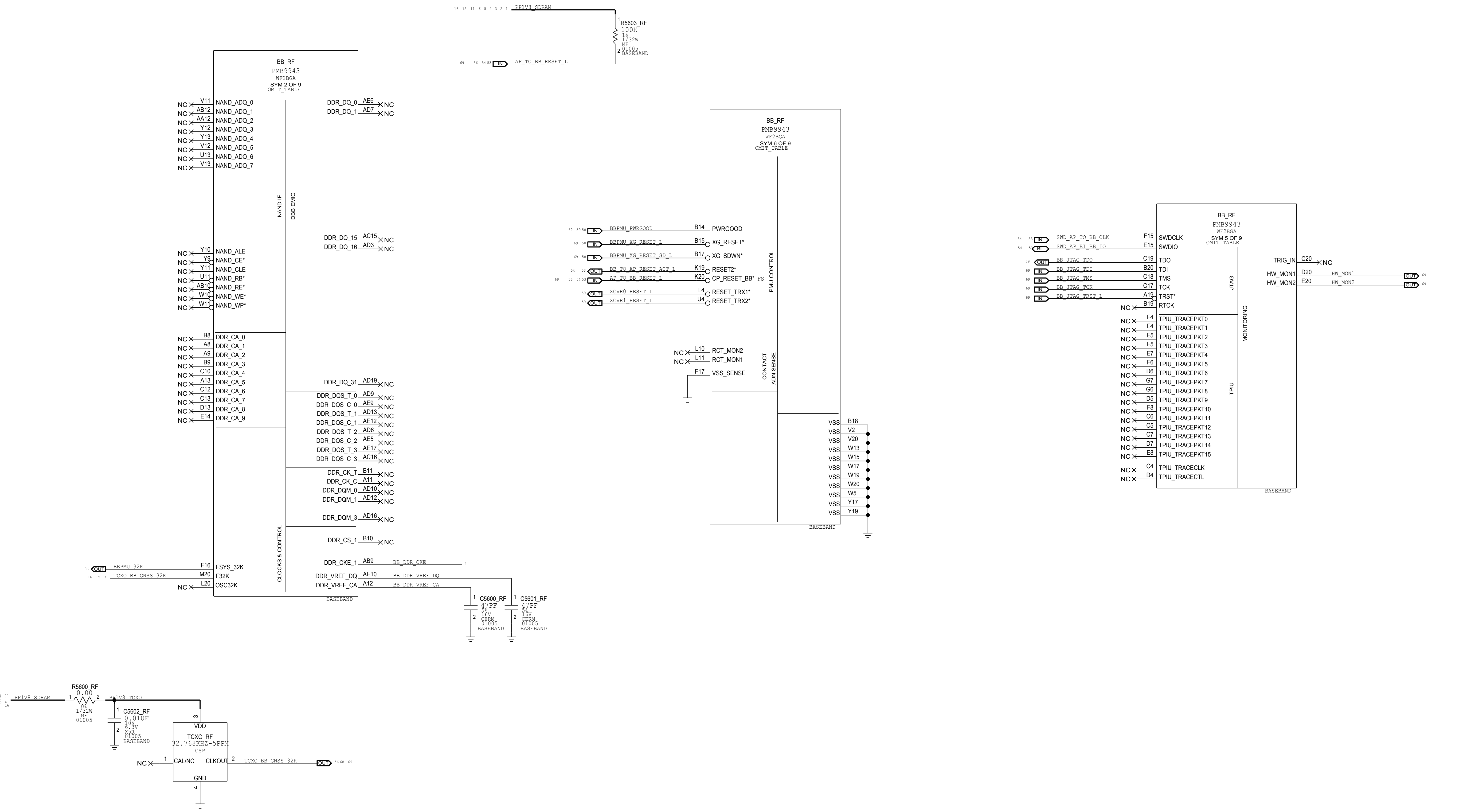
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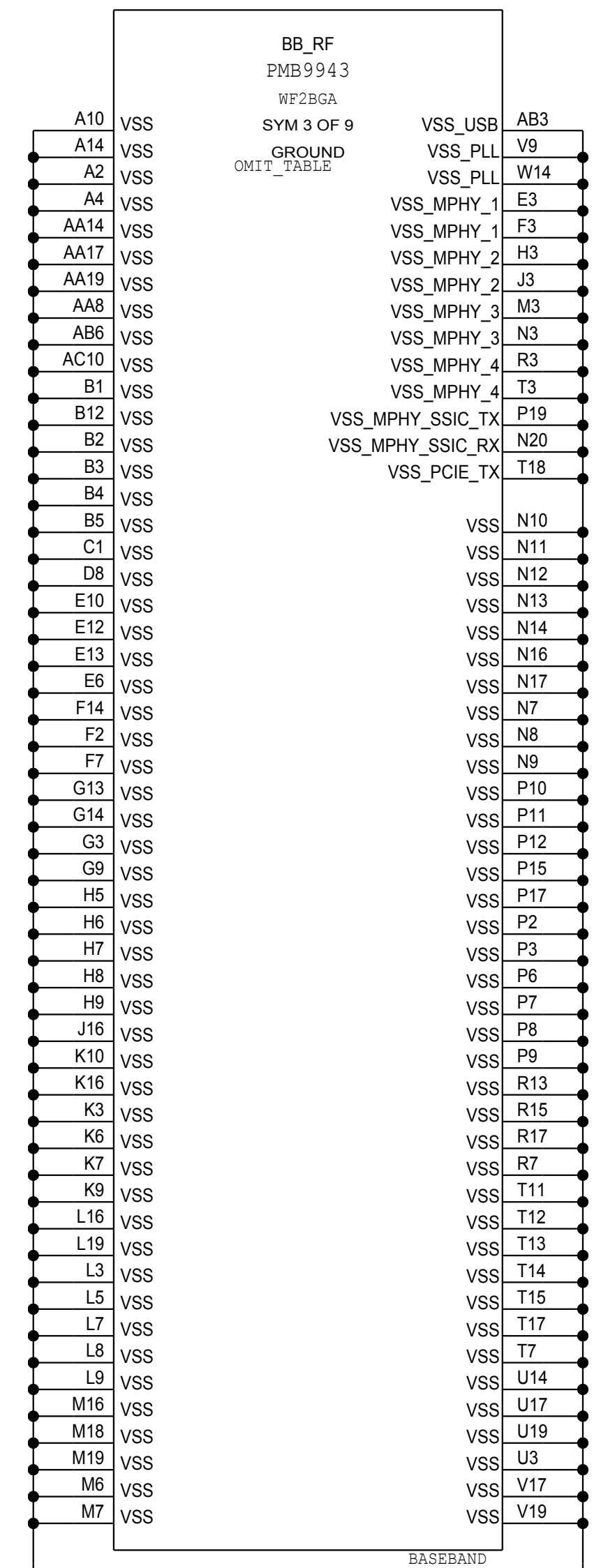
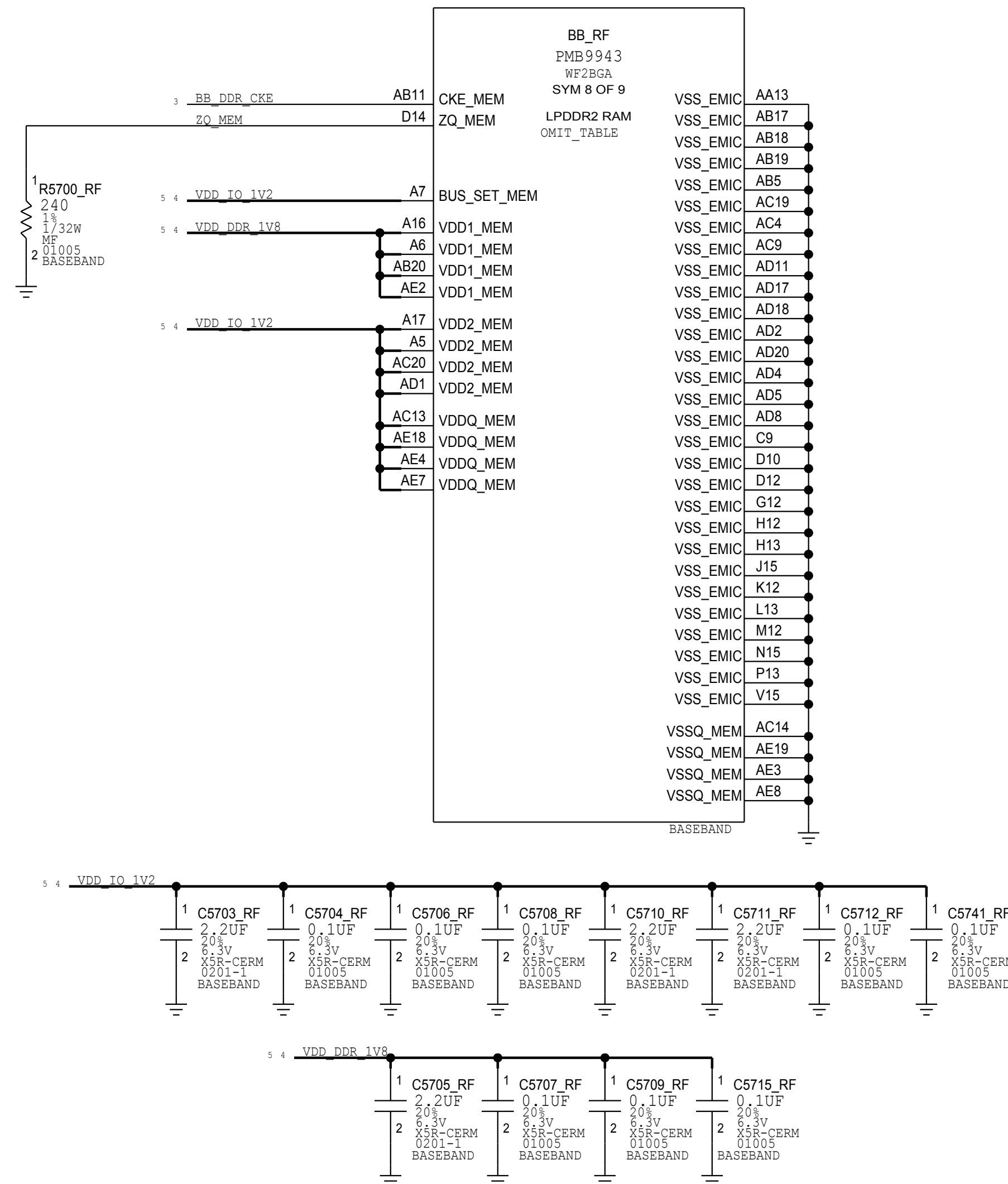
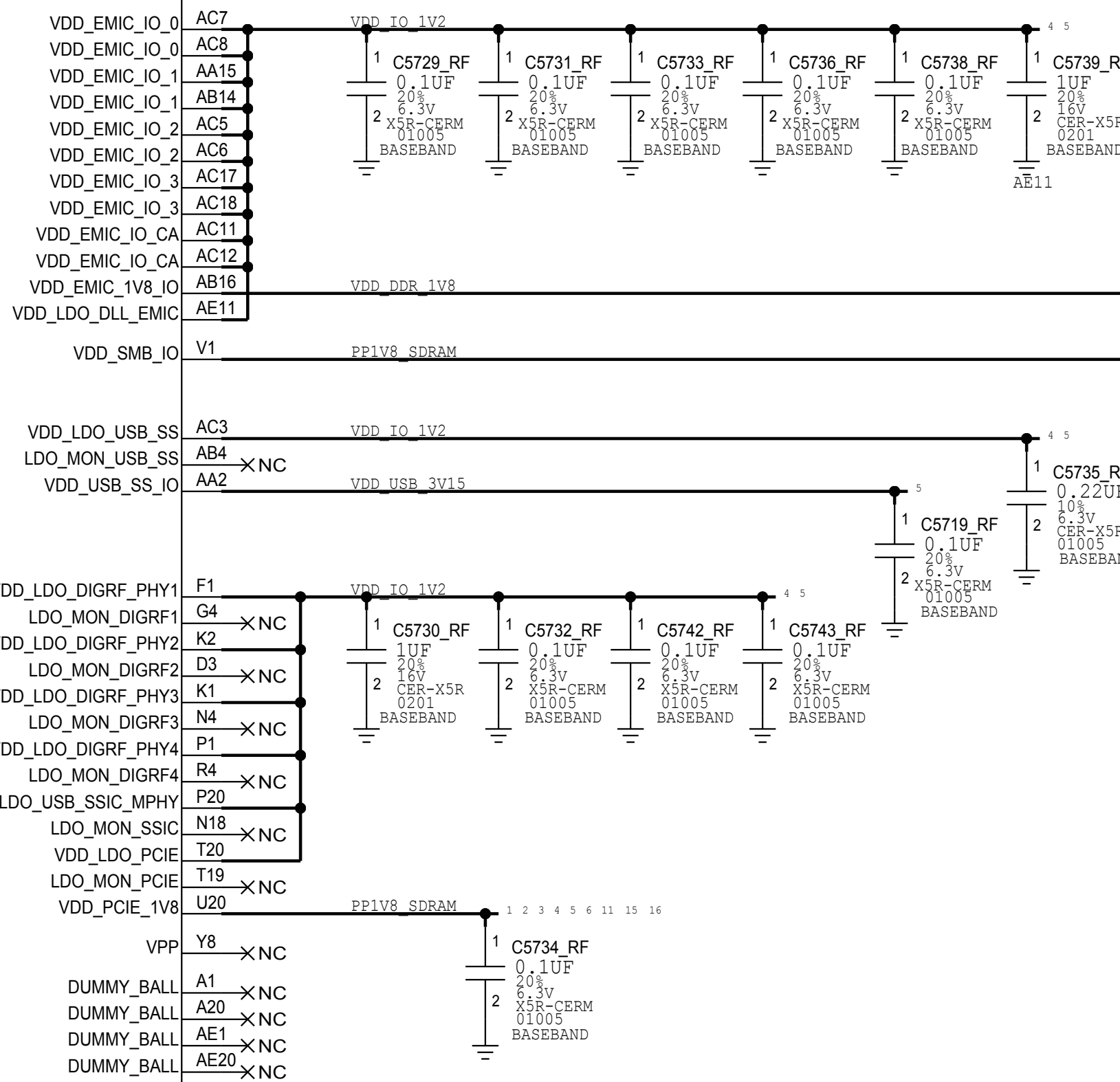
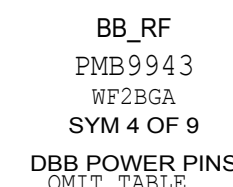
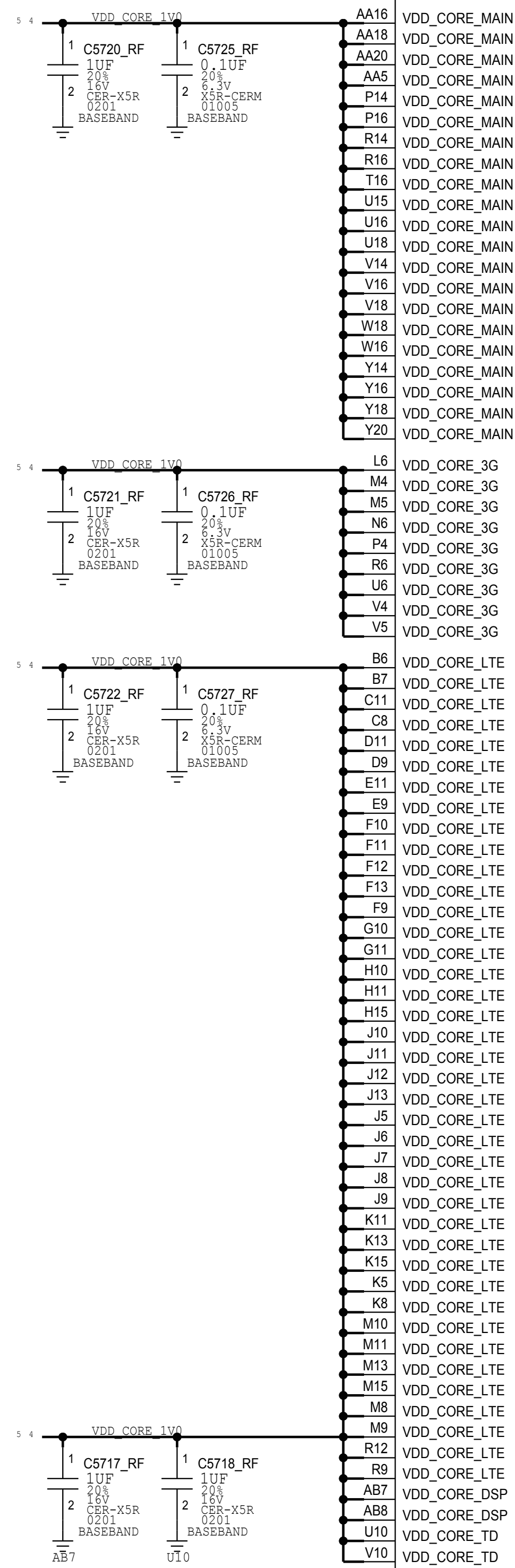
BB EEPROM



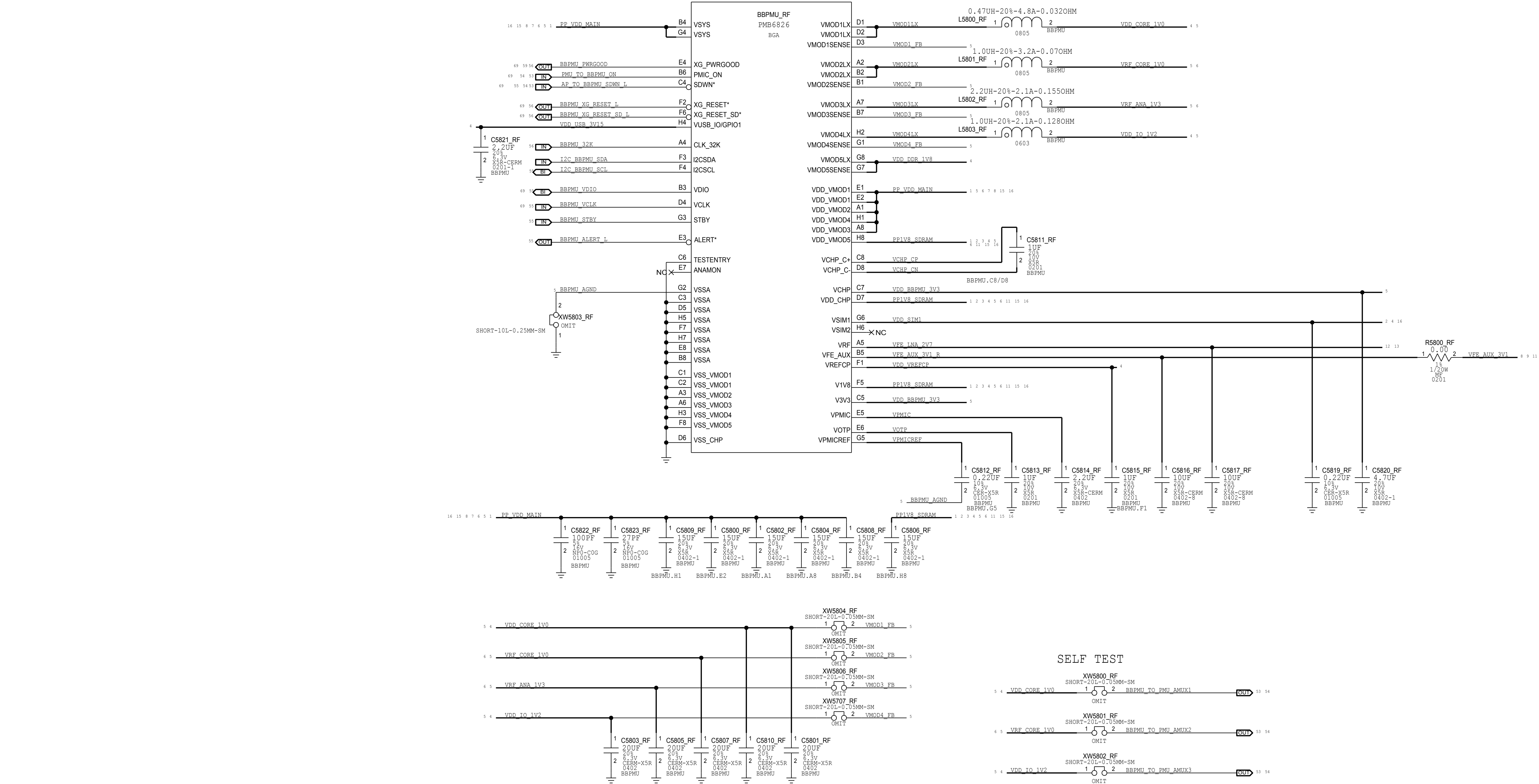
BASEBAND MEMORY/DEBUG



8	7
BASEBAND	POWER

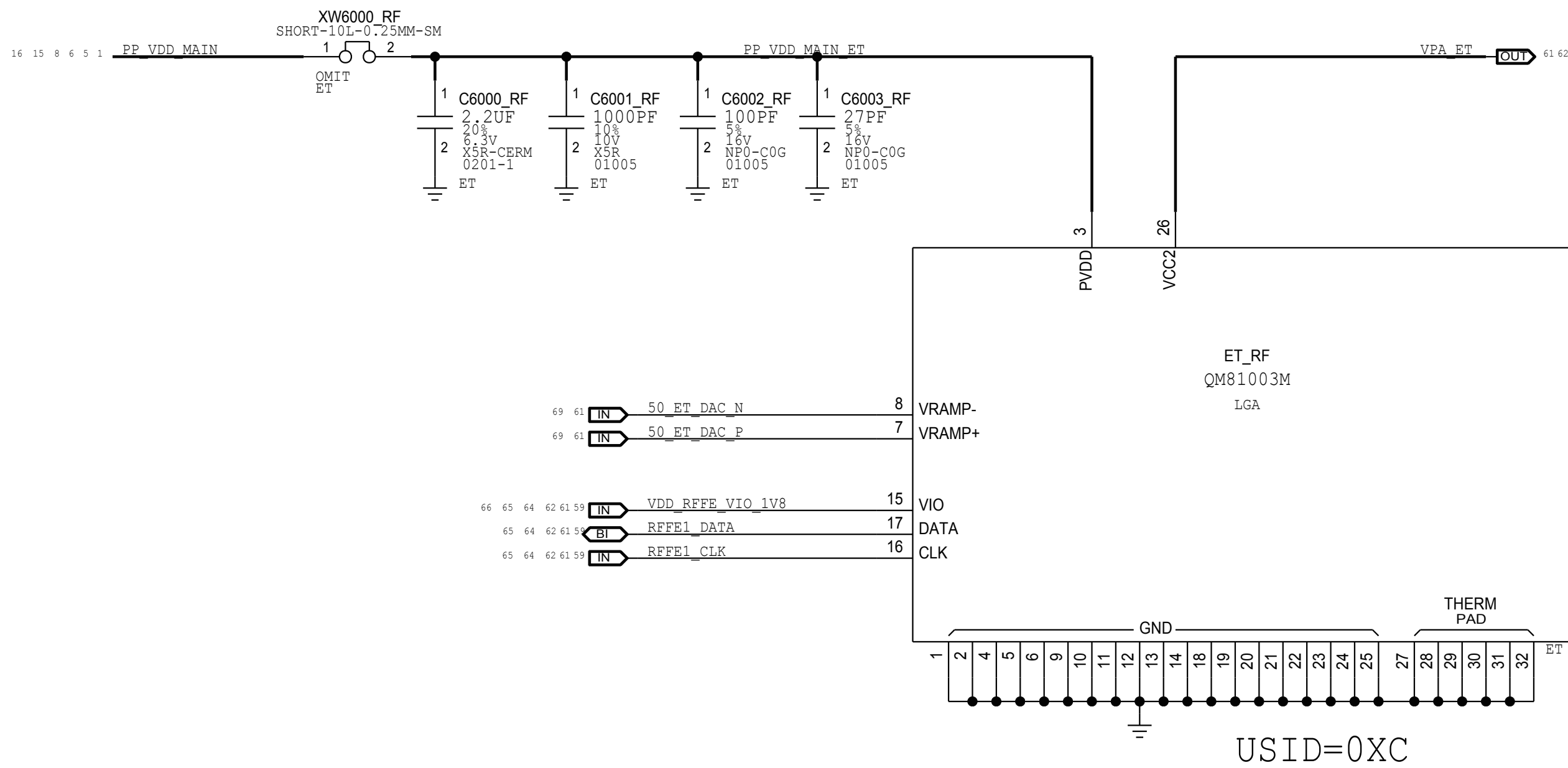


BASEBAND PMIC



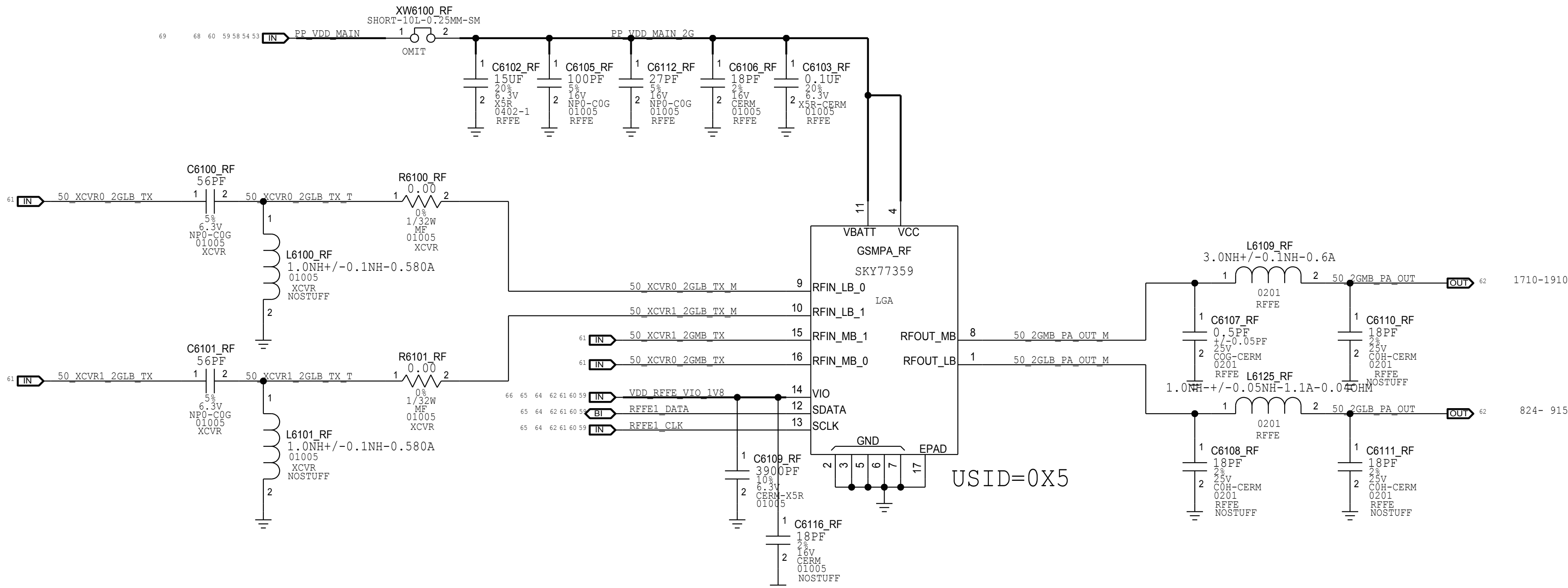
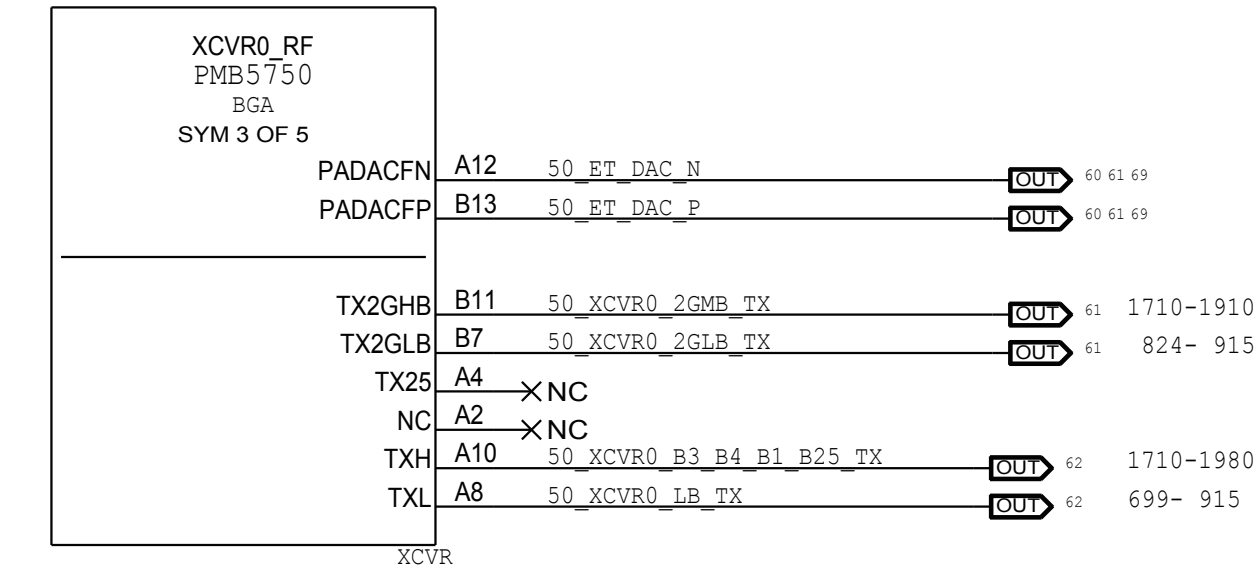
ET MODULATOR

ALPES II ES2

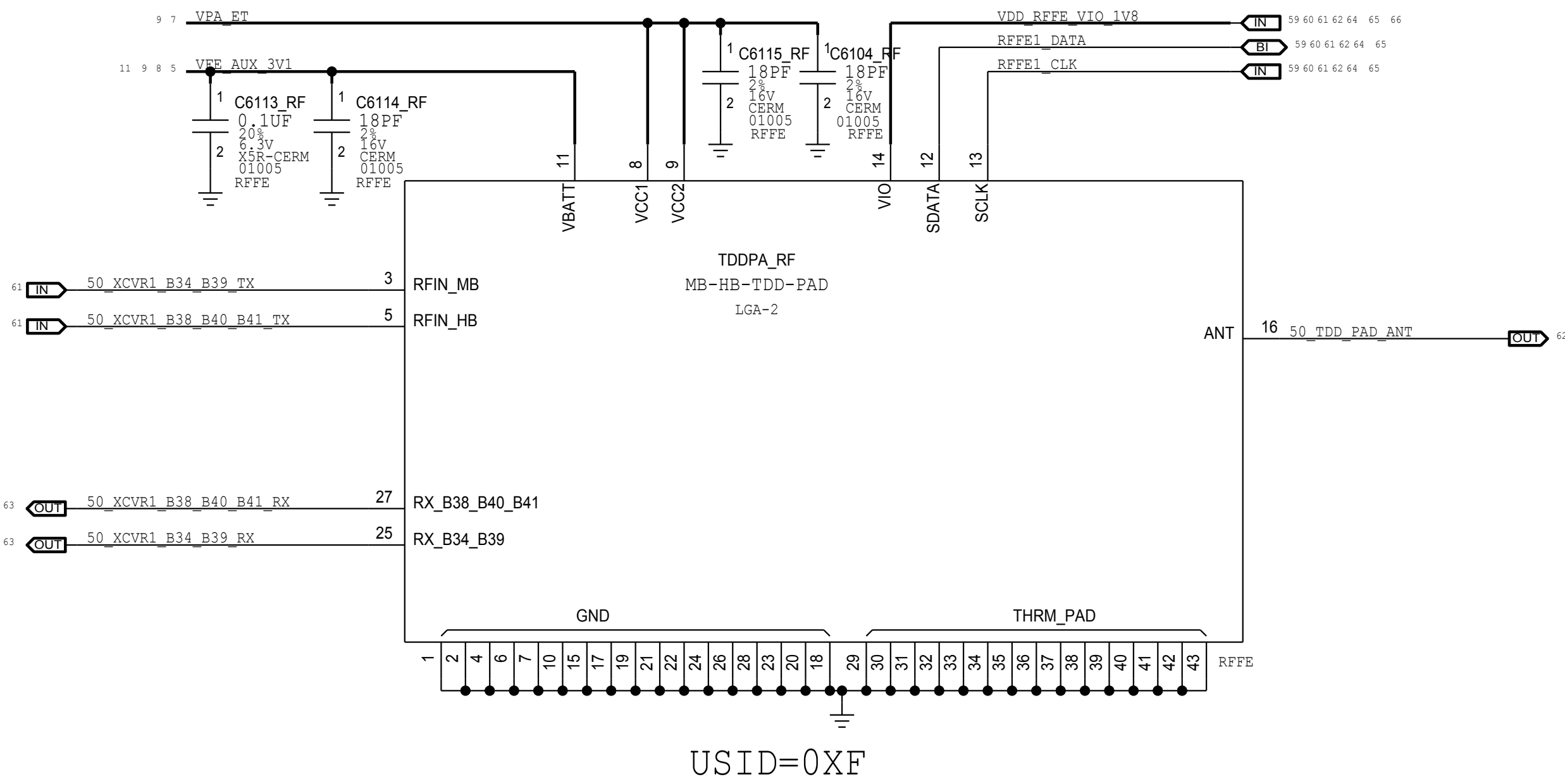
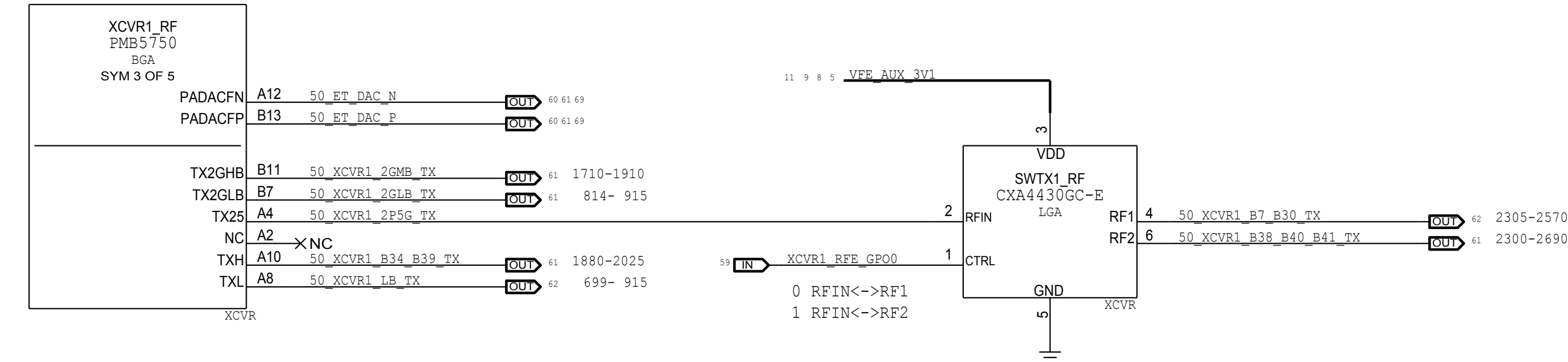


TDD TRANSMIT

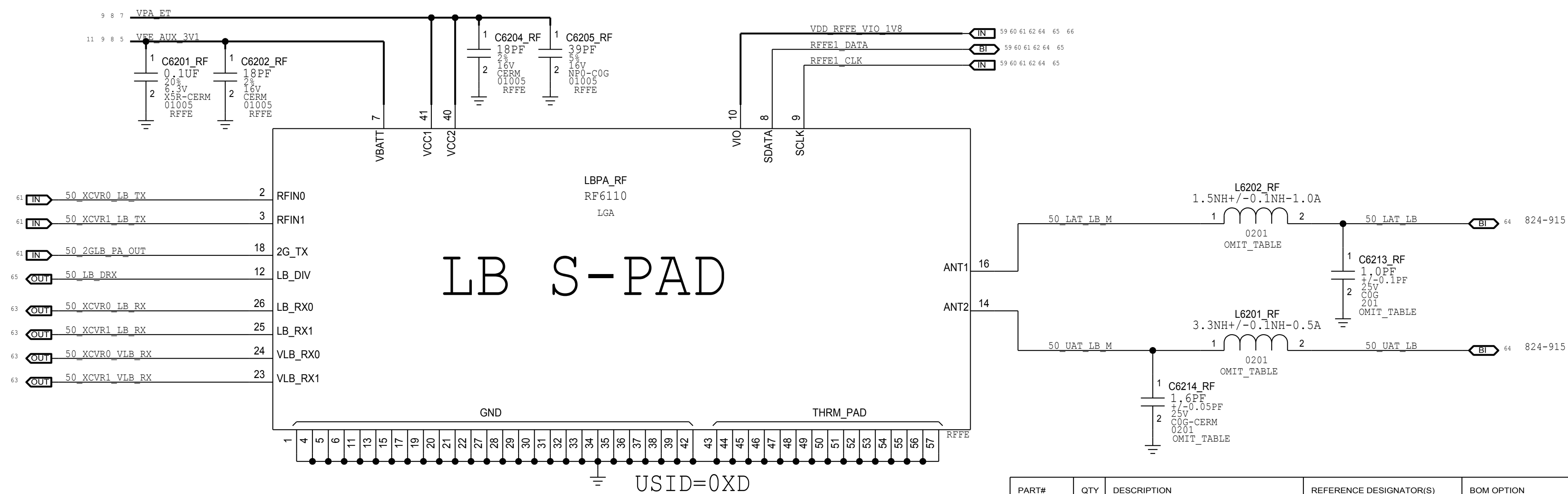
2G PA



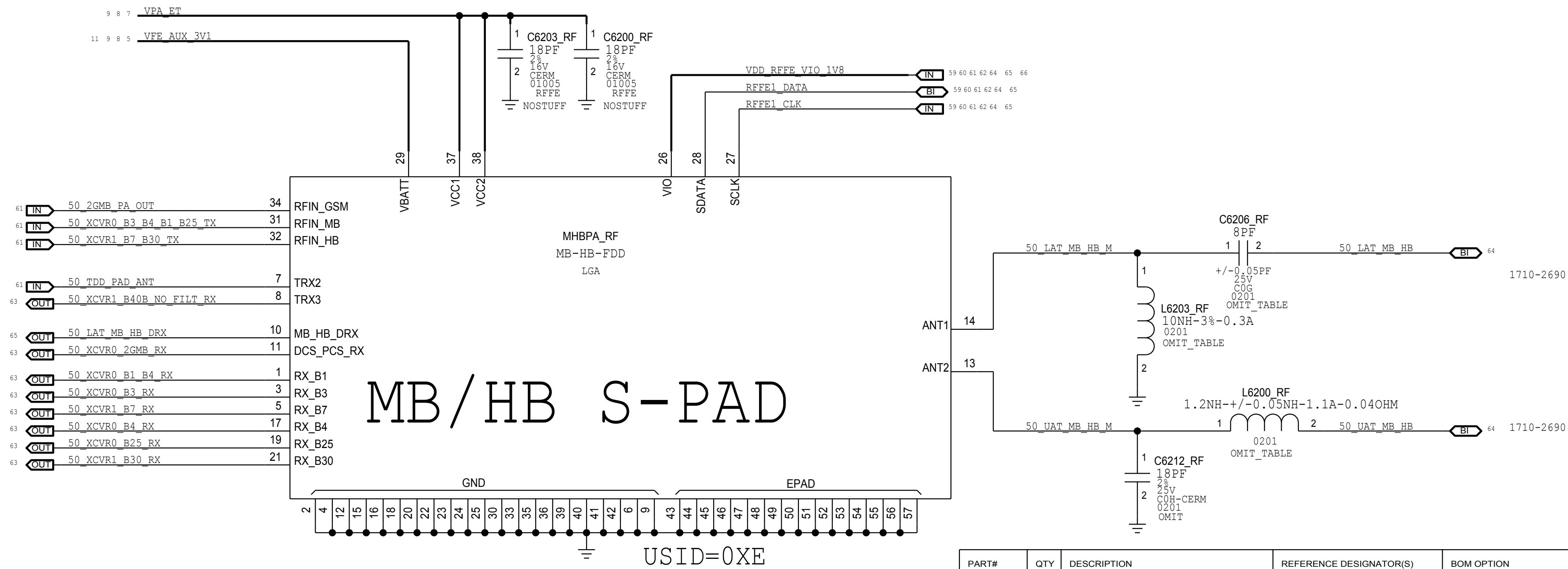
MB HB TDD S-PAD



FDD TRANSMIT



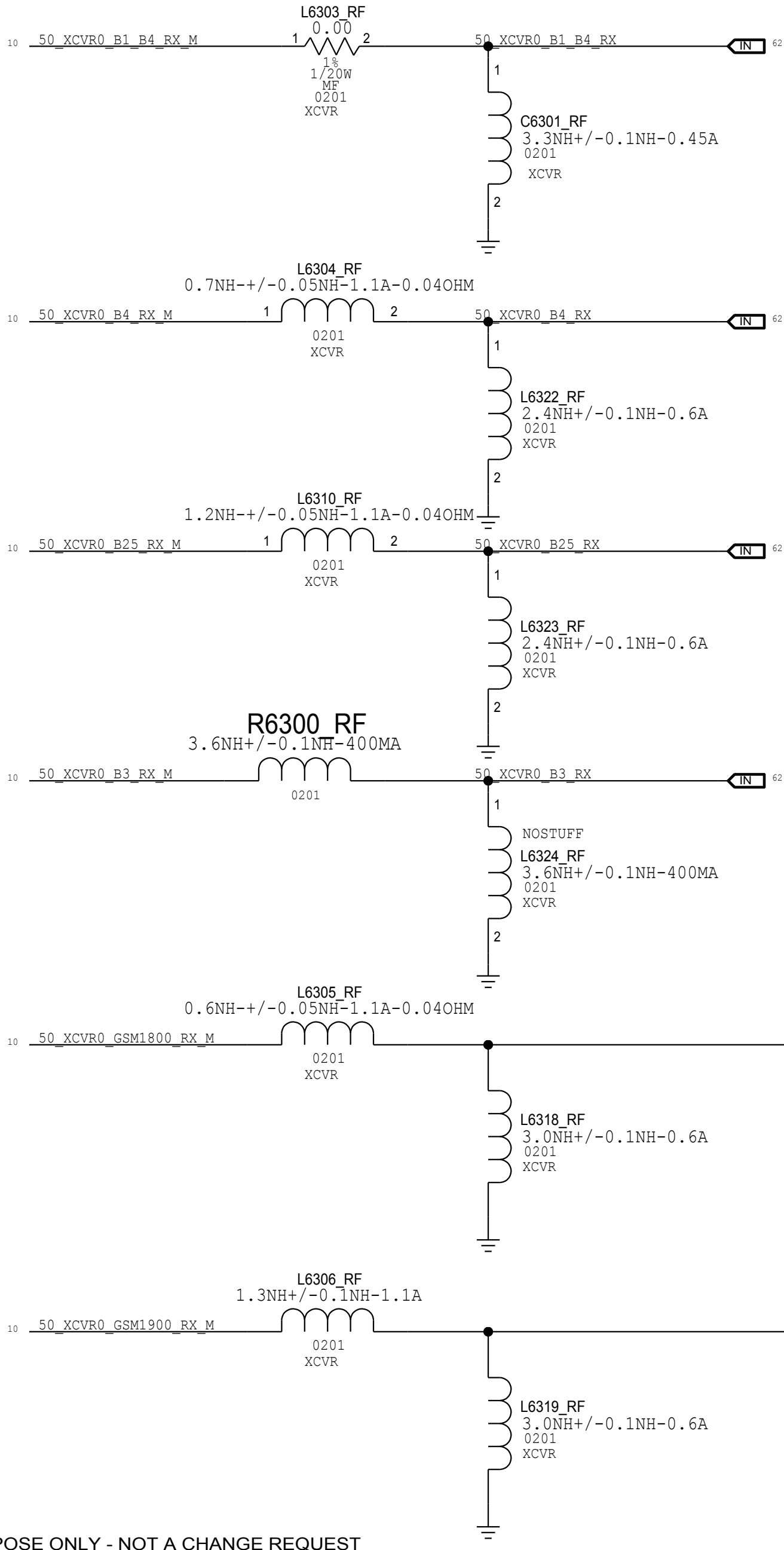
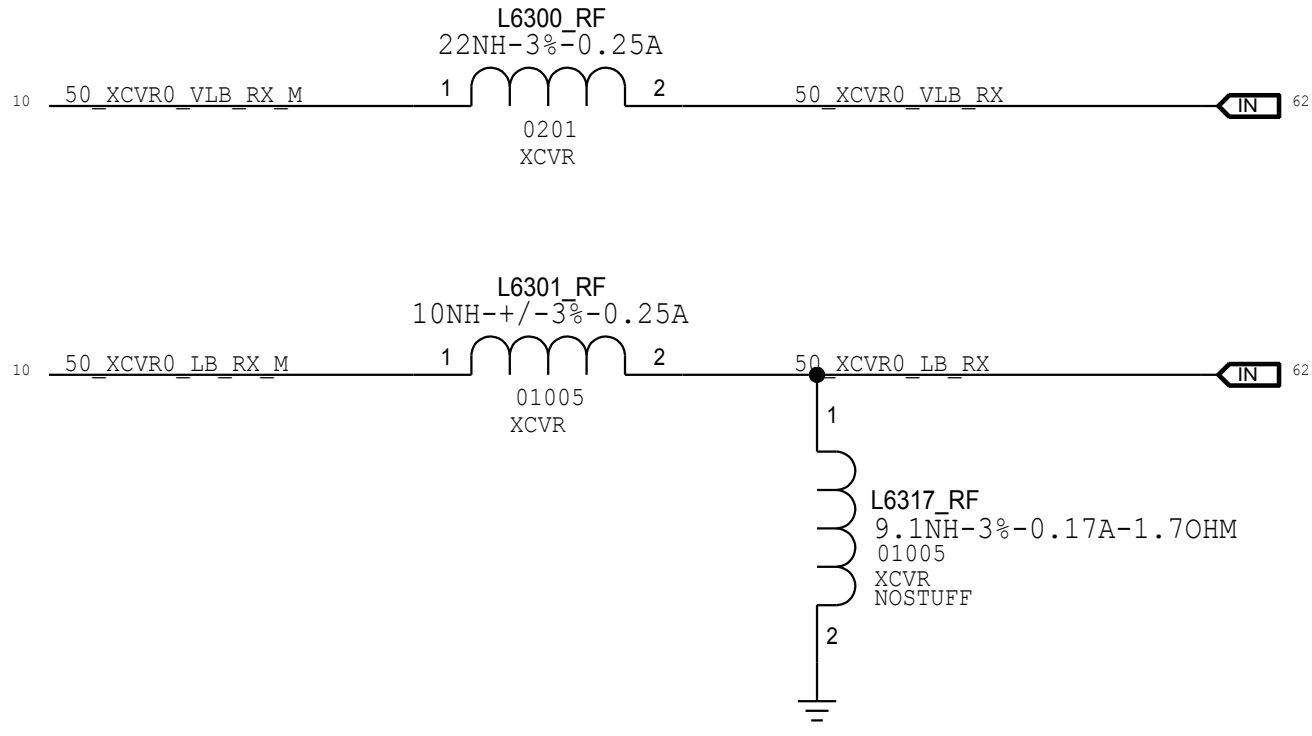
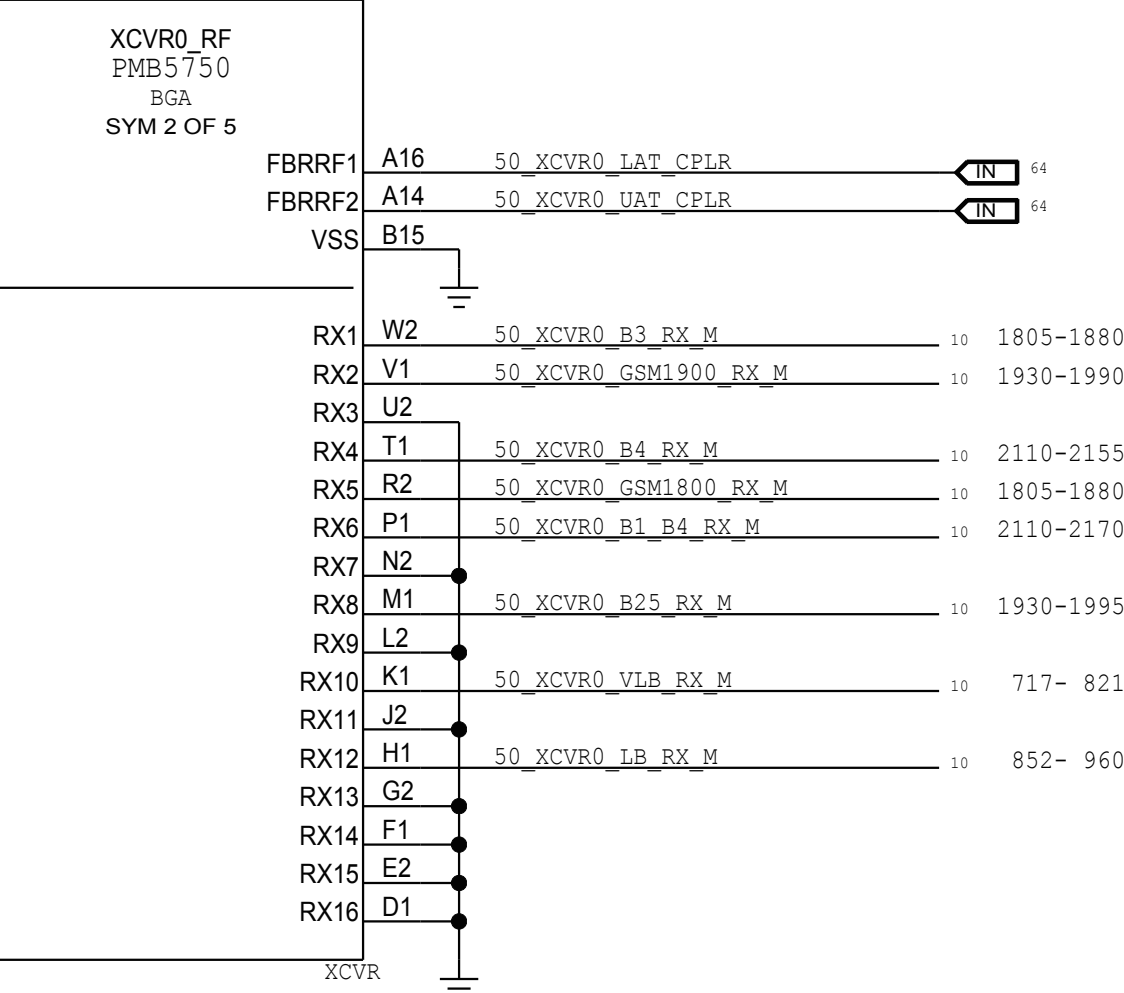
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
15282021	1	IND, 1.5NH, 0201, UH-Q	L6202_RF	D101
131S0278	1	CAP, 1.0PF, 0201, HQ	C6213_RF	D101
15282004	1	IND, 3.3NH, 0201, UH-Q	L6201_RF	D101
131S0338	1	CAP, 1.6PF, 0201, HF	C6214_RF	D101
152800151	1	IND, 0.8NH, 0201, UH-Q	L6202_RF	D111
131S0312	1	CAP, 1.2PF, 0201, HQ	C6213_RF	D111
15282002	1	IND, 2.7NH, 0201, UH-Q	L6201_RF	D111
131S0338	1	CAP, 1.6PF, 0201, HF	C6214_RF	D111



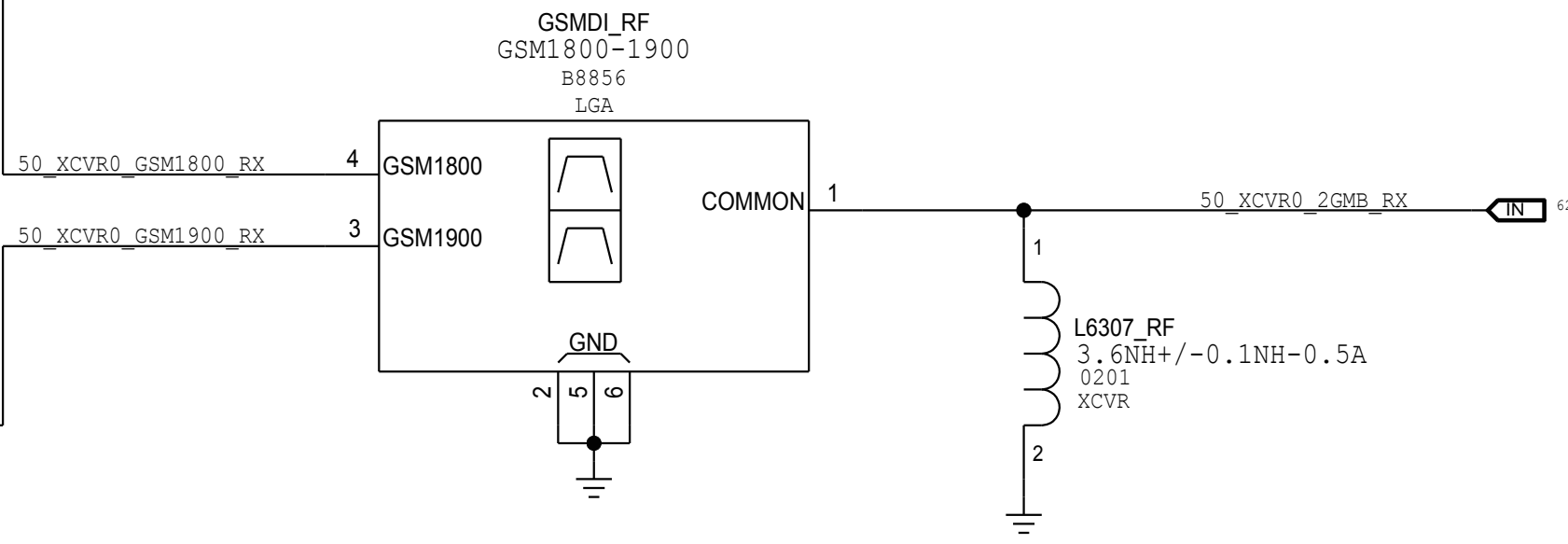
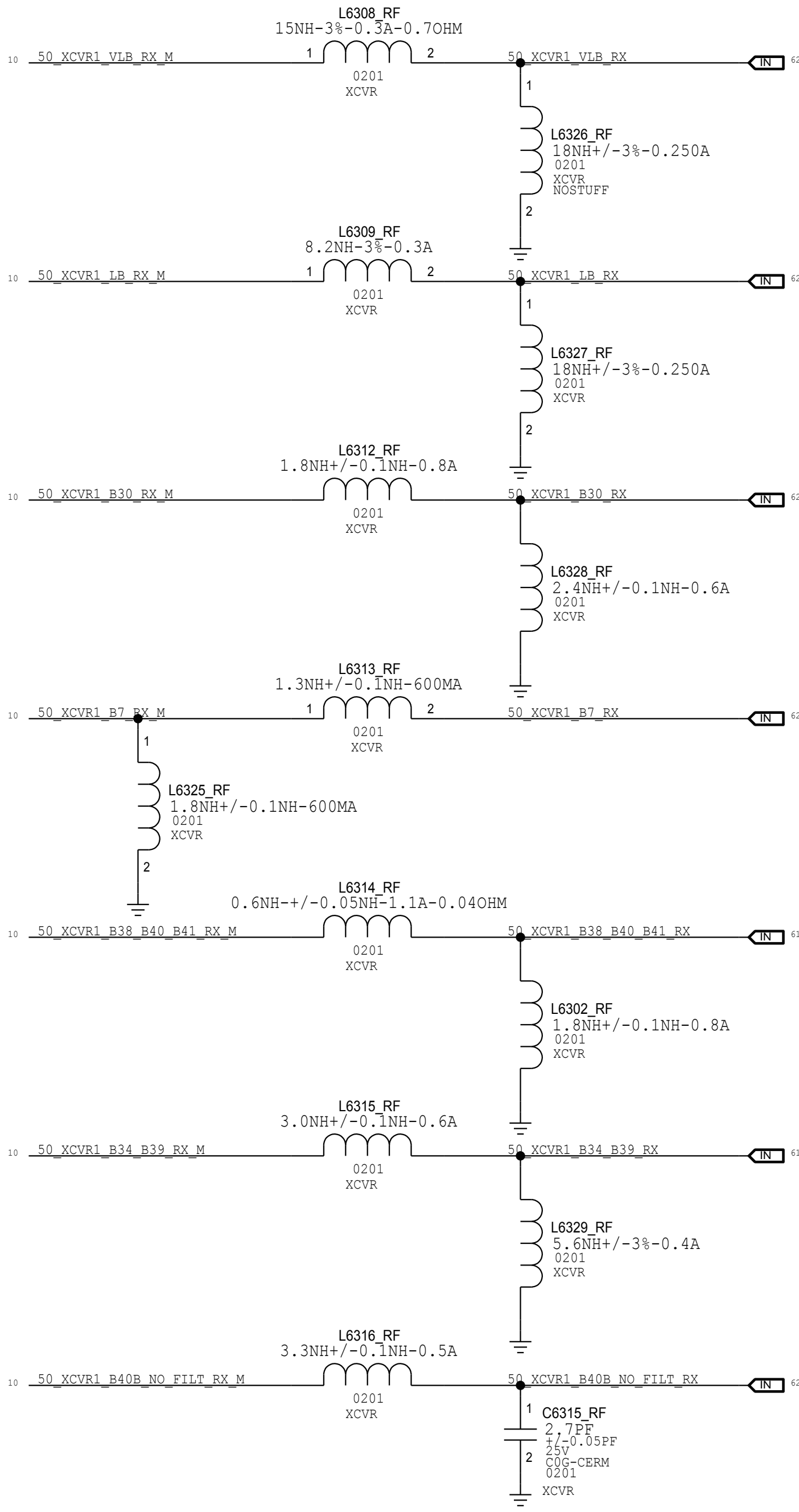
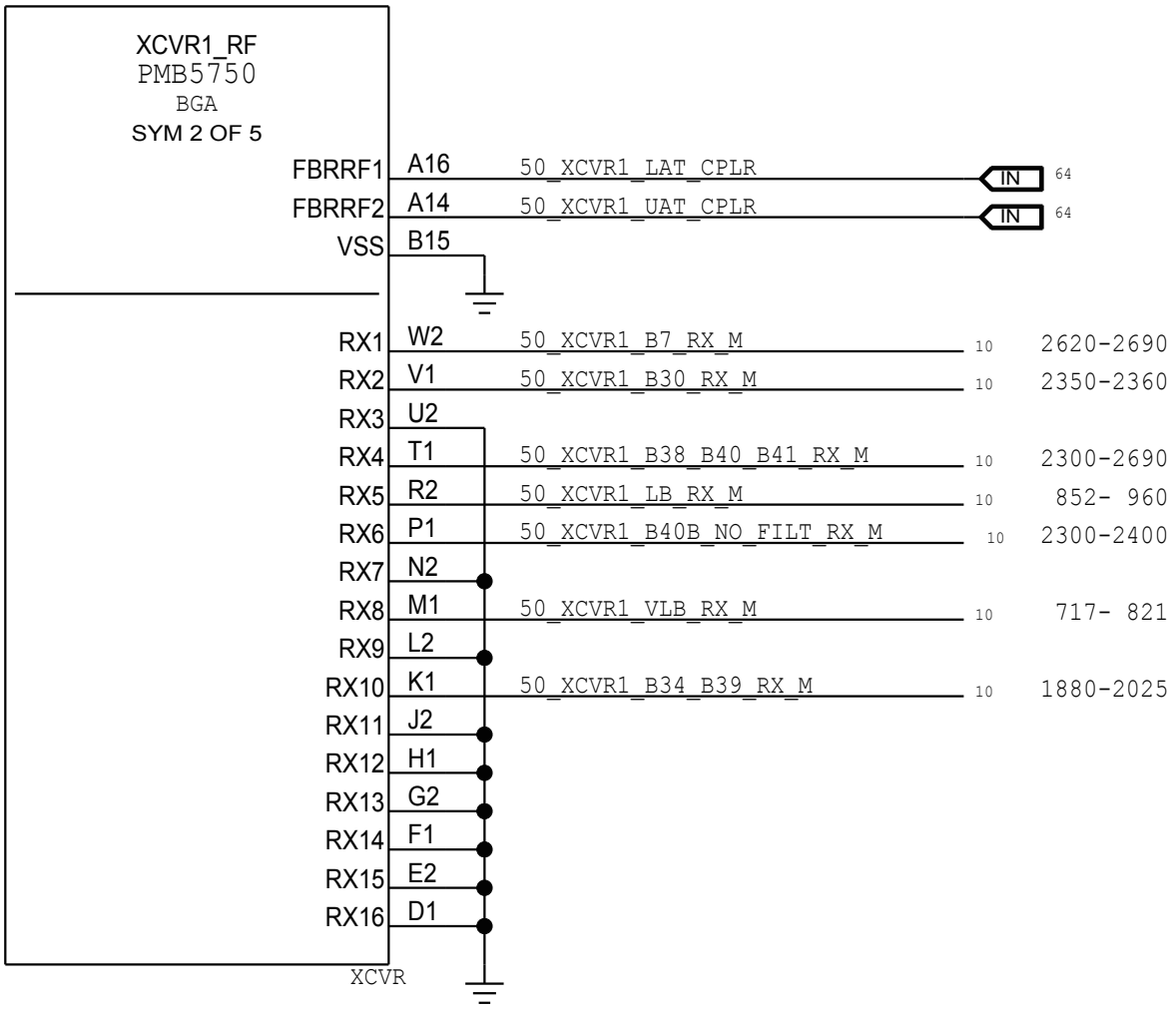
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131800068	1	CAP, 8.0PF, 0201, HQ	C6206_RF	D101
15282041	1	IND, 10NH, 0201, UH-Q	L6203_RF	D101
152800157	1	IND, 1.2NH, 0201, UH-Q	L6200_RF	D101
		NOSTUFF	C6212_RF	D101
131800068	1	CAP, 8.0PF, 0201, HQ	C6206_RF	D111
152800020	1	IND, 12NH, 0201, UH-Q	L6203_RF	D111
15282021	1	IND, 1.5NH, 0201, UH-Q	L6200_RF	D111
		NOSTUFF	C6212_RF	D111

PRIMARY RECIEVE

XCVR PRX

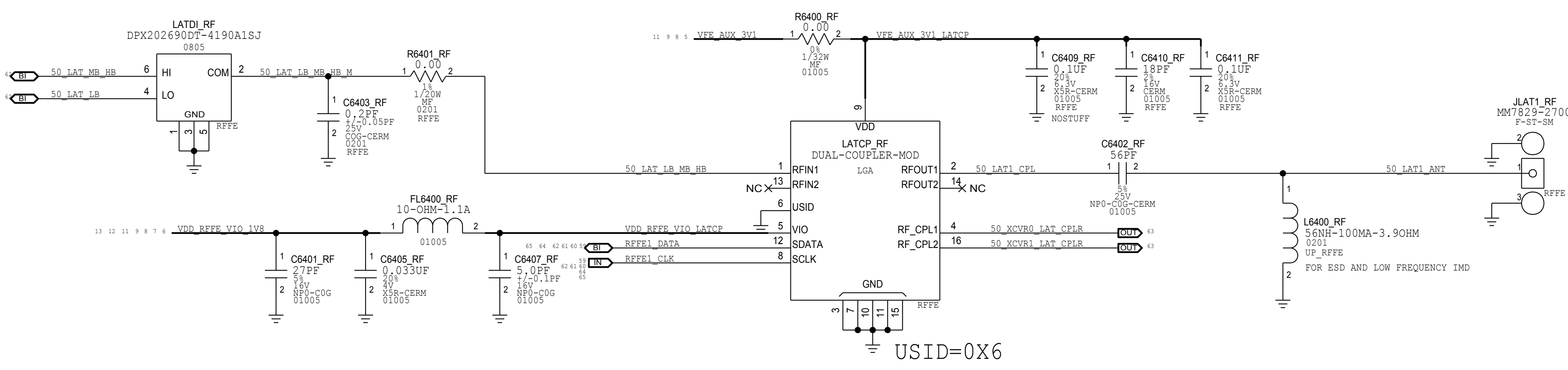


XCVR PRX

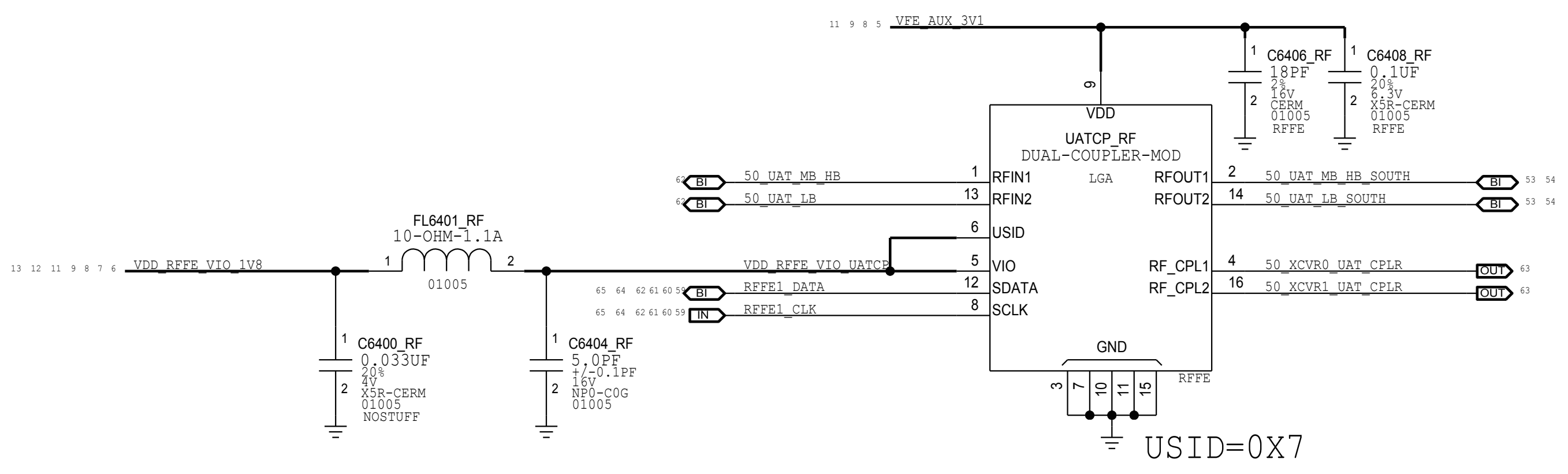


LOWER ANTENNA AND COUPLER

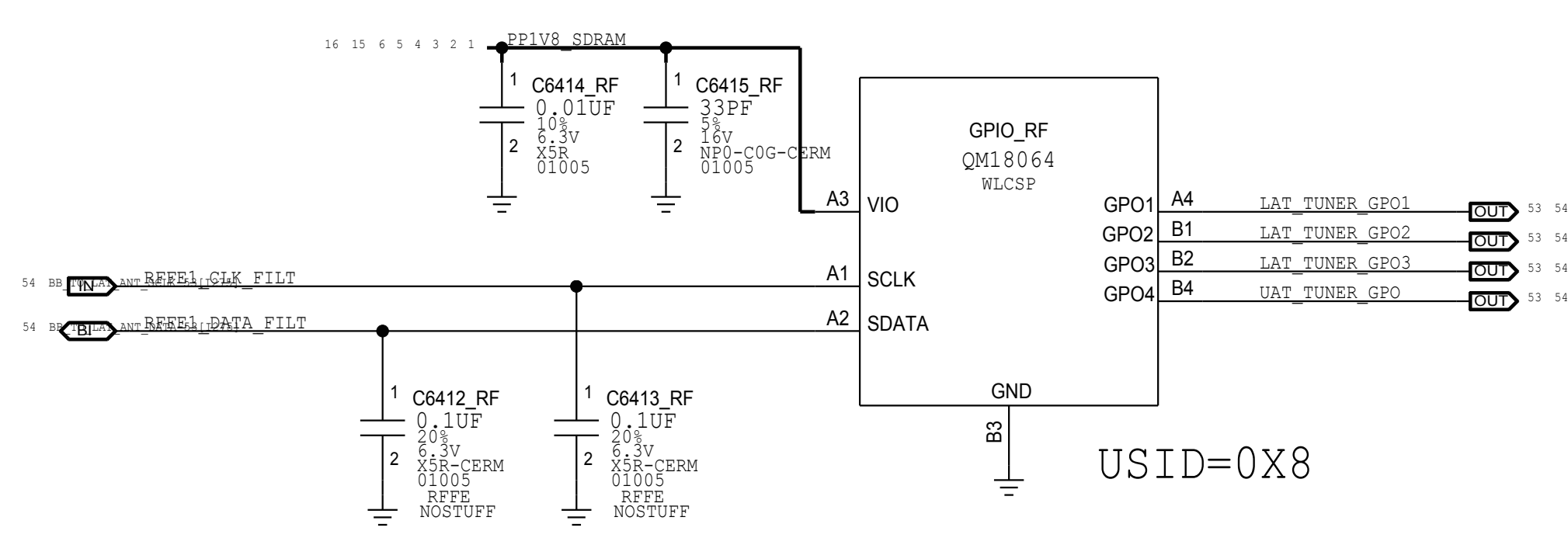
LOWER ANTENNA COUPLER



UPPER ANTENNA COUPLER

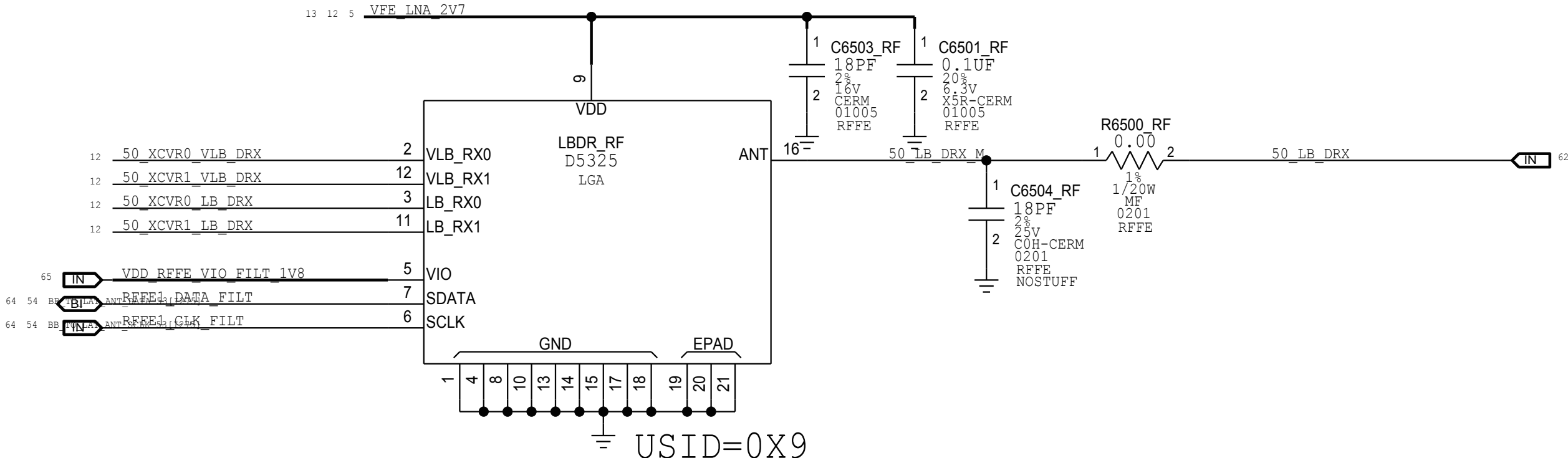
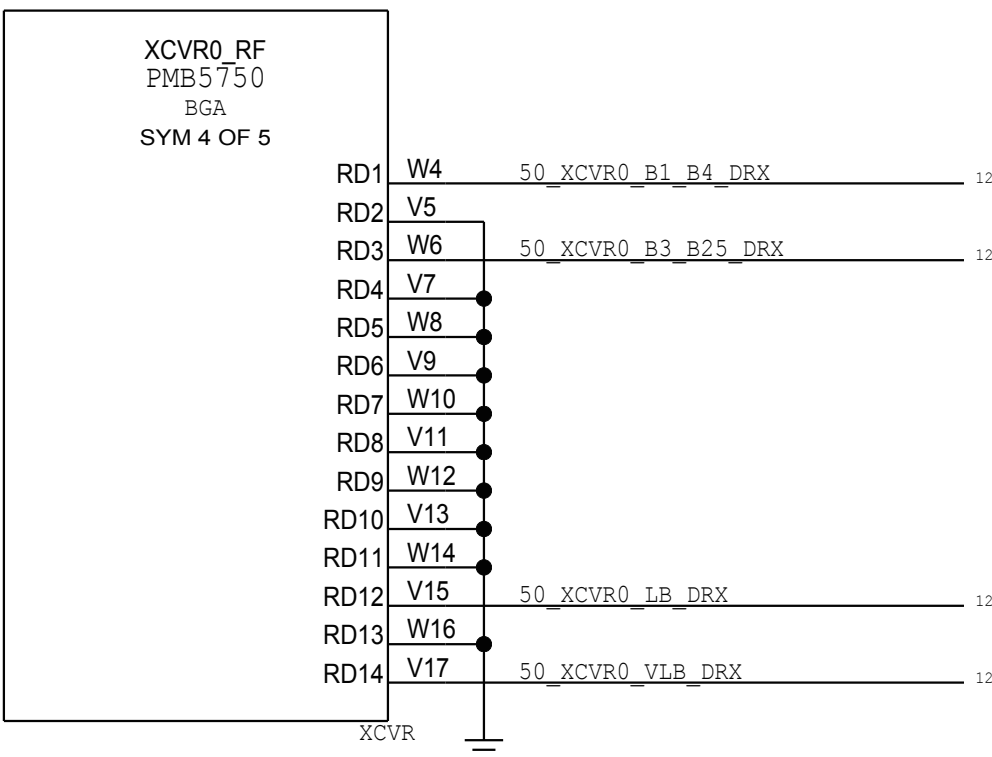


TUNER GPO

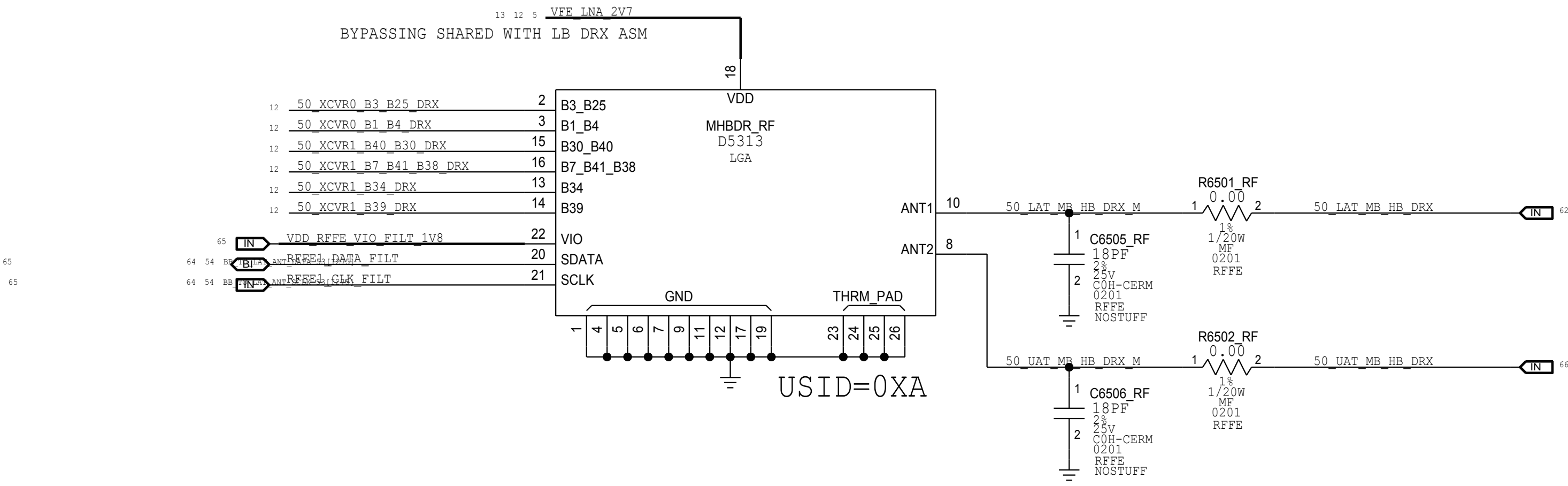
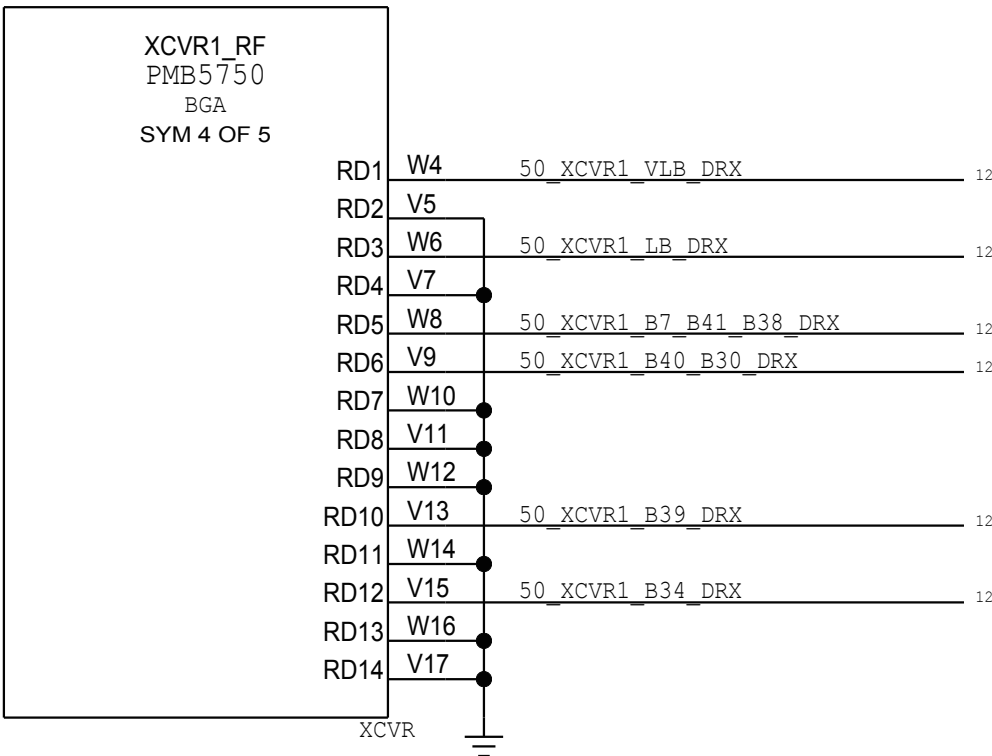


DIVERSITY RECEIVE ASM'S

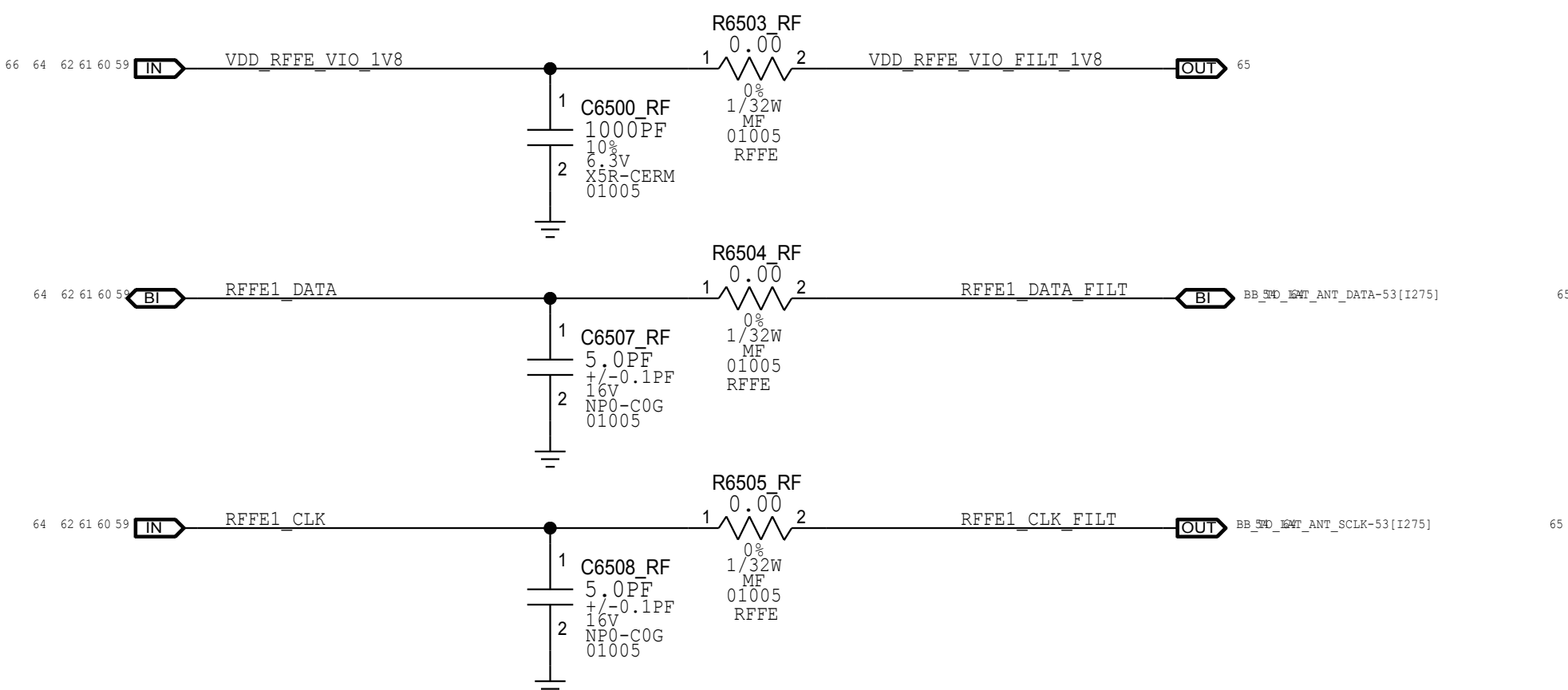
LB DRX ASM



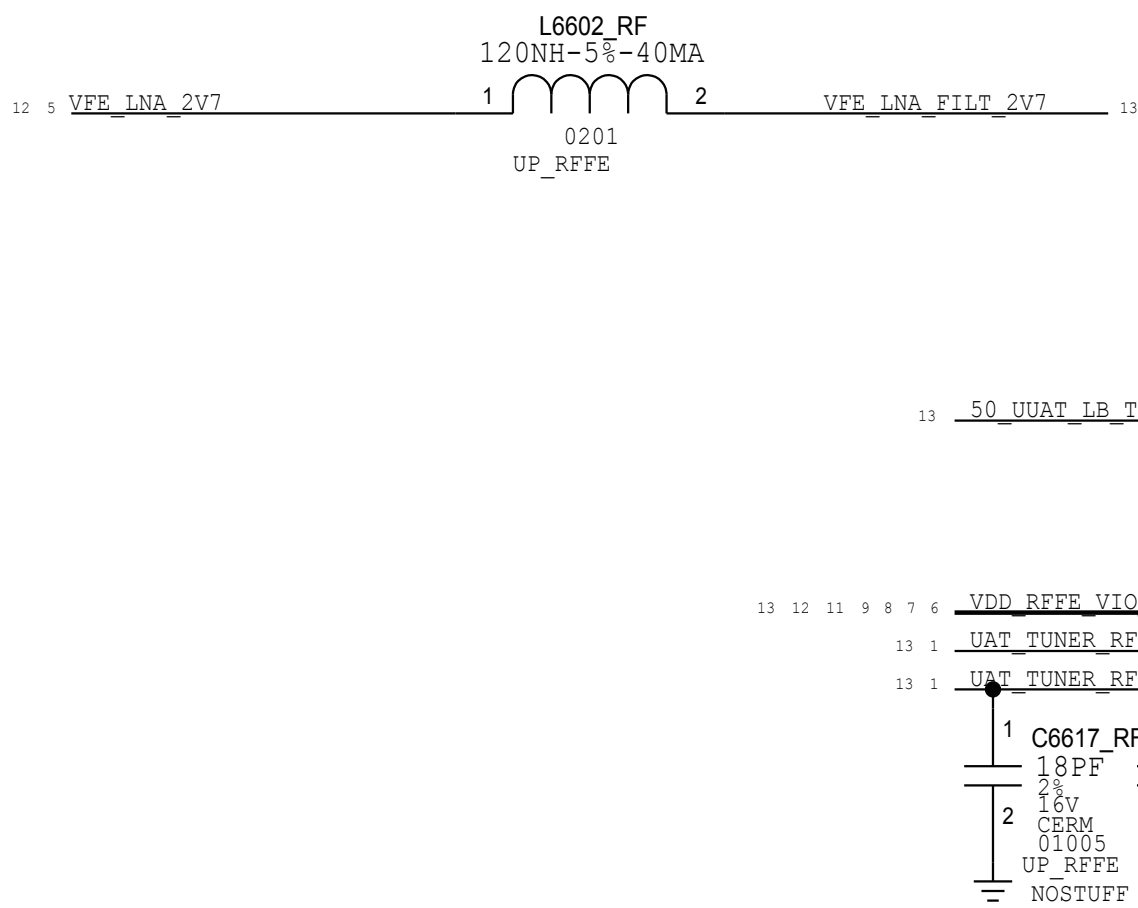
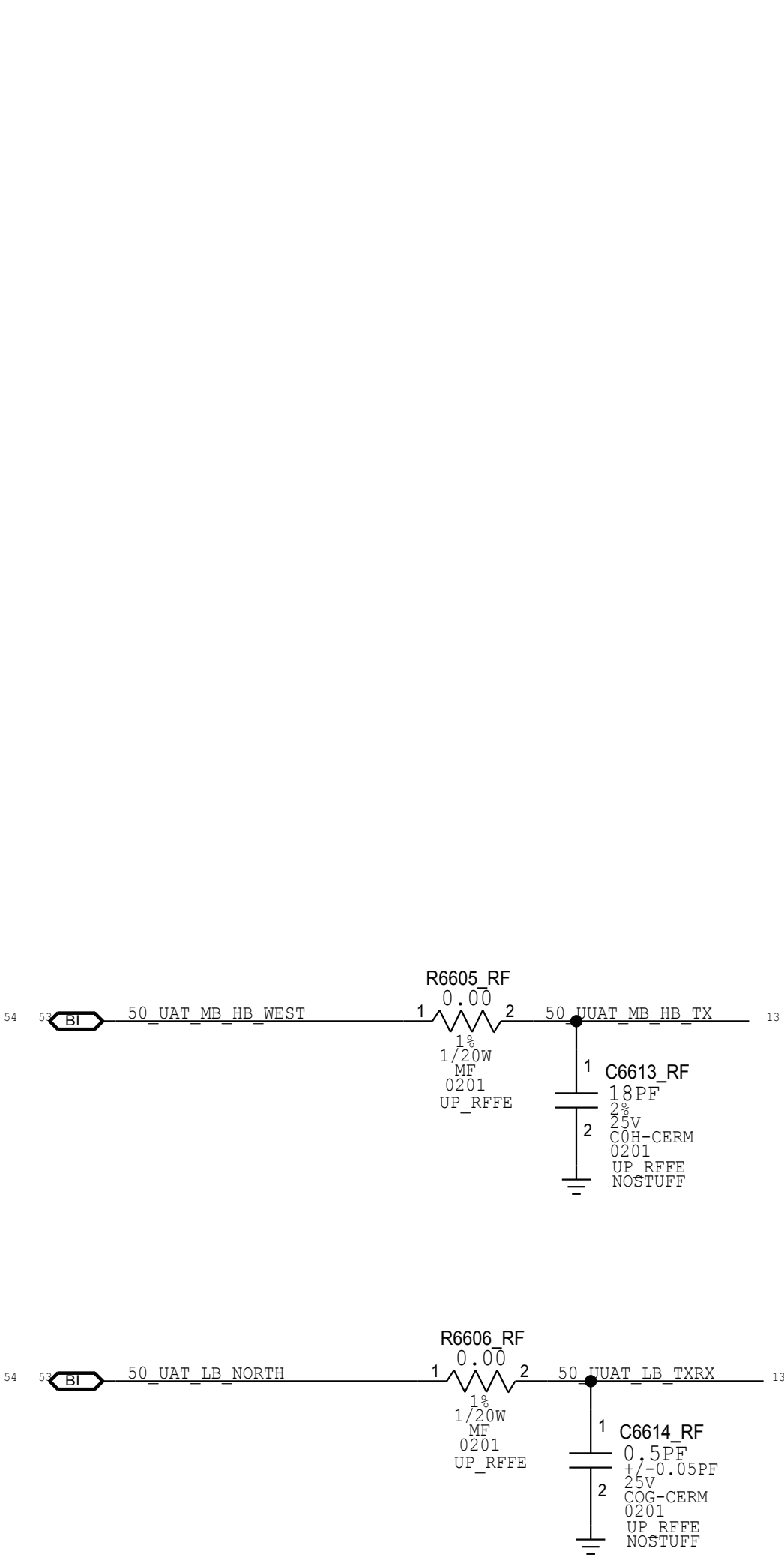
MB HB DRX ASM



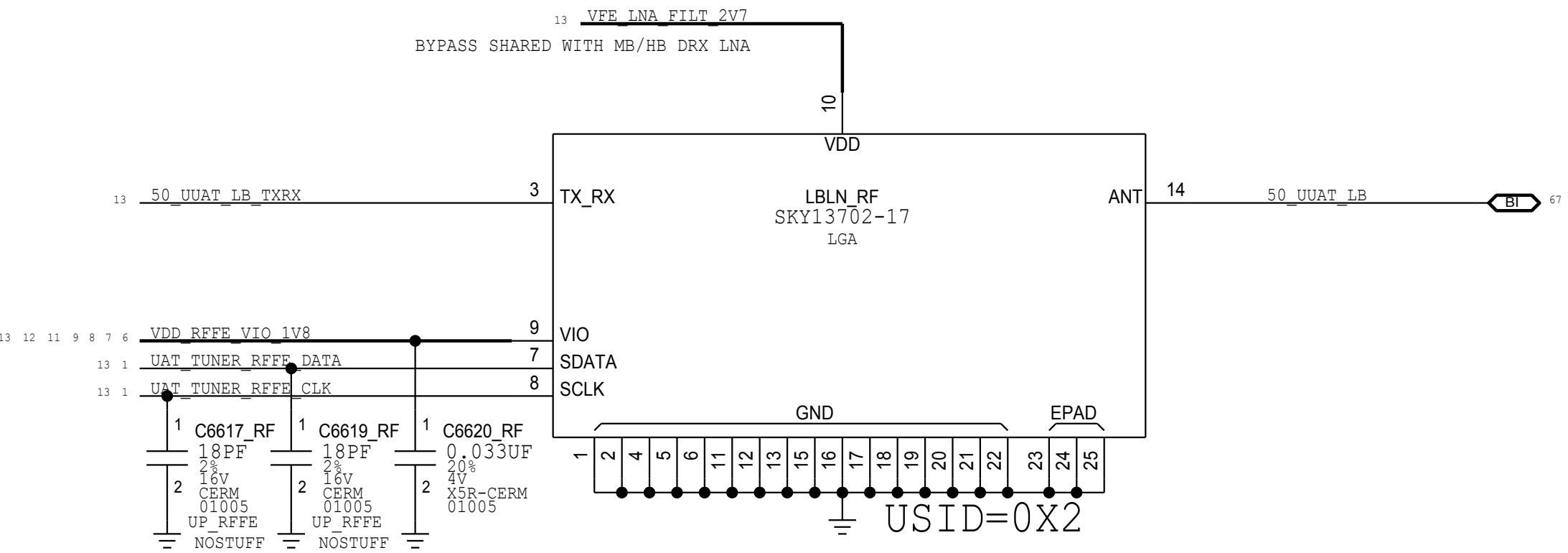
RFFE FILTERING



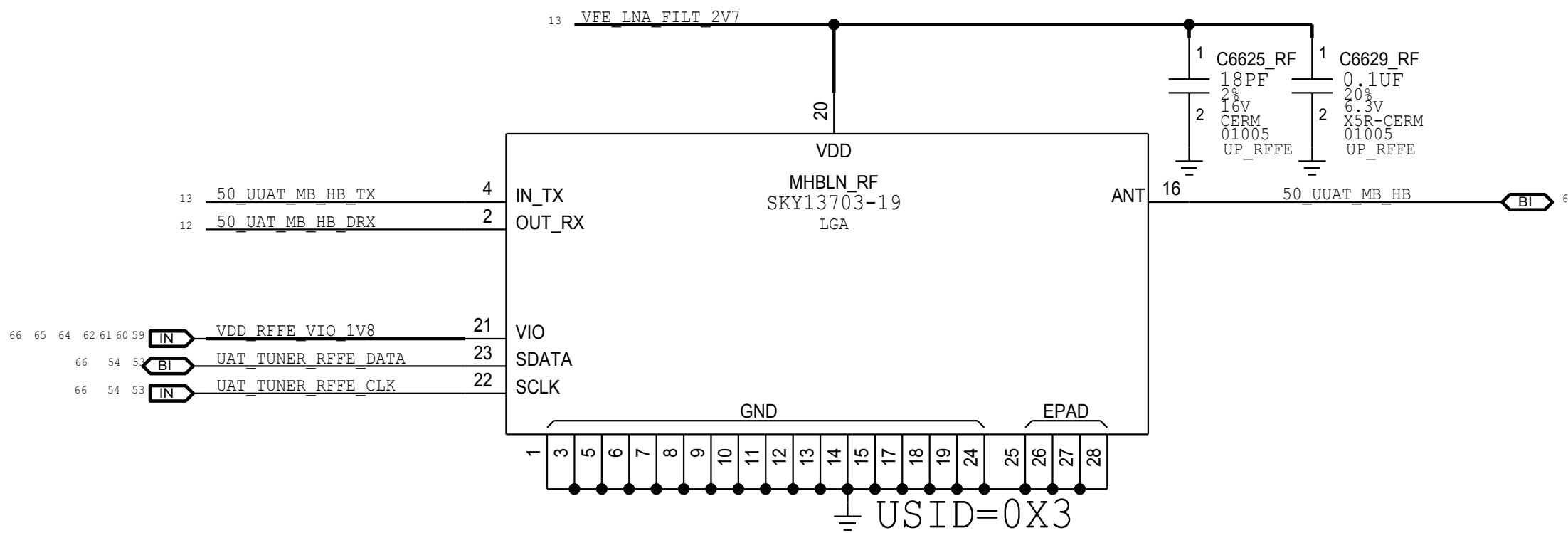
DIVERSITY RECEIVE LNAS



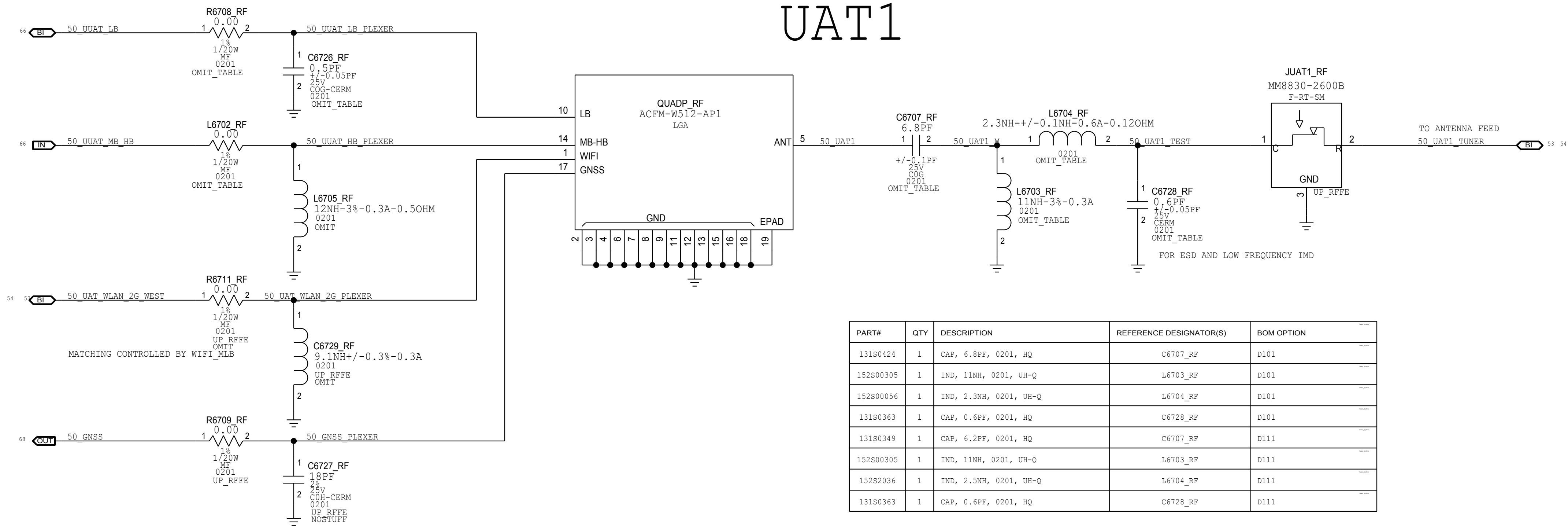
LB DRX LNA



MB/HB DRX LNA



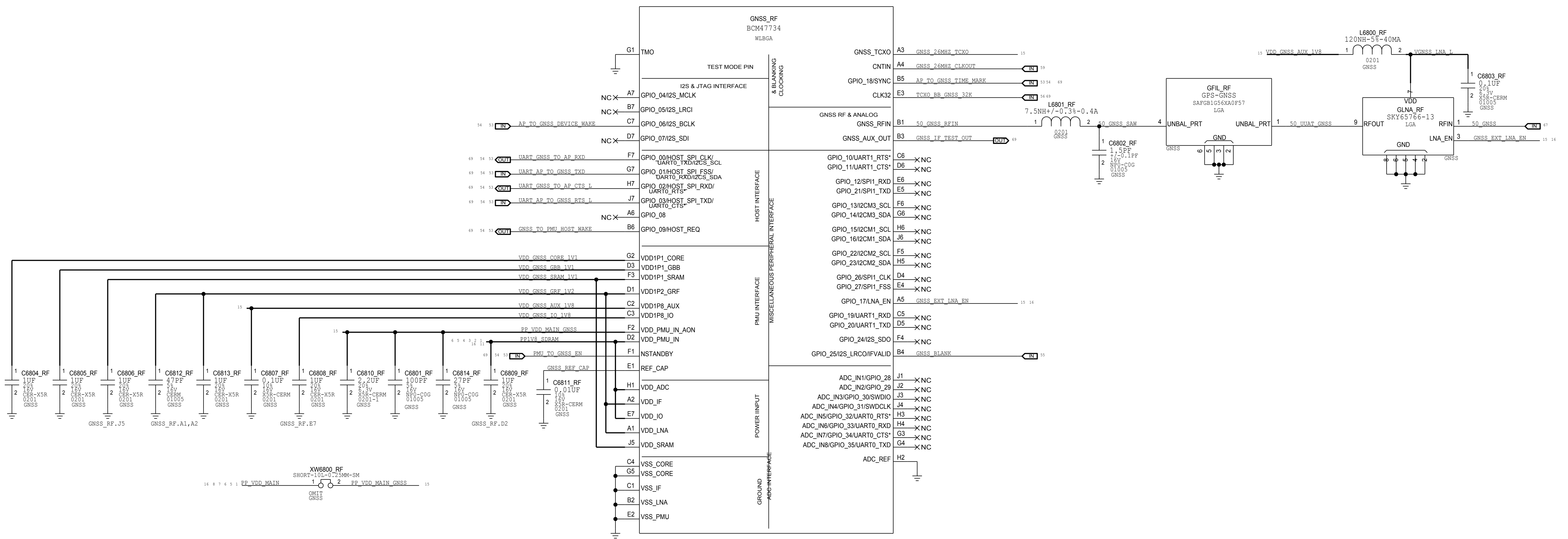
UPPER ANTENNA FEEDS



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
118S0724	1	RES, 0OHM, 0201	R6708_RF	D101
131S0425	1	CAP, 0.5PF, 0201, HQ	C6726_RF	D101
118S0724	1	RES, 0OHM, 0201	L6702_RF	D101
152S00020	1	IND, 12NH, 0201, UH-Q	L6705_RF	D101
118S0724	1	RES, 0OHM, 0201	R6708_RF	D111
131S0425	1	CAP, 0.5PF, 0201, HQ	C6726_RF	D111
118S0724	1	RES, 0OHM, 0201	L6702_RF	D111
152S00020	1	IND, 12NH, 0201, UH-Q	L6705_RF	D111

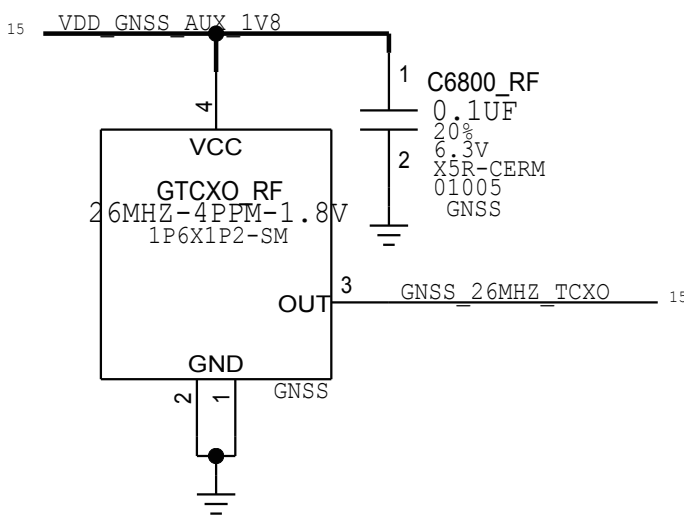
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S0424	1	CAP, 6.8PF, 0201, HQ	C6707_RF	D101
152S00305	1	IND, 11NH, 0201, UH-Q	L6703_RF	D101
152S00056	1	IND, 2.3NH, 0201, UH-Q	L6704_RF	D101
131S0363	1	CAP, 0.6PF, 0201, HQ	C6728_RF	D101
131S0349	1	CAP, 6.2PF, 0201, HQ	C6707_RF	D111
152S00305	1	IND, 11NH, 0201, UH-Q	L6703_RF	D111
152S2036	1	IND, 2.5NH, 0201, UH-Q	L6704_RF	D111
131S0363	1	CAP, 0.6PF, 0201, HQ	C6728_RF	D111

GNSS LNA + RECEIVER



TCXO

KEEP AWAY FROM HEAT SOURCES



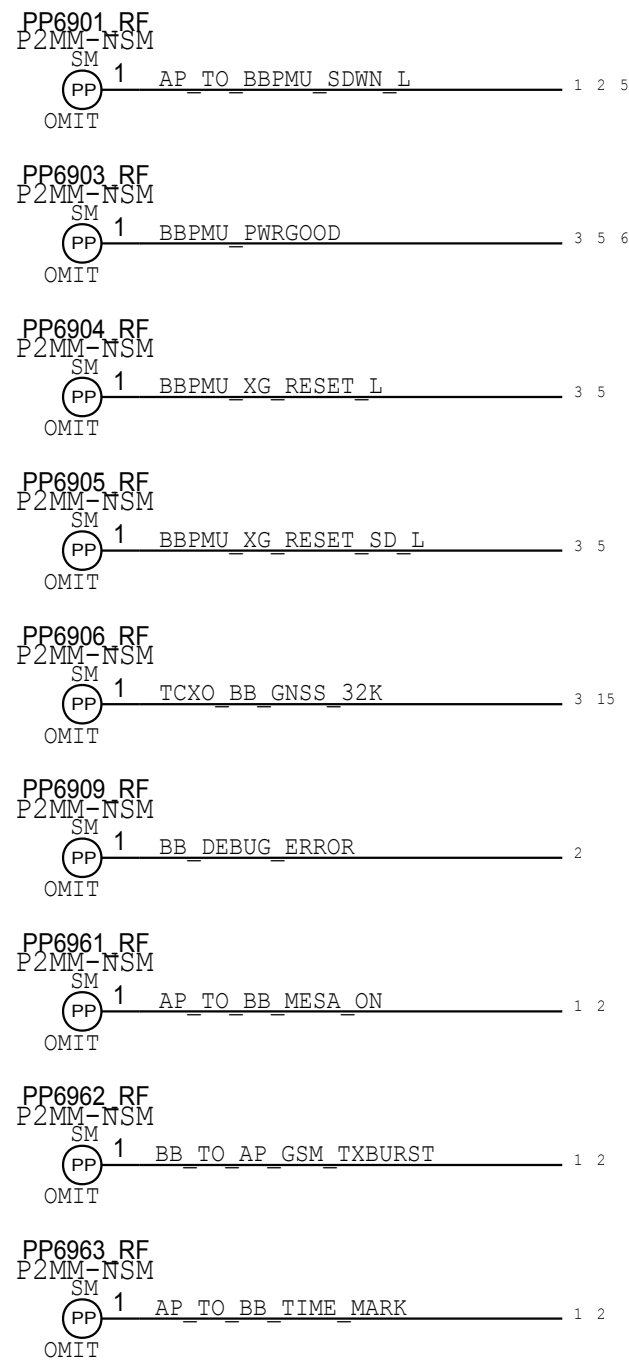
MLB TEST POINTS

SIM CARD CONNECTOR

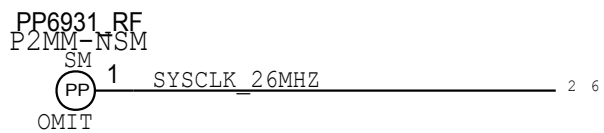
HARDWARE ID

DEV CONFIG

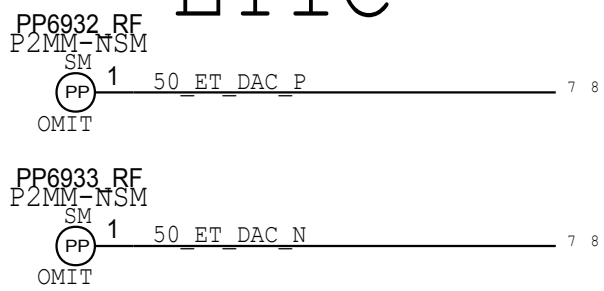
BASEBAND



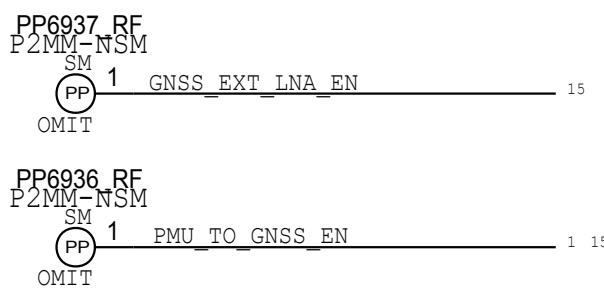
XCVR



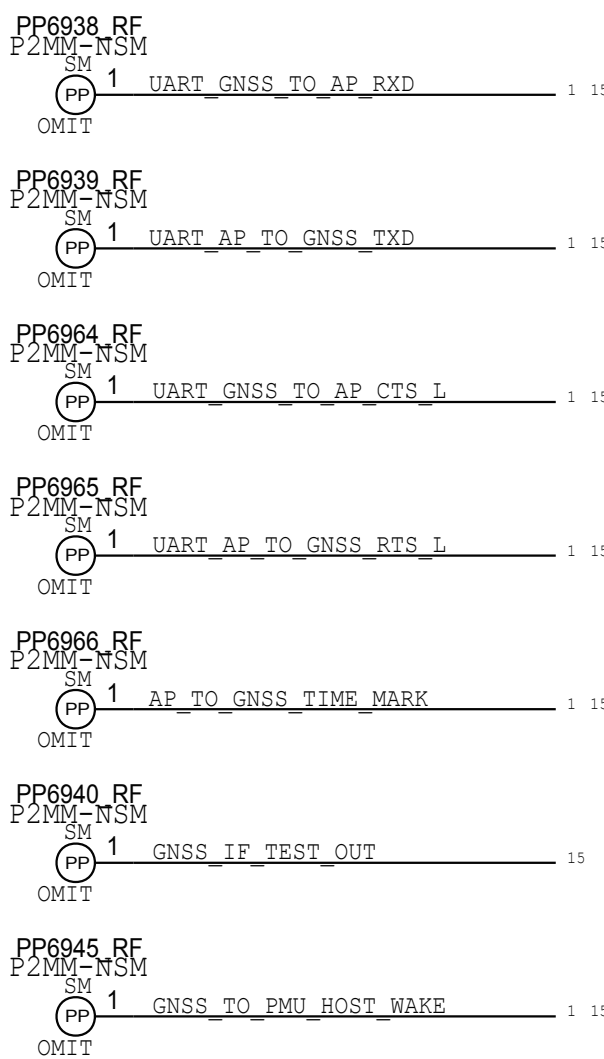
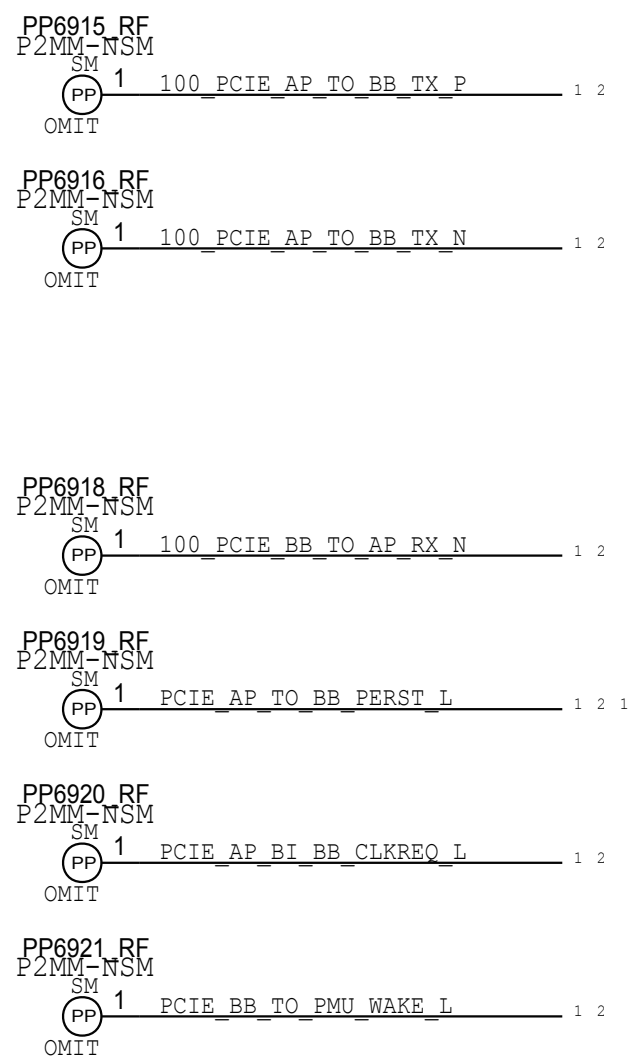
ETIC



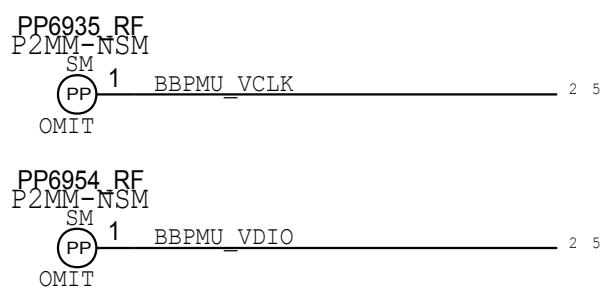
GNSS



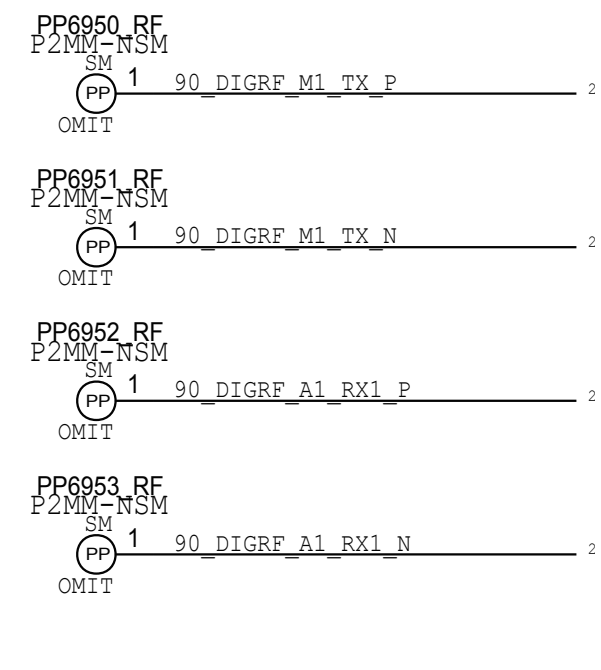
PCIE



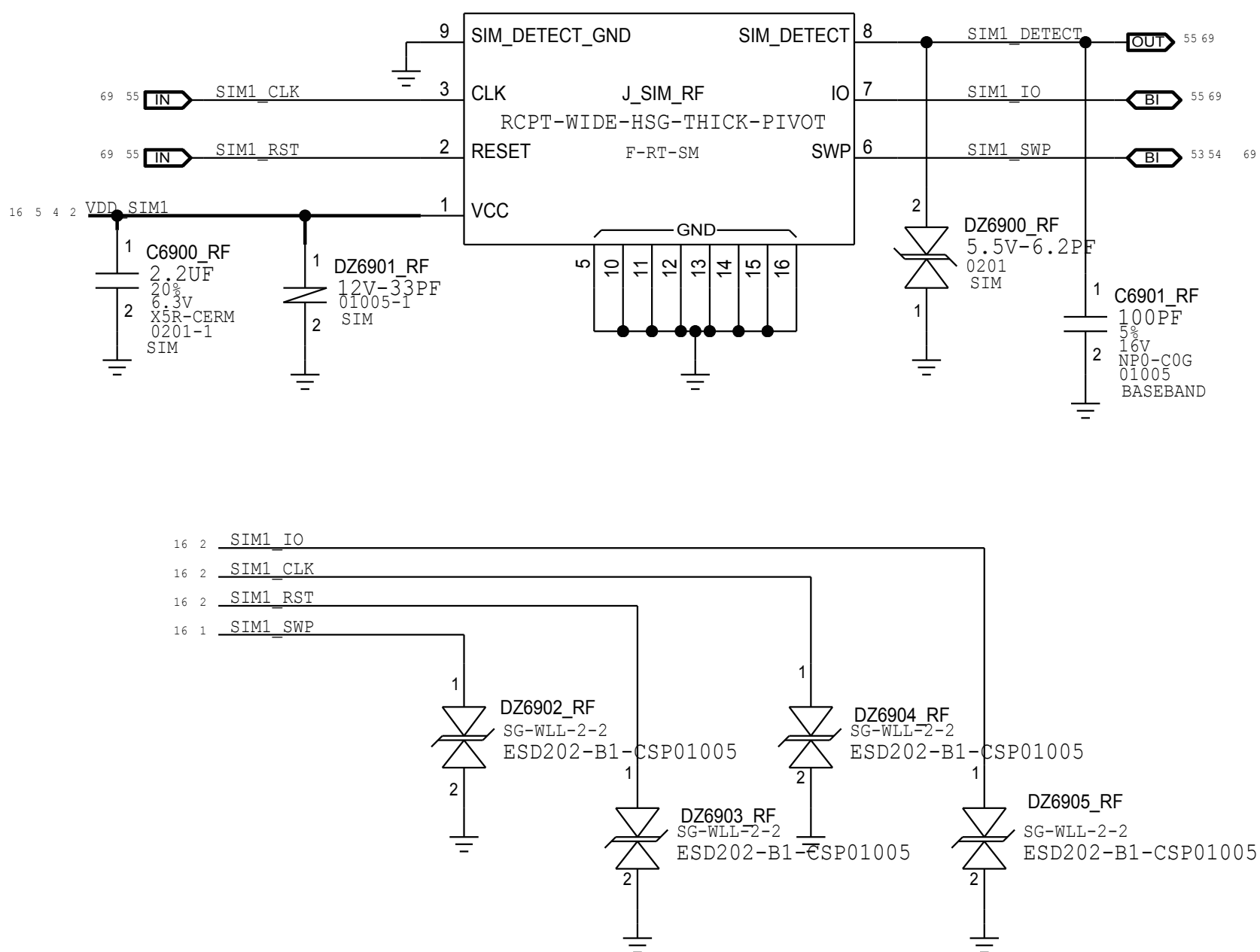
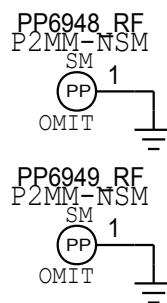
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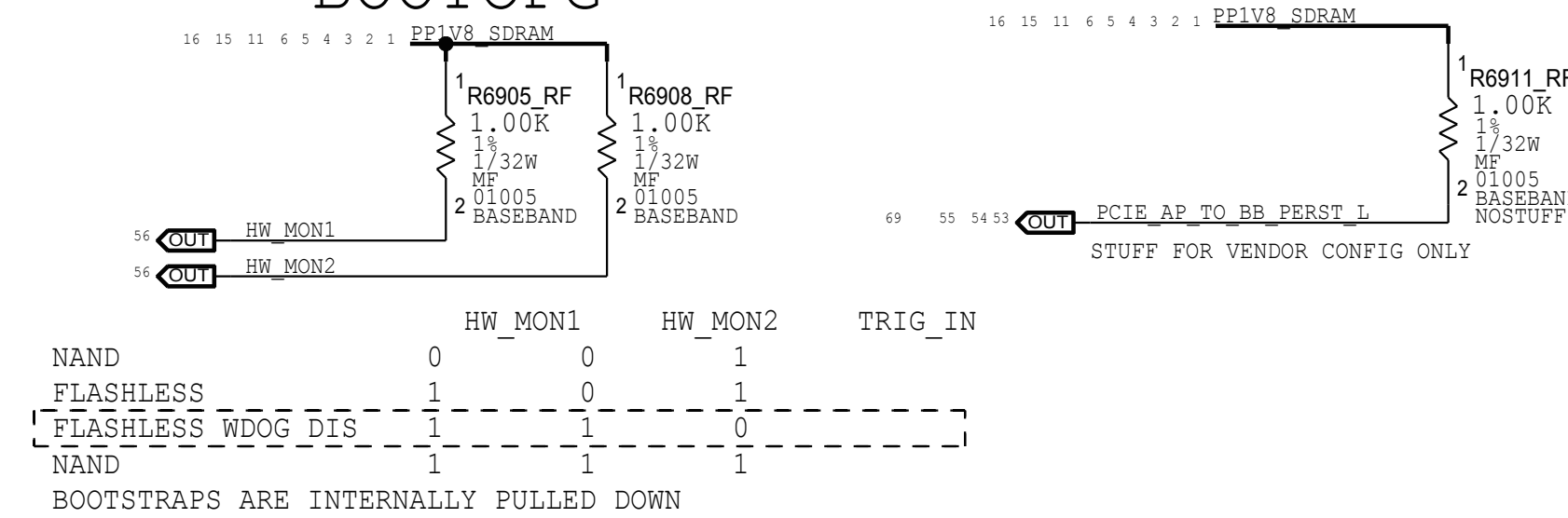
DIGRF



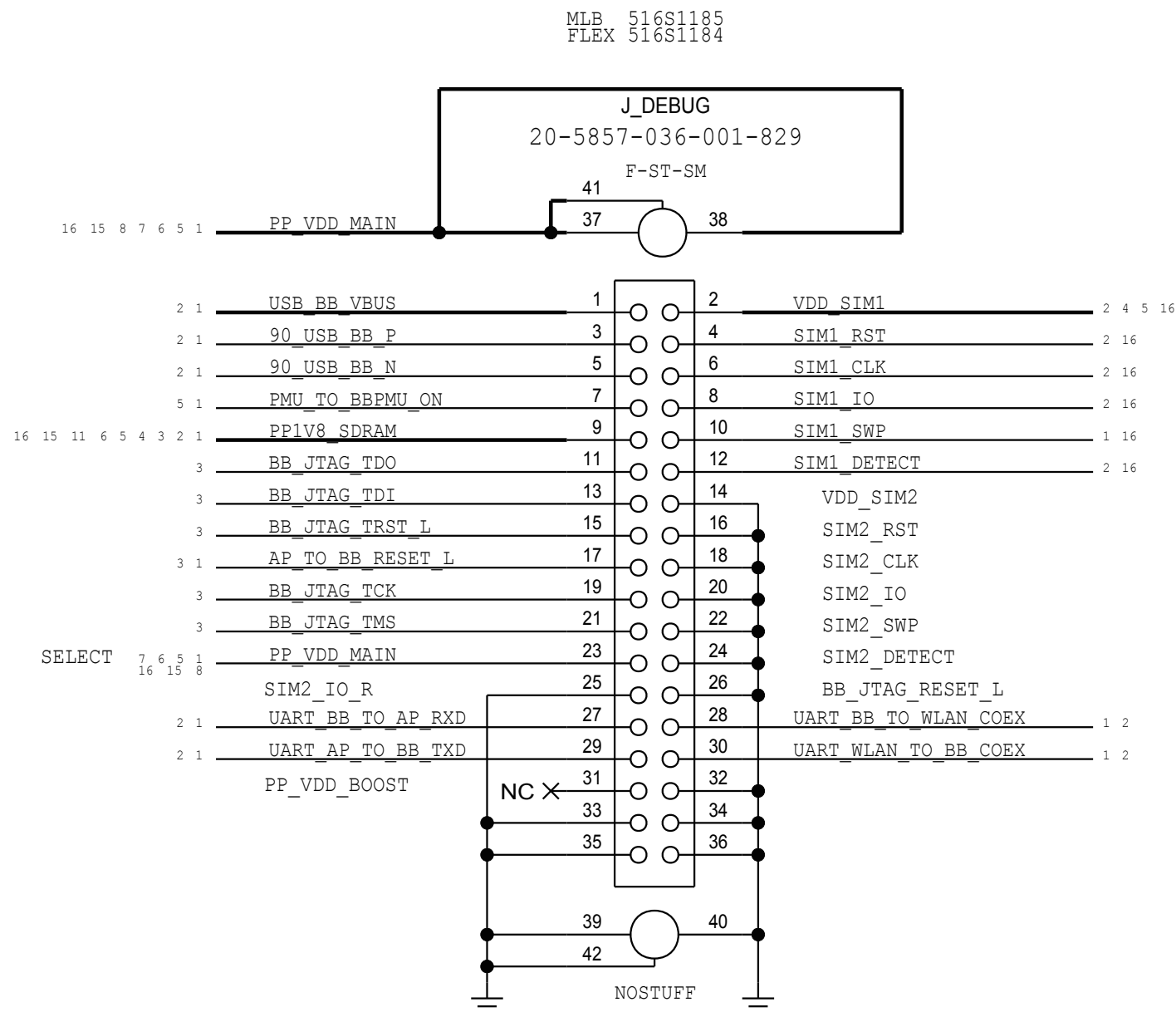
PCIE GND



BOOTCFG

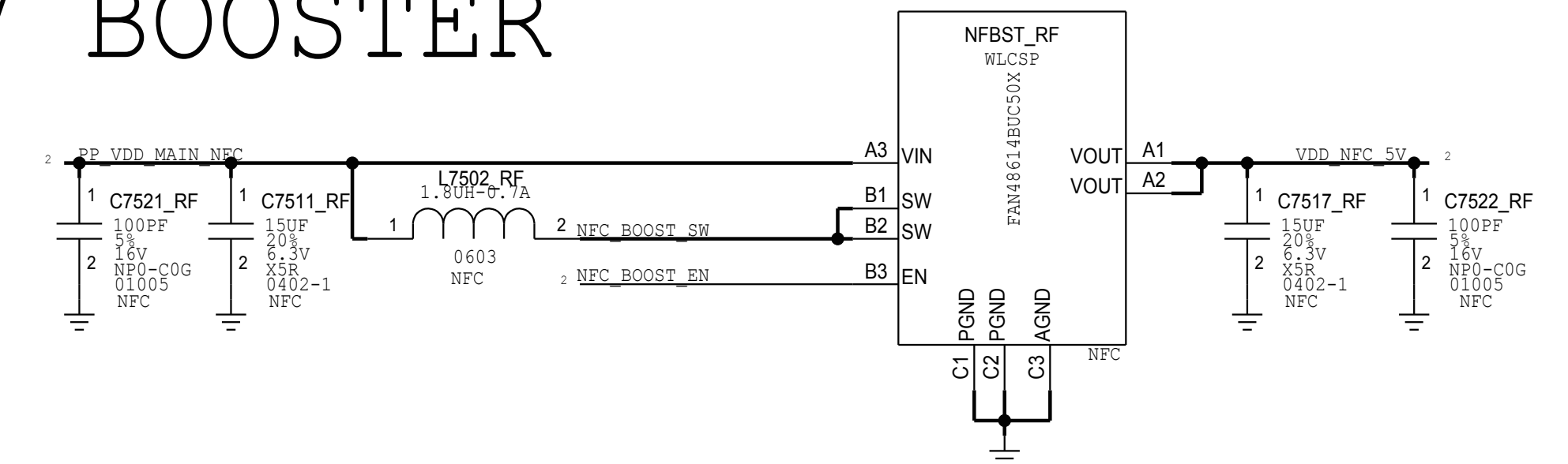


DEBUG CONNECTOR

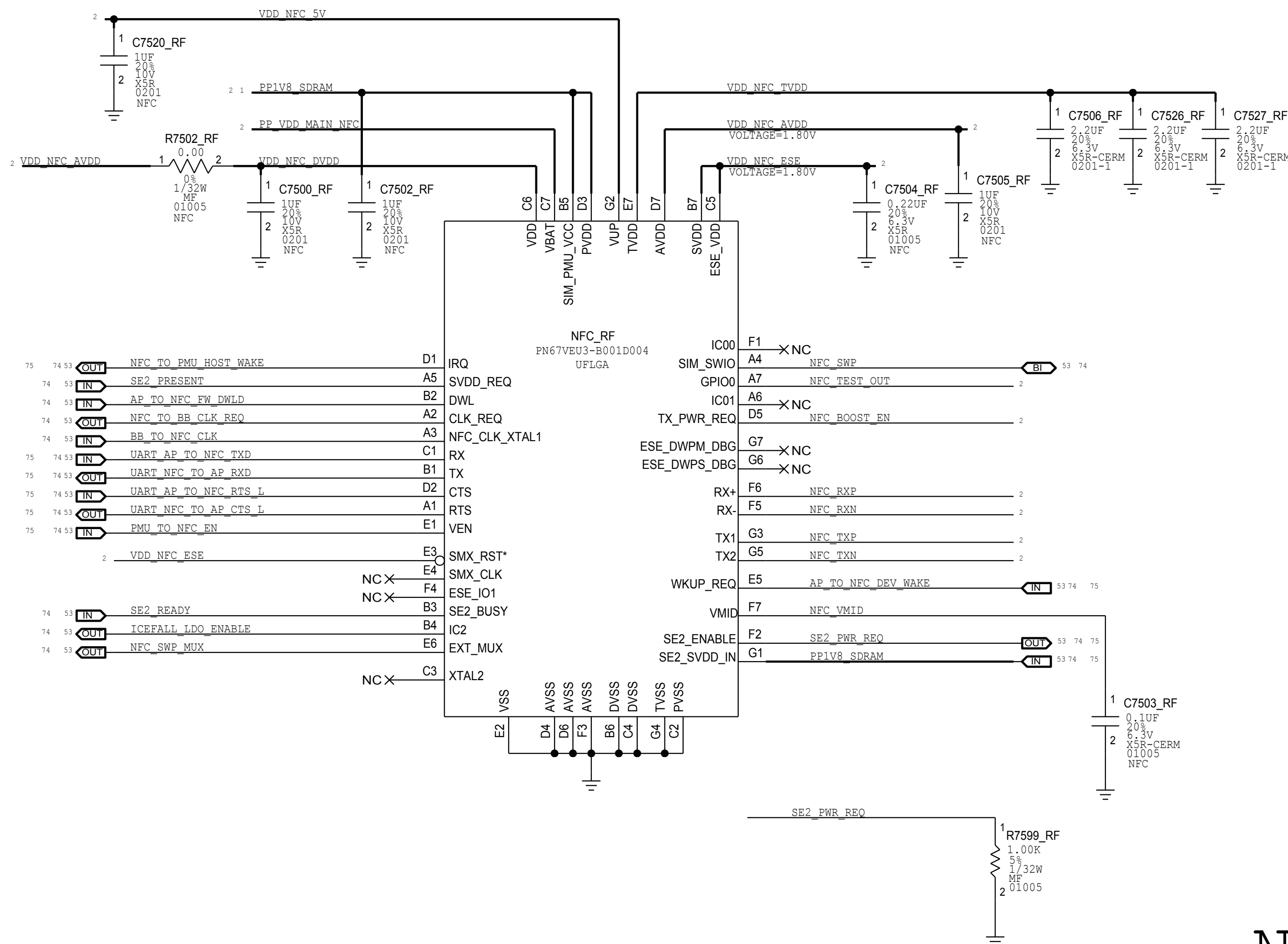


STOCKHOLM

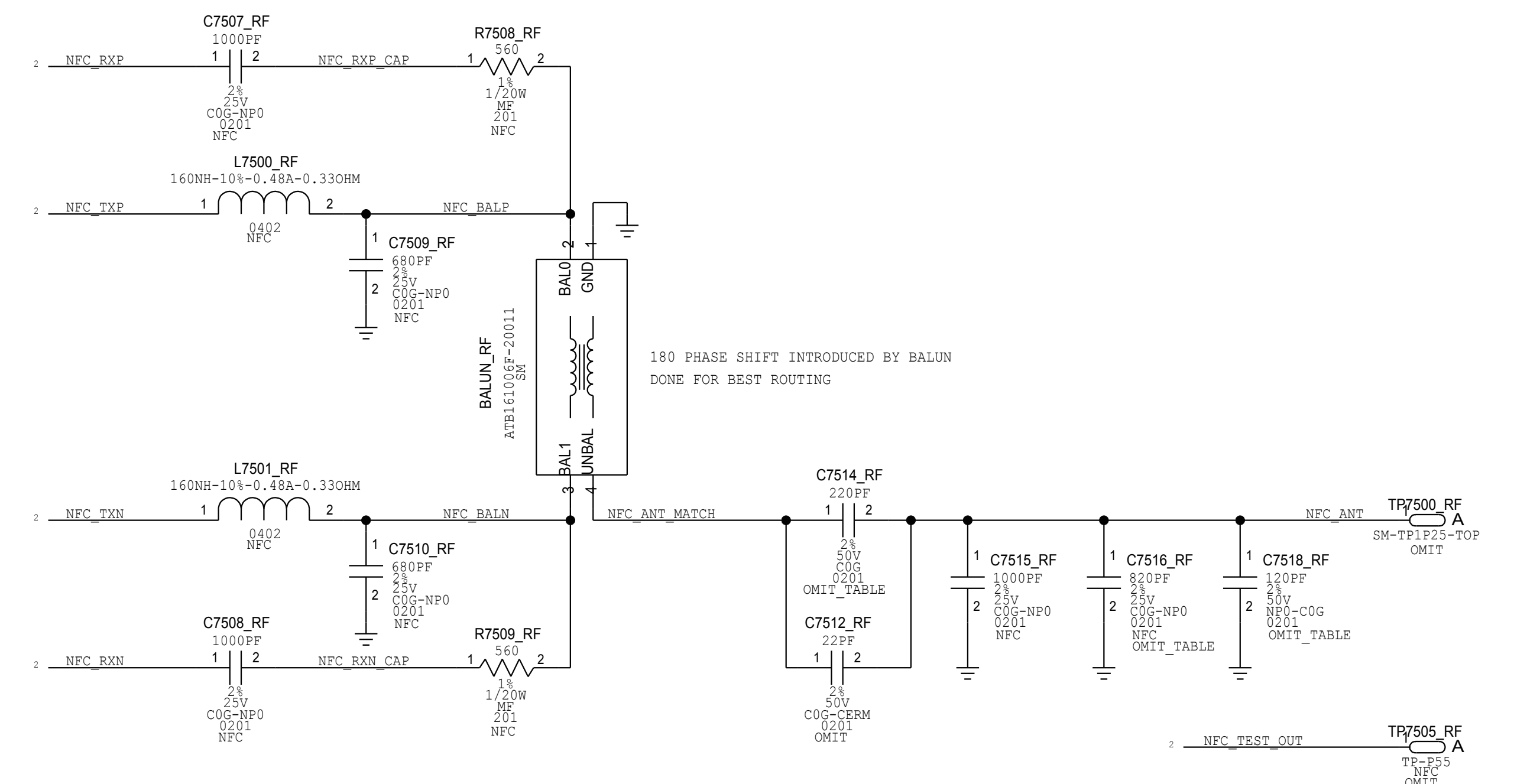
5V BOOSTER



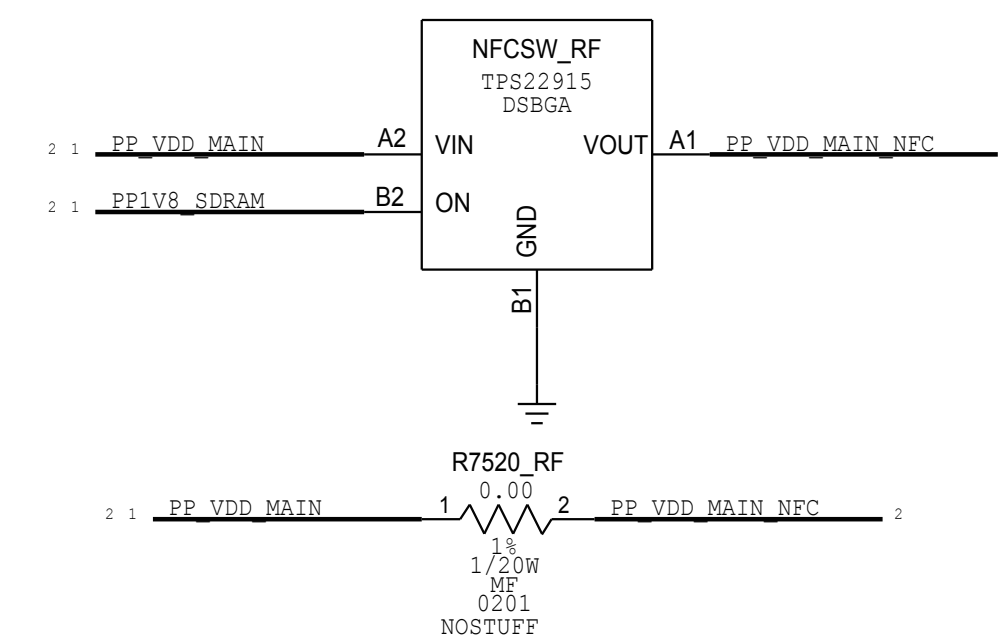
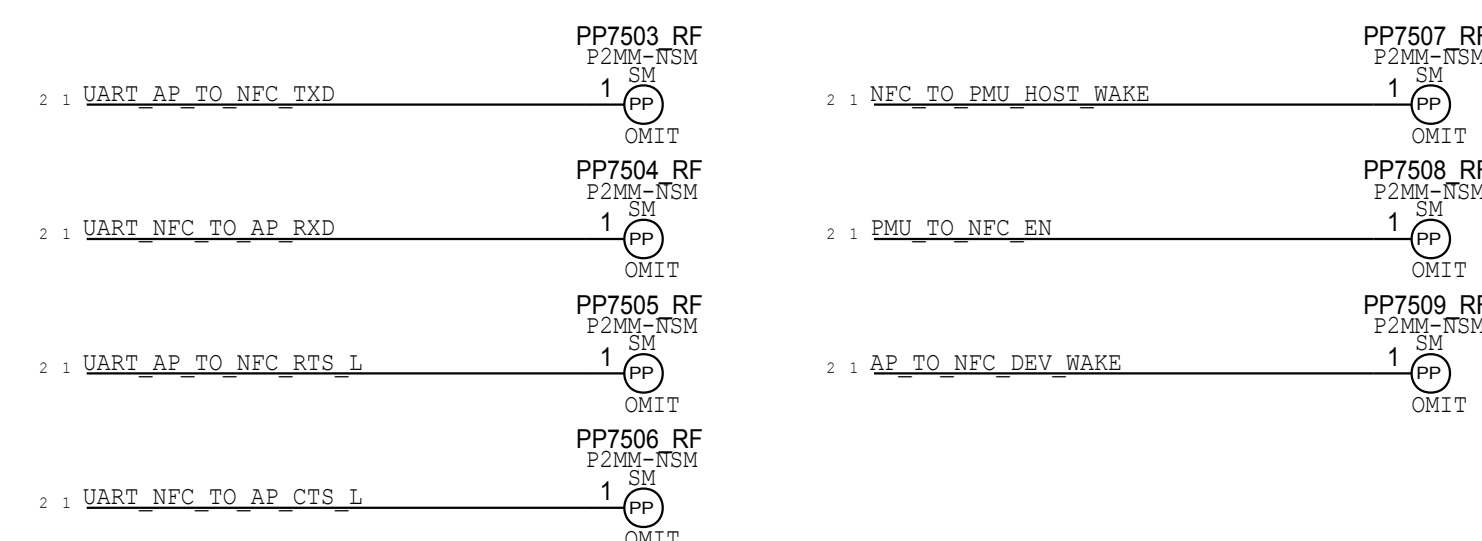
NFC CONTROLLER



NFC FRONT END



NFC LOAD SWITCH



8		7		6		5		4		3		2		1					
D	1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%. 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS. 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.											REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE				
												5	0006232160	ENGINEERING RELEASED	2016-05-16				
D111 RADIO MLB FF																			
Mon May 16 17:43:10 2016																			
<table><tr><td>PDF PAGE</td><td>CONTENTS</td></tr><tr><td>2</td><td>METROCIRC</td></tr></table>																PDF PAGE	CONTENTS	2	METROCIRC
PDF PAGE	CONTENTS																		
2	METROCIRC																		

METROCIRC

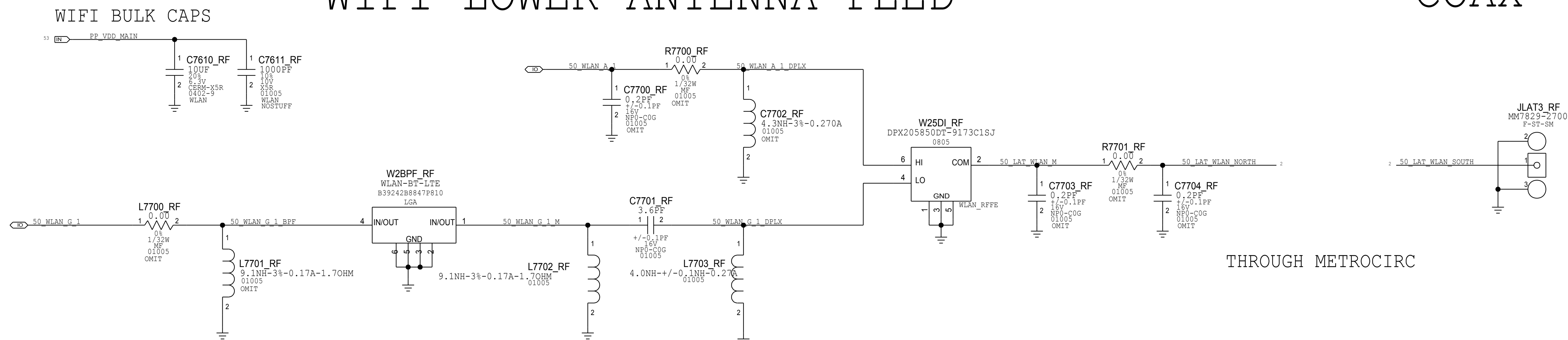
D111 NORTH-SOUTH METROCIRC
339S00147

D111 EAST-WEST METROCIRC
339S00146



WIFI LOWER ANTENNA FEED

COAX



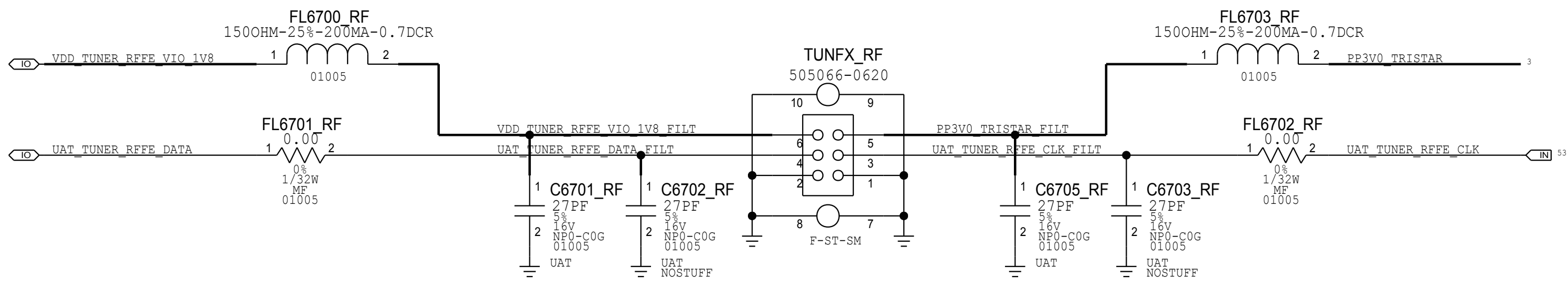
THROUGH METROCIRC

ICE16.1/2 EVT2

MATCHING CONTROLLED BY WIFI_MLB

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
118S0608	2	RES, WF, 1K OHM, 1%, 1/32W, 0205	R6901_RF, R6906_RF	

UAT TUNER FLEX

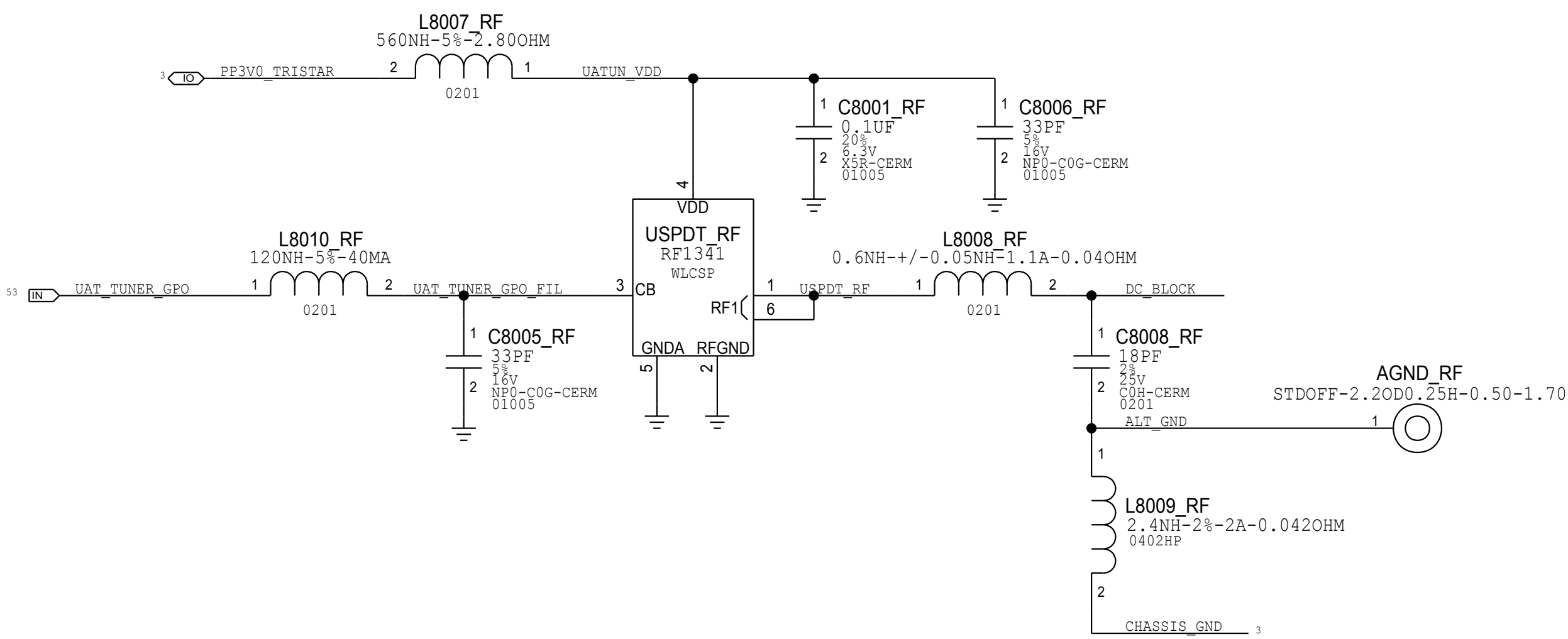


PINOUT IS ROTATED 180 COMPARED TO D101

ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152800482	152800490	ALTERNATE	L8007_RF	MURATA LQW03CAR56J00

ALT UAT1 GND

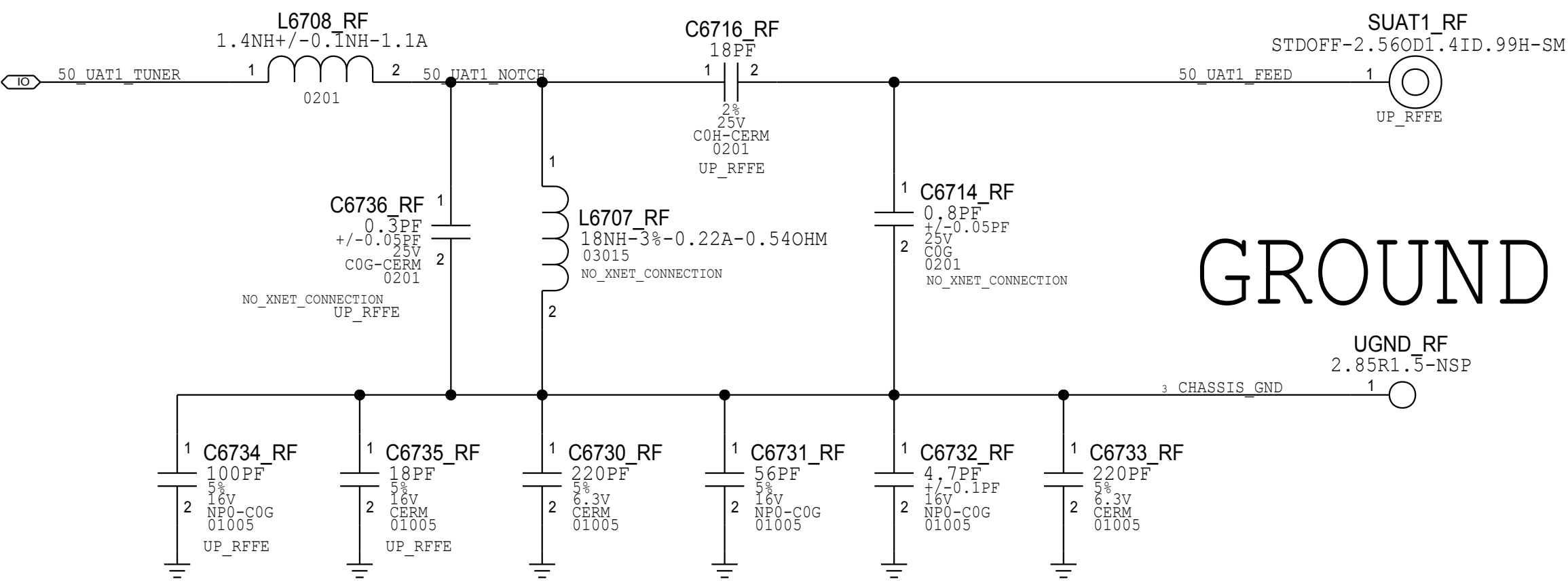


UAT TUNER

REMOVED

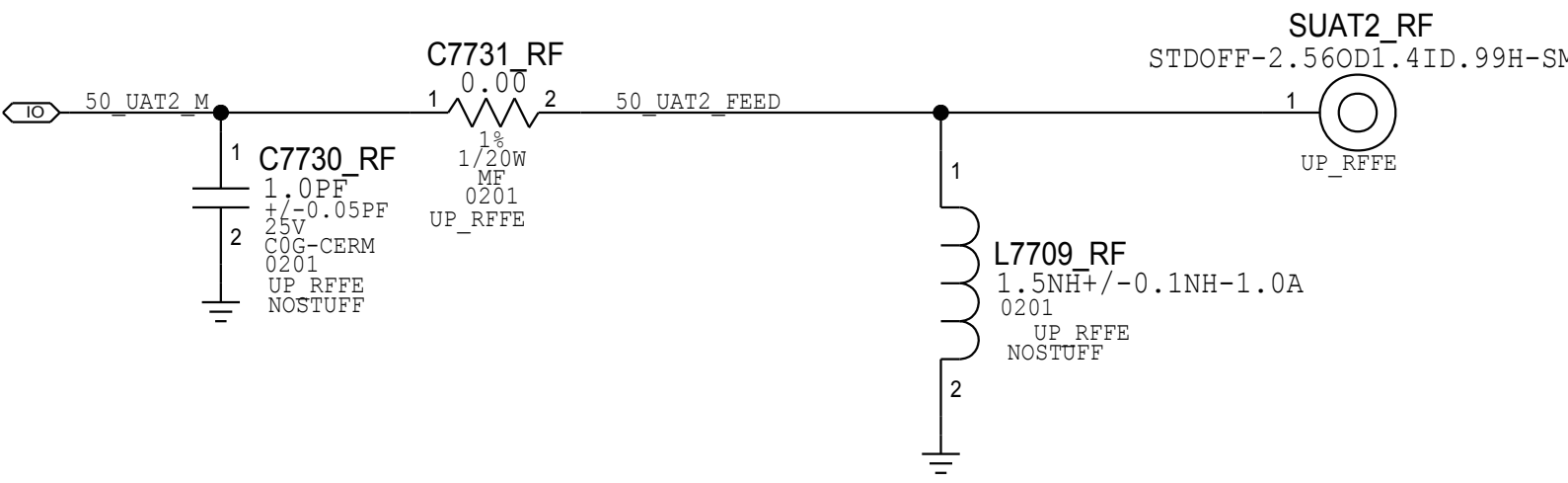
USID=0XC

LB/MLB/GNSS/MB/HB
STANDOFF

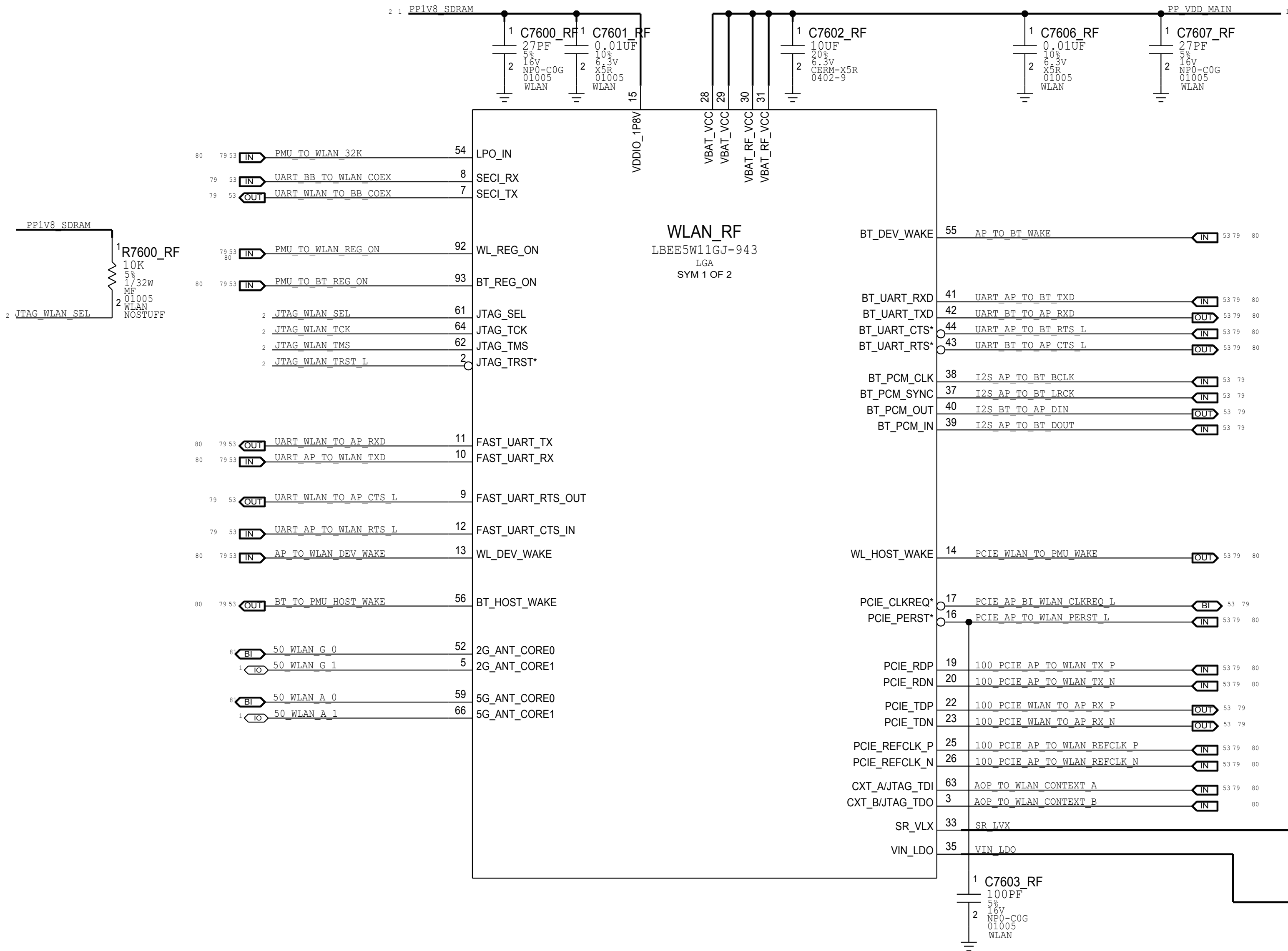


GROUND RING

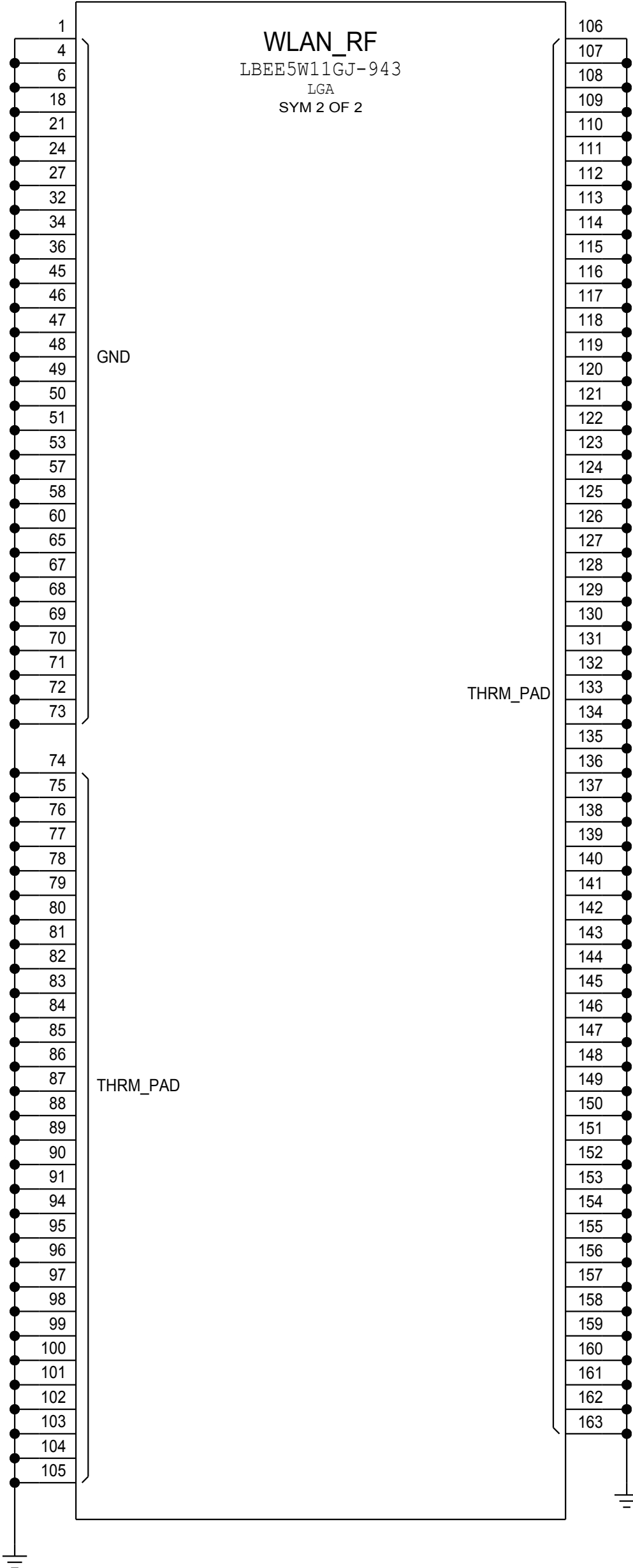
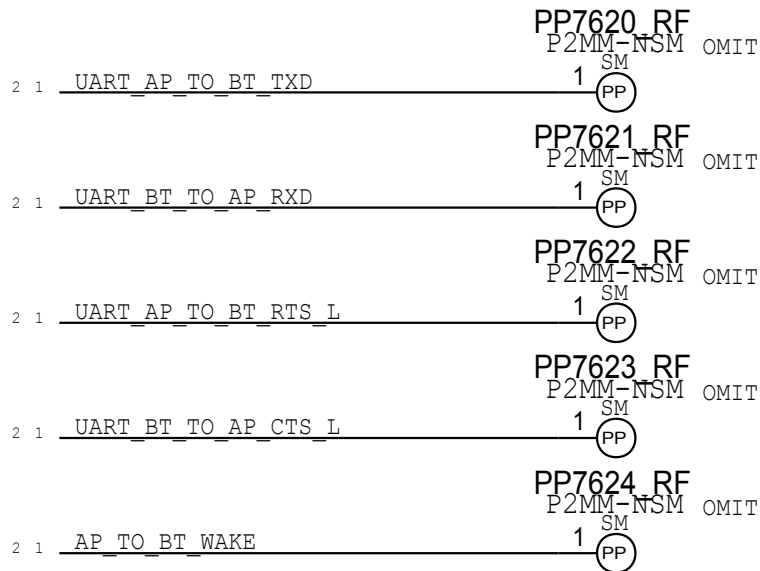
5G WIFI
STANDOFF



WIFI/BT



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339800201	339800199	ALTERNATE	WLAN_RF	ALT WIFI/BT MODULE



WIFI UPPER ANTENNA FEEDS

